

“It Takes Two to Tango”: Joint Location Choices and Local Shocks*

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September 2026

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Abstract

I examine how joint decision-making shapes adjustment to local shocks. Combining geocoded data from the Panel Study of Income Dynamics with measures of exposure to the China Trade Shock, I find that exposed individuals are more likely to leave their commuting zones, but adjustment within couples is asymmetric. Out-migration rises when the husband is exposed, but not when the wife is. Wives’ labor force participation, meanwhile, rises when husbands are exposed and falls when wives are directly exposed. These findings suggest that the added-worker effect, alongside spatial mobility, is central to adjustment to local shocks. I then extend a dynamic spatial and trade equilibrium model to include couples who jointly choose a location and a sector for each spouse and decide whether the wife participates in the labor market. Incorporating couples does not alter the aggregate effects of the shock, but it changes who bears it. Because a wife’s participation responds to her wage relative to her husband’s, female labor supply is one-third less elastic than a single-agent economy predicts, so female wages absorb more of the adjustment. Relative to an economy without couples, single women gain 20.5% more, while single men gain 5.4% less. For couples, the two adjustment margins are asymmetric substitutes: couples move more when the participation response is dampened, but wives do not work more when migration is shut down. The added-worker effect insures all couples, whereas migration insures only those directly exposed.

Keywords: Local labor market, Household structure, Migration decisions

JEL Codes: F16, J12, J61, R23

*I am deeply grateful to my advisors, Ana Reynoso, Sebastián Sotelo, Javier Cravino, and Kathryn Dominguez, for their invaluable guidance, support, and advice. I am especially grateful to José Ignacio Cristi for countless conversations. I thank Micah Baum, Ruth Cesar Heymann, Emilio Colombi, Joaquín Endara, Luciana Galeano, Paula López, Hiroshi Toma, Willem Wilken, and Julia Yates for their insightful comments and discussions. I also thank seminar and workshop participants at the University of Michigan, the Pontificia Universidad Católica de Chile, and the Central Bank of Chile. This study was approved by the University of Michigan Health Sciences and Behavioral Sciences Institutional Review Board under protocol HUM00266721. All remaining errors are my own.
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1 Introduction

A substantial body of theoretical literature shows that local labor markets adjust to local economic shocks through worker mobility (Allen and Arkolakis, 2025). However, empirical evidence generally indicates that workers migrate less in response to such shocks than theory predicts (Autor et al., 2013; Dix-Carneiro and Kovak, 2017; Borusyak et al., 2026). Migration costs are the leading explanation for this disconnect (Allen and Arkolakis, 2025; Desmet and Parro, 2025).

Nevertheless, according to the 2020 Census, roughly two-thirds of US households consisted of couples (Anderson et al., 2024). Moreover, the share of dual-earner households has risen substantially alongside female labor force participation. For most adults, therefore, responding to a local shock is a joint decision. Unlike single individuals, couples must weigh two labor-market exposures against each other. To relocate in pursuit of better opportunities, both partners must be better off as a unit, which may make couples less likely to move. At the same time, couples have an additional insurance margin: the added-worker effect (Lundberg, 1985; Attanasio et al., 2005). When one spouse loses a job or earnings, the other spouse, usually the wife, may increase their labor supply to offset the household’s lost income. This response can, in turn, shape the couple’s migration decision and adjustment to regional shocks.

In the presence of local shocks, couples face a trade-off: they must make a joint location decision, which makes moving harder, while having the option to reallocate labor within the household, which makes staying more bearable. Abstracting from couples may therefore over- or underestimate the economy’s capacity to absorb local shocks and misstate who bears their effects and where those effects occur.

In this paper, I examine how joint decision-making shapes adjustment to local shocks. I focus on two main research questions. First, how do couples respond to a local shock: by moving or by reallocating labor within the household? Second, what are the implications for the shock’s aggregate, spatial, and welfare effects?

To answer the first question, I use geocoded data from the Panel Study of Income Dynamics (PSID) to examine how single individuals and couples respond to local labor-demand shocks. The PSID’s longitudinal design allows me to track household formation, migration, and labor-market outcomes over time. I exploit the China Trade Shock, i.e., the increase in import competition faced by US manufacturing following China’s accession to the WTO in 2001 (Autor et al., 2013, 2021), as an exogenous source of variation and estimate its effects using a dynamic difference-in-differences design. Specifically, I measure individual-level exposure to the shock and compare exposed and non-exposed households within commuting

zones.

Based on this empirical design, I establish three empirical facts. First, among individuals living in the same commuting zone during the trade shock, those employed in industries adversely affected by Chinese import competition, whom I call “exposed individuals”, experienced worse labor-market outcomes and were more likely to leave their commuting zone than non-exposed individuals. On average, exposed individuals were 6.3 p.p. less likely to be employed and 3.9 p.p. less likely to participate in the labor force during the shock. They also worked 2 fewer hours per week and earned 10% lower hourly wages. These results are consistent with previous evidence that increased import competition from China harmed more exposed workers and places, even though it improved aggregate welfare in the US economy (Autor et al., 2013, 2021; Caliendo et al., 2019). Although my analysis uses a different dataset, sample definition, and empirical design—one that does not rely on a shift-share instrument (Borusyak et al., 2022), it likewise indicates that the trade shock had persistent employment effects on manufacturing workers. Exposed individuals responded by leaving their commuting zones: employment in an industry adversely affected by Chinese competition increased the probability of out-migration by an average of 4.7 p.p., with the gap widening by approximately 2 p.p. every two years and reaching 6 p.p. by the end of the shock.

The second fact concerns couples, whose out-migration responses differ according to which spouse is exposed. Relative to couples with non-exposed husbands, out-migration rises by 5 p.p. when the husband is exposed but does not change when the wife is exposed. Changes in family structure do not explain this asymmetry: the shock has no effect on marriage, divorce, or fertility. Nor is the asymmetry attributable to differences in migration responses among parents or women.

The third fact is an asymmetric response in wives’ labor force participation. When husbands are exposed, wives are 5.4% more likely to participate in the labor force, consistent with an added-worker effect. By contrast, exposed wives are 13% less likely to be employed and 14% less likely to participate in the labor force than non-exposed wives. I find no offsetting response in husbands’ extensive-margin labor supply when wives are exposed, although husbands’ hourly wages decline. Notably, exposure has no statistically significant effect on household income, regardless of which spouse is exposed. This finding suggests that spatial and intra-household insurance jointly help couples mitigate the shock.

I next extend an existing quantitative spatial and trade equilibrium model (Caliendo et al., 2019) to rationalize the empirical patterns documented above and examine how joint decision-making shapes adjustment to regional shocks. The model is dynamic, multi-region, and multi-sector. Households consist of single men, single women, or couples and choose

among location-sector bundles each period, subject to migration frictions and idiosyncratic preference shocks. Because location is the key choice couples must make jointly, I follow Rodríguez-Clare et al. (2026) and nest the preference shocks, allowing households to substitute across sectors within a location at a different rate than across locations. On the production side, each region contains multiple sectors that trade with one another, and firms combine male and female labor using a CES technology with sector-specific gender intensities. Thus, shocks can affect men and women differently, even within the same household.

The model’s main distinctive feature arises from couples’ decisions. A couple jointly chooses where to live and the sectors to which each spouse is attached. Then, the couple also decides whether the wife will work in the market or remain at home. This decision depends on both spouses’ relative wages and a home-production productivity draw, thus embedding an endogenous participation margin and the economies of scale from marriage within a dynamic spatial and trade model. This structure captures the interaction among migration, sectoral reallocation, and intra-household labor supply in response to local shocks.

To bring the model to the data, I align the quantitative environment with Caliendo et al. (2019) (hereafter CDP): 50 US states, 37 foreign trade partners, 22 sectors, and non-employment, at a quarterly frequency starting in 2000Q1. In fact, CDP’s model is a special case of mine in which every household is a single individual, male and female labor are perfect substitutes, and the two mobility elasticities coincide. Any difference between the two sets of results is therefore attributable to household heterogeneity rather than to the environment. I take the dispersion of preferences across sectors from the literature and calibrate the dispersion across locations by indirect inference, so that the model reproduces the population response to import competition estimated by Autor et al. (2013) following the same method as Rodríguez-Clare et al. (2026). Calibrated to the same target, an economy in which every couple is split into two singles implies a higher migration elasticity: ignoring couples makes individuals look more mobile than they are.

There are four parameters that govern the couple’s problem and have no counterpart in CDP: the wife’s weight in the household objective, the public-good expenditure share, and two parameters of the home-production shock. I recover the first two from consumption expenditure data and identify the slope of the home-production shock using the same trade-shock variation that drives the reduced-form evidence. Thus, the model’s participation margin is disciplined by the same shock that the quantitative exercise subsequently simulates.

The shock reduces manufacturing’s share of employment across all household types, and this decline persists in the long run. Yet the shock generates gains for US workers. Cheaper Chinese goods lower the consumption price index and increase the real return to market

work relative to non-employment, while services absorb more workers than manufacturing sheds. Consequently, the employment rate rises for every household type, and the average individual gains 0.18% in money-metric consumption-equivalent terms.

The decline in manufacturing jobs per capita is close to CDP’s estimate (-0.20 vs. -0.22 p.p.), indicating that adding couples, joint location choices, and an endogenous participation margin changes neither where the shock lands nor its magnitude. This aggregate similarity, however, masks substantial heterogeneity across household types. The decline in manufacturing employment is about twice as large for men as for women, with married men experiencing the largest decline (-0.41 p.p., compared with -0.29 p.p. for single men). Because married men are already employed, they adjust primarily by reallocating across sectors rather than by entering or leaving the labor force; consequently, their employment gains are the smallest. For married women, by contrast, the employment gain reflects an added-worker response. By disproportionately affecting male-intensive manufacturing, the shock raises wives’ wages relative to their husbands’ and draws wives into the labor force, reproducing the third empirical fact. This response occurs where husbands’ jobs are lost: across states, the increase in married women’s labor force participation is positively correlated with the loss of manufacturing employment.

Welfare gains exhibit a similar pattern. Single women gain the most (0.23%) because they are concentrated in expanding sectors, underrepresented in the manufacturing sectors most affected by the shock, and unconstrained by a joint location decision. Couples (0.17%) and single men (0.16%) gain less. Among couples, gains decline monotonically with exposure and are ordered by the husband’s exposure: they fall from 0.18% when neither spouse initially works in manufacturing to 0.16% when only the wife does, 0.14% when only the husband does, and 0.13% when both do. The husband’s wage scales the household’s entire payoff, whereas the wife’s wage matters only through her option to work in the market. Couples’ welfare gains are also the most geographically unequal because spouses cannot take jobs in different states.

To isolate the role of joint decision-making, I compare these results with those from a *single-agent economy*, in which every couple is split into two singles, all inputs are reconstructed from the data, and the location elasticity is recalibrated to match the same population response. The two economies yield similar results for the magnitude and spatial distribution of the shock: the decline in manufacturing differs by less than 1%, the aggregate welfare gain differs by about 1%, and state-level welfare gains are highly correlated (0.96). They differ sharply, however, in the incidence of the shock. Because a wife’s decision to participate depends on her wage relative to her husband’s rather than on her wage alone, female labor supply is 34% less elastic when couples are present. Consequently, the em-

ployment gain is 15% smaller and unfolds more slowly. Although couples front-load some adjustment through wives' participation, the single-agent economy reaches 95% of its long-run employment gain ten quarters earlier. With fewer women entering the labor market, female wages bear more of the adjustment, redistributing welfare among single individuals. Relative to the single-agent economy, single women gain 20.5% more, single men gain 5.4% less, and married women gain 12.3% less. Welfare gains are also one-third more dispersed across states. Thus, both economies capture the magnitude and location of the shock, but the single-agent economy misses who bears it within each location.

I next rank the margins of adjustment by shutting them down one at a time. Eliminating migration changes aggregate welfare only slightly (+1%), but reduces the employment response by 12% and makes state-level welfare twelve times more unequal. The losses fall on couples whose husbands initially work in manufacturing, whereas unexposed couples gain because exposed households no longer enter their labor markets, thereby pushing down wages. The added-worker effect does not fill this gap: a wife's participation decision depends only on the ratio of the spouses' wages, and shutting down spatial adjustment changes both wages by nearly the same proportion. Substitution therefore operates in only one direction. When the participation response is dampened, couples migrate more, the employment response falls by 24%, and aggregate welfare declines by 4.6%. All couples lose, including those in which neither spouse was initially exposed, with the largest loss (10.2%) borne by couples in which only the husband was exposed. The added-worker margin provides insurance to every couple, whereas the spatial margin provides insurance only to exposed couples. Finally, holding the gender wage ratio nearly fixed eliminates about 63% of wives' participation response and closes most of the gap between the gains of single women and single men: single men gain 18.8% more, while single women gain 10.4% less.

Contribution This project contributes to three strands of the literature. First, it contributes to the dynamic spatial general-equilibrium literature on trade and migration responses to local shocks (Redding and Rossi-Hansberg, 2017; Desmet et al., 2018; Monte et al., 2018; Allen and Donaldson, 2020, 2022; Donald et al., 2025; Desmet and Parro, 2025; Pellegrina and Sotelo, 2025). It is particularly closely related to Caliendo et al. (2019), which emphasizes the interaction between migration, productivity, and the spatial distribution of economic activity under the China Trade Shock, and to Rodríguez-Clare et al. (2026), who extend Caliendo et al. (2019) to separate mobility across sectors within a location from mobility across locations. My contribution is to extend this class of models to include heterogeneous households, joint location choices, and wives' endogenous labor supply. This extension makes it possible to study not only migration responses but also how

joint decision-making affects the spatial allocation of labor and the welfare effects of regional shocks through general equilibrium effects. In particular, the contribution of this paper is not to identify a different aggregate than the previous literature; it is to understand how that aggregate is distributed and who bears the effects of the shock.

Second, the paper contributes to the literature that studies how dual-earner households respond to shocks, highlighting the roles of risk sharing within couples (Attanasio et al., 2005; Pilossoph and Wee, 2021; Calvo et al., 2024) and collocation frictions (Guler et al., 2012; Flabbi and Mabli, 2018; Foerster and Ulbricht, 2023; Jayachandran et al., 2024; Yan, 2025)¹. My contribution is to bring these insights into the study of local shocks. In an aggregate shock, couples are always better off than singles because they can reallocate labor within the household, and because the shock is aggregate, the joint location friction does not bind. In the presence of a local shock, however, couples face a trade-off between lower mobility and an additional margin of adjustment. My contribution is therefore to compare these two mechanisms in a general equilibrium framework.

Third, the paper contributes to the literature on the China Trade Shock and its effects on US local labor markets (Autor et al., 2013, 2014, 2016, 2019, 2021, 2025; Acemoglu et al., 2016; Pierce and Schott, 2016; Greenland et al., 2019; Caliendo et al., 2019; Faber et al., 2022; Borusyak et al., 2026). This literature has documented substantial heterogeneity in the local effects of rising competition from Chinese imports. In particular, Autor et al. (2013) and subsequent papers document a significant decrease in manufacturing employment in US commuting zones with larger manufacturing shares. The loss of jobs has implied long-lasting consequences, including increases in political polarization (Autor et al., 2020), child and adult poverty, and even mortality among young men (Autor et al., 2019). This paper is closely related to Parey et al. (2025), who, using UK data, show that household responses to trade shocks can differ sharply by gender within the couple, finding an added-worker effect when wives are exposed. My contribution to this literature is twofold. First, using the Panel Study of Income Dynamics, I provide new empirical results on the effects of the shock in the US by measuring individual-level exposure and comparing individuals within the same commuting zone. Second, to the best of my knowledge, this is the first paper to provide empirical evidence on within-household responses to the shock in the US.

¹There is an extensive literature on family migration and household adjustment. Starting with Mincer (1978), this literature has emphasized that migration is often a joint decision and that such decisions can have unequal consequences within the household. Subsequent work has documented the role of “trailing wives” and the labor market costs of family migration for women (Costa and Kahn, 2000; Cooke et al., 2009; Benson, 2014; Blau and Kahn, 2017; Fadlon et al., 2020; Braun et al., 2021; Le Barbanchon et al., 2021; Venator, 2026, 2024). Even if it is not in the scope of this paper, analyzing the outcomes of the “trailing wives” under a trade shock is an open research question.

This paper is organized as follows. Section 2 presents the empirical analysis, detailing the approach and presenting the main empirical findings. Section 3 presents the theoretical model. Section 4 brings the model to the data, and Section 5 presents the results of the quantitative exercise. Finally, Section 6 concludes.

2 Empirical Analysis

This section presents the empirical analysis of how household structure shapes adjustment to regional shocks, using the China Trade Shock as a natural experiment. I first describe the data, define exposure at the individual and couple levels, and outline the empirical design. I then document three empirical facts concerning labor market outcomes, migration responses, and intra-household labor supply adjustments. Together, these findings motivate the theoretical framework developed in the next section.

2.1 Empirical Design

2.1.1 Data Source

The primary data source for the empirical analysis is the Panel Study of Income Dynamics (PSID). Most studies of the China Trade Shock use commuting zones as the unit of observation and rely on cross-sectional data. As a result, they cannot distinguish how exposure varies within households or track individuals across time and space. The panel structure of the PSID allows me to address both limitations.

The PSID has three additional advantages that make it particularly well suited to this analysis. First, the restricted geocoded version records household addresses at the census-tract level in every survey wave.² This information allows me to follow individuals across local labor markets and compare individuals within the same market. Following the previous literature, I define local labor markets as US commuting zones. The geographic detail also allows me to study migration across commuting zones.

Second, the PSID allows me to construct marital status from household composition and each person’s relationship to the household head in every period. This approach does not rely exclusively on current legal marital status, which may understate the number of couples by excluding cohabitation or overstate it by including individuals who are separated but not divorced. Given the PSID’s sampling structure, I restrict the sample to three marital-status categories:³

²The PSID staff has previously disclosed all results; access to the restricted data is under Contract 921.

³Remarried individuals are excluded from the sample. An individual who separated after a cohabitation

1. “Married”: Individuals in their first union who are legally married or cohabiting with a heterosexual partner
2. “Divorced”: Individuals who are legally divorced or separated, or whose cohabiting union ended
3. “Single”: Individuals whom I never observe as “married” to another individual

I then pool single and divorced individuals into a “unmarried” category because neither group makes joint household decisions with a current partner.⁴

Third, the PSID contains detailed information on labor market outcomes, allowing me to examine employment, unemployment, and labor force participation. For employed individuals, I observe average weekly hours, hourly wages, annual labor income, and several additional outcomes. I also observe total household income for couples.⁵

I use all PSID waves conducted in odd-numbered years between 1991 and 2013.⁶ I restrict the sample to individuals aged 20–55 who lived in the mainland US and were present in the 1999 wave. The final sample contains 4,731 individuals and 38,634 observations across 12 waves.

2.1.2 Individuals Exposed to the China Trade Shock

The China Trade Shock refers to the increase in competition from Chinese manufactured goods in the US following China’s accession to the WTO in 2001 and continuing through 2012 (Autor et al., 2013, 2021). I identify exposed individuals in two steps. First, following Autor et al. (2021) (henceforth ADH), I define the growth in Chinese import competition faced by US manufacturing industry j between 2000 and 2012 as:

$$\Delta IC_j = 100 \times \frac{\Delta M_j^{\text{China,US}}}{Y_{j,2000} + M_{j,2000} - X_{j,2000}},$$

where $\Delta M_j^{\text{China,US}}$ is the change in US imports from China in industry j between 2000 and 2012, $Y_{j,2000}$ denotes industry j ’s shipments, and $M_{j,2000} - X_{j,2000}$ denotes its net imports in 2000. A positive value of ΔIC_j therefore indicates that US manufacturing industry j experienced increased competition from Chinese imports between 2000 and 2012.

is considered “divorced”, even if their legal marital status is single.

⁴Assuming that divorced individuals do not make joint decisions follows previous literature studying life after divorce (Weiss and Willis, 1993; Del Boca and Flinn, 1995; Flinn, 2000; Reynoso, 2024; Didier and Lopez, 2026).

⁵All monetary variables are converted to 2017 dollars using the Personal Consumption Expenditures Price Index (PCE) from the US Bureau of Economic Analysis.

⁶Given that the PSID was conducted annually until 1997, to avoid double-counting migration, I consider only odd years to construct a biennial sample before and after 1997.

To isolate the foreign supply-driven component of US imports from China, I follow ADH and measure industry-level growth in Chinese exports to eight other high-income countries.⁷ I also use three-year-lagged values of Y_j , M_j , and X_j rather than base-year values. The resulting measure of import competition is

$$\widehat{\Delta IC}_j = 100 \times \frac{\Delta M_j^{\text{China, Other Countries}}}{Y_{j,1997} + M_{j,1997} - X_{j,1997}},$$

A positive value of this measure indicates that industry j experienced increased competition from Chinese imports in other high-income countries. I therefore classify the corresponding US industry j as negatively exposed to the China Trade Shock.

Next, I identify the industry in which each individual i was employed in 1999 and define the indicator Exposed_i as follows:

$$\text{Exposed}_i = \begin{cases} 1 & \text{if } i \text{ works in 1999 in industry } j \text{ with } \widehat{\Delta IC}_j > 0 \\ 0 & \text{otherwise} \end{cases} \quad (1)$$

For the analysis of within-household mechanisms, I further restrict the sample to couples.⁸ I divide couples into three groups: those in which neither spouse worked in an exposed industry in 1999, those in which the husband worked in an exposed industry, and those in which the wife worked in an exposed industry. I define couple h as exposed through the husband if

$$\text{Exposed Husband}_h = \begin{cases} 1 & \text{if he worked in 1999 in industry } j \text{ with } \widehat{\Delta IC}_j > 0 \\ 0 & \text{otherwise} \end{cases} \quad (2)$$

and as exposed through the wife if

$$\text{Exposed Wife}_h = \begin{cases} 1 & \text{if she worked in 1999 in industry } j \text{ with } \widehat{\Delta IC}_j > 0 \\ 0 & \text{otherwise} \end{cases} \quad (3)$$

2.1.3 Summary Statistics

Table 1 reports summary statistics for the pooled 1991–1999 sample. The first column describes the full sample; the second pools single and divorced individuals; and the third and fourth report statistics for husbands and wives, respectively. Married individuals consti-

⁷Autor et al. (2013) and their follow-up papers consider Australia, Denmark, Finland, Germany, Japan, New Zealand, Spain, and Switzerland. They select these countries based on the availability of data for the full sample period.

⁸For this analysis, I drop 4% of couple-year observations across the full 1991–2013 panel, where both spouses were working in exposed industries in 1999, because I cannot measure whether one spouse’s job was more important to the couple than the other.

tute approximately 60% of the sample, single individuals 24%, and divorced individuals the remainder.

The average individual is 37 years old; two-fifths of the sample are women, one-third are college graduates, and 86% are white. Compared with married individuals, unmarried individuals are slightly younger and less likely to be college graduates or white. Half of the sample have children under 18, including nearly one-quarter of unmarried individuals. The youngest child in the household is, on average, 4 years old and tends to be younger among unmarried individuals, consistent with the younger average age of unmarried individuals. According to ADH's measure, one in ten individuals lives in a commuting zone (CZ) in the top two deciles of exposure to Chinese import competition. Exposure does not vary meaningfully by marital status, suggesting little spatial sorting along this dimension. The average migration rate is 8%, with unmarried individuals migrating at nearly twice the rate of married individuals.

In 1999, 15% of the sample worked in an industry exposed to competition from Chinese imports, with substantial heterogeneity across groups. Exposure rates were 14% among unmarried individuals, 19% among husbands, and 10% among wives. At the household level, 20% of couples were exposed. As noted by Autor et al. (2013, 2021), this period also featured two related economic forces: increasing automation and broader globalization. Following Autor and Dorn (2013), I calculate routine-task intensity and offshorability indices for the occupations represented in the sample. Slightly less than one-quarter of observations correspond to individuals employed in routine-task-intensive occupations, while approximately one in ten workers is employed in an offshorable occupation. Offshorability varies substantially by gender, reflecting the occupational sorting of men and women.

Because the sample consists of prime-age individuals, employment and labor force participation rates are high, at 91% and 95%, respectively. Labor market status nevertheless differs by marital status. Among non-employed individuals, unemployment is the most common status for unmarried individuals and husbands, whereas labor force nonparticipation is most common among wives. This pattern is consistent with unmarried individuals relying more heavily on reemployment after job loss, whereas married individuals can adjust labor supply within the household, thereby drawing on intra-household insurance.

The average individual works 40 hours per week, earns \$19 per hour in 2017 USD, and receives approximately \$42,000 in annual labor income. Husbands work the most hours, earn the highest hourly wages, and consequently receive the highest annual labor income; wives have the lowest values for all three outcomes. Married and unmarried individuals work similar hours on average, but married individuals earn approximately 20% higher hourly wages and therefore receive more labor income per capita. This pattern is consistent with

Table 1: Summary Statistics Before the China Trade Shock

	(1)	(2)	(3)	(4)
	All	Unmarried	Couples	
			Husbands	Wives
Panel A: Distribution by "Marital Status"				
Share of Single Individuals	0.24			
Share of Married Individuals	0.59			
Share of Divorced Individuals	0.16			
Panel B: Demographic Characteristics				
Age ⁺	36.86	34.69	38.92	37.19
Share of Female Individuals	0.42	0.55		
Share of College Graduates	0.33	0.29	0.37	0.32
Share of White Individuals	0.86	0.75	0.93	0.93
Share of Individuals with Kids	0.50	0.23		0.69
Age of Youngest Kid ⁺	3.67	1.75		4.98
Share in Highly Exposed CZs ⁺⁺	0.11	0.12		0.11
Migration Rate	0.08	0.11		0.06
Panel C: Labor Market Outcomes				
Share of Exposed Individuals	0.15	0.14	0.19	0.10
Share in Routine Occupations	0.23	0.22	0.23	0.26
Share in Offshorable Occupations	0.12	0.14	0.08	0.16
Share Employed	0.91	0.89	0.97	0.85
Share Unemployed	0.03	0.06	0.02	0.02
Share Out of the Labor Force	0.05	0.05	0.01	0.13
Weekly Hours ⁺	40.80	39.89	45.34	33.60
Hourly Wage ⁺	19.54	16.98	23.70	16.45
Labor Income ⁺	42671.70	35953.36	57073.74	27617.31
Number of Observations	16,645	4,343	6,151	6,151

Notes. The unmarried category pools single and divorced individuals. Monetary values are expressed in 2017 USD using the PCE price index. ⁺: Indicates the sample mean. ⁺⁺: Highly exposed CZs are those in the top 20% of exposure according to ADH.

couples using intra-household labor supply adjustments to allocate resources more effectively.

2.1.4 Empirical Design

To estimate the overall effect of exposure to the China Trade Shock, I use the following event-study specification for individual i in commuting zone c and year t :

$$\begin{aligned} Outcome_{i,c,t} = & \alpha \cdot \mathbb{1}[Exposed_i = 1] + \sum_{\tau \neq 1999} \beta_{\tau} \cdot \mathbb{1}[Exposed_i = 1] \cdot \mathbb{1}[t = \tau] \\ & + \delta X' + \gamma_t + \gamma_c + \varepsilon_{i,c,t} \end{aligned} \quad (4)$$

The coefficients of interest, β_{τ} , measure the difference in outcomes between exposed and non-exposed individuals in year τ , relative to the corresponding difference in 1999. The control vector includes age and indicators for being female, holding a college degree, having children younger than 18, working in manufacturing in 1999, working in a routine occupation, and working in an offshorable occupation (Autor and Dorn, 2013). The specification also includes commuting-zone and year fixed effects. All regressions are weighted by PSID sampling weights, and standard errors are clustered at the individual and industry levels.

To examine joint household decision-making, I estimate the following specifications for spouse s in couple h , commuting zone c , and year t :

$$\begin{aligned} Outcome_{s,h,c,t} = & \beta_1 \cdot \mathbb{1}[Exposed Husband_h = 1] + \beta_2 \cdot \mathbb{1}[Exp. Husband_h = 1] \cdot \mathbb{1}[t \geq 2001] \\ & + \delta X' + \gamma_c + \gamma_t + \varepsilon_{s,h,c,t} \end{aligned} \quad (5)$$

$$\begin{aligned} Outcome_{s,h,c,t} = & \beta_1 \cdot \mathbb{1}[Exposed Wife_h = 1] + \beta_2 \cdot \mathbb{1}[Exposed Wife_h = 1] \cdot \mathbb{1}[t \geq 2001] \\ & + \delta X' + \gamma_c + \gamma_t + \varepsilon_{s,h,c,t}. \end{aligned} \quad (6)$$

In these specifications, the coefficient of interest is β_2 , which captures the post-2001 difference in outcomes between couples with an exposed spouse and those without. Both equations include the controls used in Equation 4. For controls measured at the individual level, I include separate values for each spouse, such as the husband's and wife's ages. In Equation 5, standard errors are two-way clustered at the household and husband-industry levels; in Equation 6, they are clustered at the household and wife-industry levels.

2.2 Empirical Facts

The three facts proceed from individual-level effects to within-household adjustment. The first establishes the labor market effects of the shock and examines whether exposed individuals respond by leaving their commuting zone. The second shows that couples migrate in response only when the husband is the exposed spouse. The third examines the wife's

labor supply response, addressing the margin left unexplained by the migration asymmetry in the second fact. Across all three analyses, identification is based on comparing individuals whose exposure differs by their 1999 industry, with commuting zone and year held fixed. The relevant comparison is therefore between exposed and non-exposed individuals facing the same local labor market, rather than between local labor markets with different levels of exposure.

Fact 1: Exposure to the China Trade Shock worsened labor market outcomes and increased out-migration

Figure 1 presents estimates of Equation 4 for employment, unemployment, labor force participation, weekly hours worked, hourly wages, and annual labor income. Table A.2 reports estimates from the corresponding difference-in-differences specification, which summarizes the average effect during the China Trade Shock.

Panel A of Figure 1 shows that, during the first seven years of the shock, exposed individuals were less likely to be employed than non-exposed individuals. On average, exposure was associated with a 6.3-percentage-point (7%) decline in the probability of employment. After 2009, the decline stopped deepening, although the estimates are not statistically significant. Most of the decline in employment between 2001 and 2007 reflects individuals leaving the labor force (Panel C) rather than entering unemployment (Panel B). Between 2001 and 2013, exposed individuals were, on average, 4% less likely to participate in the labor force than non-exposed individuals.

Weekly hours worked follow a pattern similar to employment (Panel D): individuals employed in exposed industries in 1999 worked 5% fewer hours per week during the shock than individuals employed in other industries that year. Hourly wages exhibit a different pattern (Panel E). Although the estimate is positive in 2001, the hourly wages of exposed individuals decline through the final year of the sample. On average, exposed individuals earned 10% lower hourly wages during the shock than non-exposed individuals. Combined with the reduction in hours worked, this decline resulted in 11% lower annual labor income among exposed individuals (Panel F).

To compare my estimates more directly with the previous literature, Table A.3 replaces the binary exposure measure in Equation 1 with the continuous import-competition measure used by ADH.⁹ The average decadalized growth in Chinese import competition implies a 0.5-percentage-point average decline in the probability of employment among exposed individuals

⁹One potential concern is that, given the small PSID sample, the estimates may be driven by the particular set of CZs represented. To address this concern, Section A.1.1 replicates ADH's findings using the same CZs included in my sample, ruling out this alternative explanation.

relative to non-exposed individuals between 2001 and 2013. For comparison, converting ADH’s cumulative estimate for the employment-to-population ratio between 2001 and 2019 into an evenly distributed effect and retaining only odd-numbered years, as in the PSID sample, implies an average decline of 0.55 percentage points.

These findings are consistent with previous evidence on the adverse effects of the China Trade Shock in the US (Autor et al., 2013, 2014, 2016, 2021; Acemoglu et al., 2016; Pierce and Schott, 2016). Despite using a different dataset, sample restriction, and empirical approach, I similarly find a negative employment response among individuals working in exposed industries.

Given the deterioration in labor market outcomes among individuals employed in industries exposed to Chinese import competition, particularly the U-shaped employment response, I next examine whether exposed individuals moved to other CZs in search of better opportunities. For each individual i , I define out-migration in year t as follows:

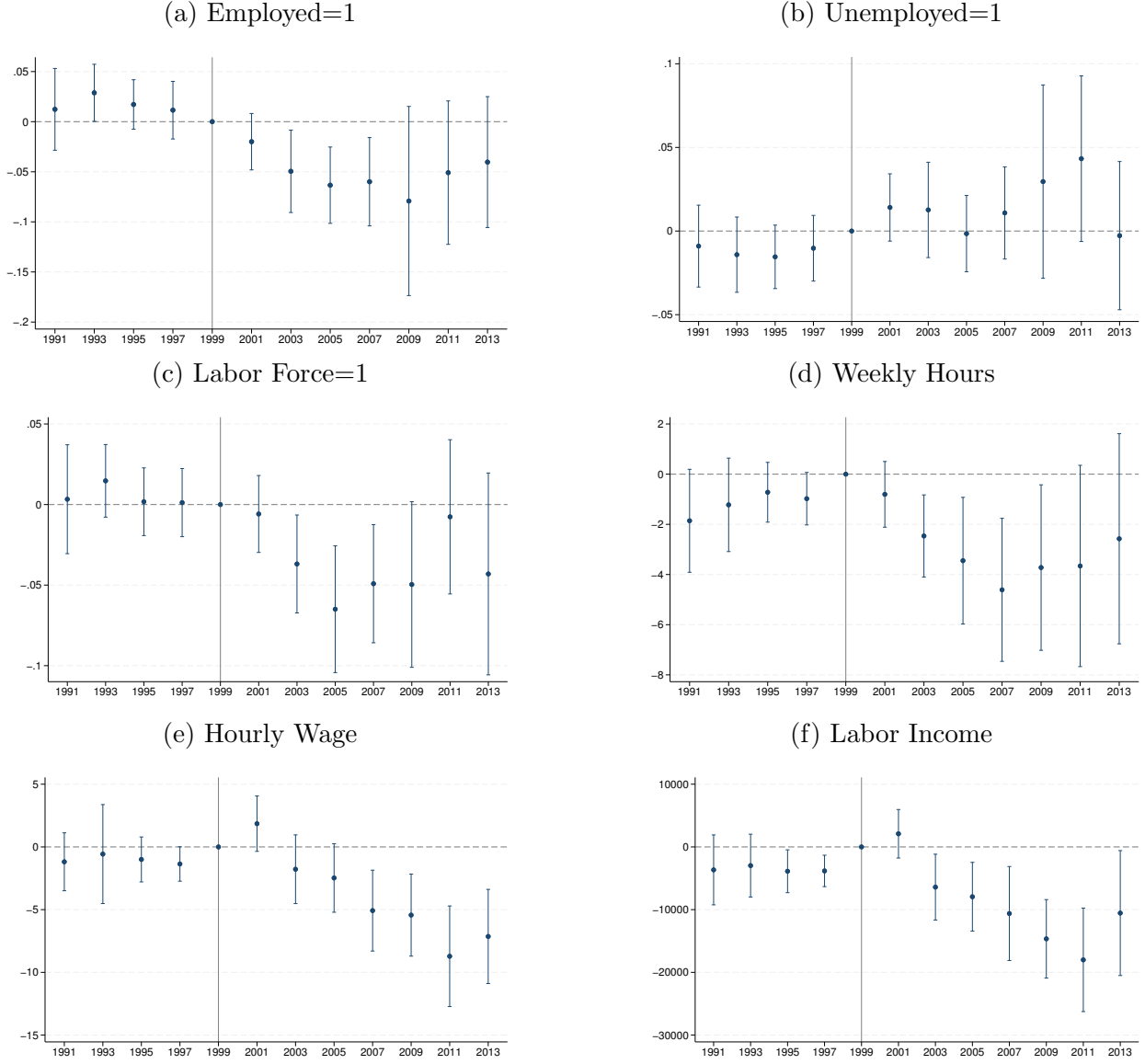
$$\text{Out-Migration}_{i,t} = \begin{cases} 1 & \text{if } CZ_{i,t-2} \neq CZ_{i,t} \\ 0 & \text{if } CZ_{i,t-2} = CZ_{i,t} \end{cases}$$

Figure 2 presents estimates of Equation 4 using out-migration as the outcome. During the shock, exposed individuals became increasingly likely to leave their CZ: the estimated difference rose by approximately 2 percentage points every two years, reaching 6 percentage points in 2013. On average, exposed individuals were 4.7 percentage points more likely to out-migrate than non-exposed individuals within the same commuting zone, as reported in Table A.4. This finding is consistent with ADH’s evidence of a statistically significant decline in the population aged 25 to 39 and a negative but statistically insignificant decline in the overall population of more exposed commuting zones. Moreover, Borusyak et al. (2026) argue that shift-share measures of industry-level shocks may attenuate estimated migration responses when they abstract from workers’ outside options, as in ADH. Accordingly, the individual-level out-migration response estimated here should be larger than their aggregate estimate. Most of the response emerges with a lag of seven to ten years, consistent with Greenland et al. (2019)’s evidence on the effects of the China Trade Shock on internal migration in the US.

Fact 2: Couples out-migrate only when the husband is exposed.

To examine how couples’ migration responses differ from those of individuals, I estimate Equations 5 and 6 using out-migration as the outcome. Column 1 of Table 2 shows that couples in which the husband was exposed to the shock were 5 percentage points more likely

Figure 1: Exposure to the China Trade Shock and Labor Market Outcomes

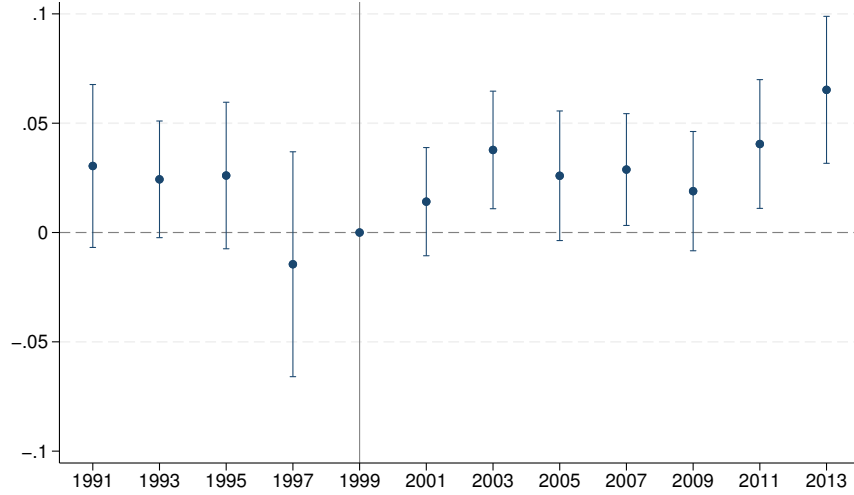


Notes. Each panel reports estimates from Equation 4, using the variable named in the panel title as the outcome. Monetary outcomes are expressed in 2017 USD using the PCE price index and include zero values. Robust standard errors are clustered at the industry and individual levels. The bars indicate 95% confidence intervals.

to leave their commuting zone than couples in which the husband was not exposed.¹⁰ By contrast, exposure of the wife produces no statistically significant migration response, and the point estimate in Column 2 is negative. Although this asymmetry may initially appear

¹⁰The negative coefficient in the first row of Table 2 reflects the relatively favorable employment conditions in manufacturing before the shock. Accordingly, couples in which the husband worked in manufacturing were less likely to out-migrate before the shock.

Figure 2: Exposure to the China Trade Shock and Out-Migration Responses



Notes. Robust standard errors are two-way clustered by industry and individual. The bars indicate 95% confidence intervals.

counterintuitive, it is consistent with evidence that husbands' employment plays a larger role in family migration decisions (Mincer, 1978; Costa and Kahn, 2000; Cooke et al., 2009; Benson, 2014; Blau and Kahn, 2017; Fadlon et al., 2020; Braun et al., 2021; Le Barbanchon et al., 2021; Venator, 2026, 2024).

There are two potential alternative explanations for this finding that are worth addressing. First, the asymmetry could be driven by changes in family structure. If, for example, couples divorce when the wife is exposed but not when the husband is exposed, my findings would be biased. In Section A.1.2, I study the effect of the trade shock on the probabilities of marriage, divorce, and fertility, finding null results for all of them. Second, the asymmetry could reflect nothing more than different migration responses among parents or among exposed women. In Section A.1.3, I expand Equation 4 to include a triple interaction for having children and for being a woman. In both cases, I can reject that parents or women responded differently from non-parents or men. Then, I can rule out both explanations as alternative mechanisms driving my results.

Fact 3: Wives' labor force participation falls when they are directly exposed and rises when husbands are exposed.

Couples have an additional margin of adjustment, the so-called "added-worker effect" (Lundberg, 1985), which refers to the fact that if one spouse specializes in home production, the other can enter the labor market upon job loss to offset the household's income loss. I therefore study labor responses to the shock, distinguishing between direct and indirect exposure.

Table 2: Exposure to the China Trade Shock and Couples' Out-Migration Responses

	(1)	(2)
	Out-Migration=1	
Exposed Husband=1	-0.121***	
	(0.043)	
(Exposed Husband=1)x(Post=1)	0.050**	
	(0.022)	
Exposed Wife=1		0.064
		(0.074)
(Exposed Wife=1)x(Post=1)		-0.010
		(0.028)
Observations	13,695	13,695
R^2	0.10	0.09
Baseline Mean	0.05	0.05

Notes. Robust standard errors are two-way clustered by household and the husband's industry in Column 1, and by household and the wife's industry in Column 2. * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

When wives are directly exposed to the shock, they are 13% less likely to be employed and 14% less likely to be in the labor force than non-exposed wives (Panel A of Table 3). However, when wives are indirectly exposed because their husbands worked in an exposed industry, they are 5.4% more likely to be in the labor force relative to the wives of non-exposed husbands (Panel B of Table 3). This provides suggestive evidence of the added-worker effect among wives indirectly exposed to the China Trade Shock through their husbands. When husbands are indirectly exposed to the shock, they nonetheless have the same probability of employment as non-exposed husbands, while earning a lower hourly wage (Table A.5), suggesting a lower reservation wage when indirectly exposed.

This result is the exact opposite of what Parey et al. (2025) find for England and Wales. They show that when English husbands are indirectly exposed to the trade shock, older workers increase their labor force participation and move into self-employment. They find a positive, though not statistically significant, response among indirectly exposed wives. The difference in the results for husbands could be driven by the fact that 93% of husbands in their sample are in the labor force, compared with 99% in my sample (Table 1). As a result,

Table 3: Wife's Exposure to the China Trade Shock and Labor Market Outcomes

	Wife Employed=1	Wife Labor Force=1
Exposed Wife=1	0.041 (0.099)	0.016 (0.110)
(Exposed Wife=1)x(Post=1)	-0.099** (0.045)	-0.110** (0.048)
Exposed Husband=1	0.027 (0.077)	0.027 (0.068)
(Exposed Husband=1)x(Post=1)	0.041*** (0.013)	0.042*** (0.013)
Observations	14,083	14,083
Baseline Mean	0.77	0.78

Notes. Robust standard errors are two-way clustered at the wife's industry and individual levels in Panel A and at the husband's industry and individual levels in Panel B. * for $p < .10$, ** for $p < 0.05$, *** for $p < 0.01$.

husbands in my sample face a “ceiling effect” that works against identifying an added-worker effect for husbands.

Given the asymmetric responses of couples depending on which spouse is exposed, an important question is whether these responses are welfare-improving. To provide suggestive evidence on this question, I estimate Equations 5 and 6 using a measure of household income from the PSID. In both columns 1 and 2 of Table A.6, I find no statistically significant difference in household income between exposed and non-exposed households during the China Trade Shock, regardless of whether the husband or the wife was employed in an exposed industry. This suggests that, using the husband's exposure as an example, the combination of out-migration and the wife's labor force participation responses helps maintain household income stability during the shock.

Therefore, the empirical evidence points to three central patterns. First, exposure to the China Trade Shock worsened labor market outcomes for directly exposed individuals, who responded by leaving their commuting zones. Second, among couples, this migration response was concentrated in households where the husband was exposed. Third, I find evidence of an added-worker effect during the trade shock, with wives' labor force participation responding differently depending on which spouse was directly exposed to the shock. Taken together,

these findings support the existence of a trade-off for couples: joint location decisions make moving harder, and this occurs only when the household’s breadwinner is affected. At the same time, they can reallocate labor within the household, making it more bearable to stay in the same location.

However, the empirical evidence cannot inform us about which margin of adjustment is more important, why, or what the general equilibrium effects are of abstracting from couples. In the next section, I develop a framework in which household decisions about location and labor supply are jointly determined, within a spatial and trade quantitative model.

3 Dynamic Spatial Model of Trade and Migration with Heterogeneous Households

This section develops the theoretical model used to rationalize the empirical facts documented in the previous section and quantify their welfare and spatial implications. The model extends the framework of Caliendo et al. (2019) (henceforth CDP) by incorporating heterogeneous households and joint decision-making.

The model is structured as follows. There are L locations, indexed by i as origins and j as destinations. In each location, a population mass N is born at $t = 0$, comprising $N/2$ males and $N/2$ females. A fraction $\bar{s}_i$ of individuals are single, while the remaining fraction, $1 - \bar{s}_i$, belong to heterosexual couples. Time is discrete and indexed by $t \in \{0, 1, 2, \dots, T\}$.

The model includes three household types, indexed by h : single men ($h = s_m$), single women ($h = s_w$), and couples ($h = c$). All households are forward-looking. Couples are further differentiated by the home-production productivity shock β realized after relocation. Households respond to local conditions by choosing their sector of employment and place of residence and, for couples, whether the wife participates in market work after relocation.

A central feature of the model is the interaction between migration and married women’s labor force participation. Together, these decisions shape how sectoral productivity shocks affect the spatial distribution of individuals and output through gender-differentiated exposure to sectors.

3.1 Environment

The environment consists of three components. First, technology and trade determine equilibrium wages in each region-sector cell. Relative to the standard model, labor is a CES composite of male and female inputs, implying that sectoral shocks have differential effects on men’s and women’s wages. Second, preferences and mobility determine households’ res-

idential and employment choices. Couples jointly choose a location and a pair of sectors rather than making these decisions sequentially. Finally, conditional on relocation, the wife chooses between market work and home production.

3.1.1 Technology

In each period t , every region i contains K productive sectors, indexed by $k, n \in \{1, \dots, K\}$. Let $A_{i,k,t} > 0$ denote Hicks-neutral productivity in sector k of region i . Let $\psi_{k,m}, \psi_{k,f} > 0$ denote sector-specific gender-intensity parameters that are common across locations and satisfy $\psi_{k,m} + \psi_{k,f} = 1$.

A representative intermediate-good firm in sector k of origin i combines male labor, $M_{i,k,t}$, and female labor, $F_{i,k,t}$, into an effective labor aggregate using a CES production function with elasticity of substitution $\sigma > 1$:

$$\mathcal{L}_{i,k,t} = \left(\psi_{k,m} M_{i,k,t}^{(\sigma-1)/\sigma} + \psi_{k,f} F_{i,k,t}^{(\sigma-1)/\sigma} \right)^{\sigma/(\sigma-1)}. \quad (7)$$

Output is then given by

$$Y_{i,k,t} = A_{i,k,t} \mathcal{L}_{i,k,t}.$$

The sectoral intermediate good is sold at producer price $p_{i,k,t}$. Shipping the good from origin-sector pair ik to destination-sector pair jn entails iceberg trade costs $\tau_{ik,jn} \geq 1$. Thus, the delivered price paid by sector n in destination region j is $p_{ik,jn,t} = p_{i,k,t} \tau_{ik,jn}$. Under perfect competition and constant returns to scale, the producer price equals the marginal cost of producing the sectoral intermediate good:

$$p_{i,k,t} = \frac{\bar{w}_{i,k,t}}{A_{i,k,t}} \implies \bar{w}_{i,k,t} = p_{i,k,t} A_{i,k,t}.$$

Here, $\bar{w}_{i,k,t}$ is the CES wage index, which gives the cost-minimizing unit cost of effective labor:

$$\bar{w}_{i,k,t} = \left(\psi_{k,m}^\sigma (w_{i,k,t}^M)^{1-\sigma} + \psi_{k,f}^\sigma (w_{i,k,t}^F)^{1-\sigma} \right)^{\frac{1}{1-\sigma}}.$$

Here, $w_{i,k,t}^M$ and $w_{i,k,t}^F$ denote the average wages paid to men and women, respectively, employed in sector k of location i at time t .

The first-order conditions of the cost-minimization problem yield optimal demands for male and female labor as functions of the effective labor aggregate and relative wages:

$$M_{i,k,t} = \psi_{k,m}^\sigma \left(\frac{w_{i,k,t}^M}{\bar{w}_{i,k,t}} \right)^{-\sigma} \mathcal{L}_{i,k,t}, \quad (8)$$

$$F_{i,k,t} = \psi_{k,f}^\sigma \left(\frac{w_{i,k,t}^F}{\bar{w}_{i,k,t}} \right)^{-\sigma} \mathcal{L}_{i,k,t}. \quad (9)$$

Inverting the labor demand functions in Equations 8 and 9 yields the equilibrium gender-specific factor prices within each region-sector pair:

$$w_{i,k,t}^M = \bar{w}_{i,k,t} \psi_{k,m} \left(\frac{\mathcal{L}_{i,k,t}}{M_{i,k,t}} \right)^{1/\sigma}, \quad w_{i,k,t}^F = \bar{w}_{i,k,t} \psi_{k,f} \left(\frac{\mathcal{L}_{i,k,t}}{F_{i,k,t}} \right)^{1/\sigma}, \quad (10)$$

In each destination region j and sector n , a representative sectoral final-good firm combines goods from all origin-sector pairs into a sector- n composite using a CES aggregator with elasticity of substitution $\eta_n > 1$. Let $Q_{j,n,t}$ denote the quantity of this composite produced at time t . The corresponding unit-cost function is

$$P_{j,n,t} = \left(\sum_{i=1}^L \sum_{k=1}^K p_{ik,jn,t}^{1-\eta_n} \right)^{1/(1-\eta_n)}. \quad (11)$$

The corresponding share of expenditure by sector n in destination region j on goods from sector k in origin region i is

$$\pi_{ik,jn,t} = \left(\frac{p_{ik,jn,t}}{P_{j,n,t}} \right)^{1-\eta_n}.$$

At the regional level, a representative final-good firm combines sectoral composites into a regional consumption good using a Cobb-Douglas aggregator with expenditure shares $\iota_k > 0$, where $\sum_{k=1}^K \iota_k = 1$:

$$Q_{j,t} = \prod_{k=1}^K \left(\frac{Q_{j,k,t}}{\iota_k} \right)^{\iota_k}$$

where $Q_{j,t}$ denotes the quantity of the regional final good produced in region j at time t . The associated regional price index is

$$P_{j,t} = \prod_{k=1}^K P_{j,k,t}^{\iota_k}. \quad (12)$$

Cobb-Douglas aggregation implies that expenditure on sector n is a constant share of total expenditure in region j : $P_{j,n,t} Q_{j,n,t} = \iota_n P_{j,t} Q_{j,t}$.

Each location j also contains a non-employment sector, indexed by 0, that is available to single men, single women, and married men. Married women do not choose sector 0; instead, they choose a productive sector and subsequently decide whether to work in the market or engage in home production. The non-employment sector produces no output. An individual who chooses sector 0 in region j receives an exogenous nominal benefit b_j , which is constant

over time:

$$w_{j,0,t}^M = w_{j,0,t}^F = b_j.$$

3.1.2 Preferences

Mobility decisions. The household's mobility problem is dynamic. Each household $h \in \{s_m, s_f, c\}$ is forward-looking and discounts the future at rate $\delta \geq 0$. At the start of every period, households begin in a particular location-sector position and must decide whether to switch sectors and whether to stay or move. Every household occupies a single location, but the sector component differs by household type: for single individuals, it is one sector in $\{0, 1, \dots, K\}$, while for couples, it is the pair formed by the husband's and wife's sectors. The husband's current and destination sectors satisfy $k_m, n_m \in \{0, 1, \dots, K\}$, while the wife's current and destination market sectors satisfy $k_f, n_f \in \{1, \dots, K\}$.

It is convenient to collapse a household's location-sector position into a single *bundle*. I write the origin (current) bundle as $\mathbf{o} \equiv (i, k_m, k_f)$, the destination (next-period) bundle as $\mathbf{d} \equiv (j, n_m, n_f)$, and a generic bundle summed over as $\mathbf{z} \equiv (\ell, q_m, q_f)$. For single men, a bundle is (i, k_m) ; for single women, (i, k_f) ; and for couples, (i, k_m, k_f) ; the same symbols $\mathbf{o}, \mathbf{d}, \mathbf{z}$ denote the bundle for every household type, and I unbundle them into components only where a formula refers to a particular location or sector.

To decide their bundle for the next period, each household h in current bundle $\mathbf{o}$ draws two idiosyncratic preference shocks: a *location* shock $\varepsilon_{\mathbf{o}j,t}^{h,L}$ for each destination location j , and a *sector* shock $\varepsilon_{\mathbf{o}\mathbf{d},t}^{h,S}$ for each destination sector bundle within that location. The two enter additively, so the total preference shock attached to destination bundle $\mathbf{d}$, located in j , is

$$\varepsilon_{\mathbf{o}\mathbf{d},t}^h = \varepsilon_{\mathbf{o}j,t}^{h,L} + \varepsilon_{\mathbf{o}\mathbf{d},t}^{h,S}. \quad (13)$$

Following Rodríguez-Clare et al. (2026), I assume that the vector $\varepsilon_{\mathbf{o},t}^h \equiv (\varepsilon_{\mathbf{o}\mathbf{d},t}^h)_{\mathbf{d} \in \mathcal{S}_h}$ is i.i.d. over time and households and has a nested Gumbel joint distribution,

$$F(\varepsilon_{\mathbf{o},t}^h) = \exp \left(- \sum_{j=1}^L \left(\sum_{\mathbf{d} \in \mathcal{S}_h(j)} \exp(-\varepsilon_{\mathbf{o}\mathbf{d},t}^h / \nu) \right)^{\nu/\vartheta} \right), \quad (14)$$

with $0 < \nu \leq \vartheta$.¹¹ The parameter ν governs the dispersion of preferences across sector

¹¹Rodríguez-Clare et al. (2026) extend Caliendo et al. (2019) by allowing the elasticities of mobility across sectors within a region and across regions to differ, adding a margin that better matches the empirical evidence in Autor et al. (2013). In this setting, the extension distinguishes between mobility across sectors and mobility across regions, with the latter being the relevant margin for couples.

bundles within a location, so $1/v$ determines substitution across sectors. Similarly, ϑ governs preference dispersion across locations, so $1/\vartheta$ determines substitution across locations. The restriction $v \leq \vartheta$ implies that the sectoral elasticity $1/v$ is at least as large as the locational elasticity $1/\vartheta$. Thus, households substitute at least as readily across sectors within a location as they do across locations.¹²

Let $\mathcal{S}_h$ denote the set of feasible bundles for household type h , and define

$$\mathcal{S}_h(j) \equiv \{\mathbf{d} \in \mathcal{S}_h : \text{the location of } \mathbf{d} \text{ is } j\}$$

denotes the feasible bundles available for type h that are located in j . For singles, $\mathcal{S}_h(j)$ contains the $K+1$ sectors available in j ; for couples, it contains the $(K+1) \times K$ sector pairs available in j .

A household moving from origin bundle $\mathbf{o}$ to destination bundle $\mathbf{d}$ incurs mobility cost $\mu_{\mathbf{o}\mathbf{d}} \geq 0$, with $\mu_{\mathbf{o}\mathbf{o}} = 0$. The lifetime utility of a household of type h occupying bundle $\mathbf{o}$ at time t is

$$v_{\mathbf{o},t}^h = \log(W_{\mathbf{o},t}^h) + \max_{\mathbf{d} \in \mathcal{S}_h} \left\{ \delta \mathbb{E}(v_{\mathbf{d},t+1}^h) - \mu_{\mathbf{o}\mathbf{d}} + \varepsilon_{\mathbf{o}\mathbf{d},t}^h \right\}, \quad (15)$$

where $W_{\mathbf{o},t}^h$ denotes the period- t indirect utility of household type h in current bundle $\mathbf{o}$. For singles $h \in \{s_m, s_f\}$, $W_{\mathbf{o},t}^{s_g} = w_{i,k,t}^g / P_{i,t}$ for $g \in \{m, f\}$, as in the standard model but allowing for gender-specific wages. For couples, $W_{\mathbf{o},t}^c$ also depends on the wife's market-work decision, as described in the next section.

Let $V_{\mathbf{o},t}^h \equiv \mathbb{E}(v_{\mathbf{o},t}^h)$. Then, after substitutions, I can write the value function as

$$V_{\mathbf{o},t}^h = \log(W_{\mathbf{o},t}^h) + \vartheta \log \left(\sum_{\ell=1}^L \left(\sum_{\mathbf{z} \in \mathcal{S}_h(\ell)} \exp[\delta V_{\mathbf{z},t+1}^h - \mu_{\mathbf{o}\mathbf{z}}]^{1/v} \right)^{v/\vartheta} \right). \quad (16)$$

Let $\Omega_{\mathbf{o}j,t}^h \equiv \sum_{\mathbf{z} \in \mathcal{S}_h(j)} \exp[\delta V_{\mathbf{z},t+1}^h - \mu_{\mathbf{o}\mathbf{z}}]^{1/v}$. The choice probability can be decomposed into a location choice and a conditional sector-bundle choice. The share of type- h households in bundle $\mathbf{o}$ that choose location j is

$$\lambda_{\mathbf{o}j,t}^h = \frac{(\Omega_{\mathbf{o}j,t}^h)^{v/\vartheta}}{\sum_{\ell=1}^L (\Omega_{\mathbf{o}\ell,t}^h)^{v/\vartheta}}. \quad (17)$$

Conditional on choosing location j , the share that selects destination bundle $\mathbf{d} \in \mathcal{S}_h(j)$ is

$$\lambda_{\mathbf{o}\mathbf{d}|j,t}^h = \frac{\exp[\delta V_{\mathbf{d},t+1}^h - \mu_{\mathbf{o}\mathbf{d}}]^{1/v}}{\Omega_{\mathbf{o}j,t}^h}. \quad (18)$$

¹²When $v = \vartheta$, Equation 14 reduces to the product of independent Gumbel distributions with scale ϑ .

The unconditional share of type- h households in bundle $\mathbf{o}$ that choose destination bundle $\mathbf{d} \in \mathcal{S}_h(j)$ is therefore

$$\lambda_{\mathbf{o}\mathbf{d},t}^h = \lambda_{\mathbf{o}\mathbf{d}|j,t}^h \lambda_{\mathbf{o}j,t}^h. \quad (19)$$

Thus, the location elasticity $1/\vartheta$ governs the response of location shares to each location's inclusive value, whereas the sectoral elasticity $1/\nu$ governs the response of within-location shares to the value of each sector bundle.

Couple's problem. After relocating to $\mathbf{d}$, each couple draws an independently and identically distributed home-production productivity shock $\beta \sim \text{Pareto}(\chi, \kappa)$, with support $\beta \geq \chi > 0$ and shape parameter $\kappa > 0$.¹³ The couple then jointly chooses the wife's labor-supply status $e_{\mathbf{d},t} \in \{0, 1\}$, where $e_{\mathbf{d},t} = 1$ indicates that the wife stays at home, along with private consumption levels $c_{\mathbf{d},f,t}$ and $c_{\mathbf{d},m,t}$ and public-good expenditure $q_{\mathbf{d},t}$. When the wife stays at home, the productivity shock increases the effective level of the household public good, $Q_{\mathbf{d},t} = \beta^{e_{\mathbf{d},t}} q_{\mathbf{d},t}$.

Conditional on choosing bundle $\mathbf{d}$, the couple therefore solves

$$\begin{aligned} \max_{c_{\mathbf{d},f,t}, c_{\mathbf{d},m,t}, q_{\mathbf{d},t}, e_{\mathbf{d},t} \in \{0,1\}} \quad & (Q_{\mathbf{d},t}^\gamma c_{\mathbf{d},f,t}^{1-\gamma})^\alpha (Q_{\mathbf{d},t}^\gamma c_{\mathbf{d},m,t}^{1-\gamma})^{1-\alpha} \\ \text{s.t.} \quad & P_{j,t} (c_{\mathbf{d},f,t} + c_{\mathbf{d},m,t} + q_{\mathbf{d},t}) = \underbrace{w_{j,n_m,t}^M + w_{j,n_f,t}^F (1 - e_{\mathbf{d},t})}_{I(e)}, \\ & Q_{\mathbf{d},t} = \beta^{e_{\mathbf{d},t}} q_{\mathbf{d},t}, \end{aligned} \quad (20)$$

with $0 < \alpha < 1$ and $0 < \gamma < 1$.

Following Chiappori and Mazzocco (2017), the optimal allocations conditional on $\mathbf{d}$ and $e_{\mathbf{d},t}$ are

$$\begin{aligned} c_{\mathbf{d},f,t}^*(e) &= \alpha(1 - \gamma) \frac{I(e)}{P_{j,t}}, \\ c_{\mathbf{d},m,t}^*(e) &= (1 - \alpha)(1 - \gamma) \frac{I(e)}{P_{j,t}}, \\ q_{\mathbf{d},t}^*(e) &= \gamma \frac{I(e)}{P_{j,t}} \implies Q_{\mathbf{d},t}^*(e) = \beta^e \gamma \frac{I(e)}{P_{j,t}} \end{aligned} \quad (21)$$

To find the decision rule for $e_{\mathbf{d},t}$, let $\mathcal{A} \equiv \alpha^{\alpha(1-\gamma)}(1 - \alpha)^{(1-\alpha)(1-\gamma)}\gamma^\gamma(1 - \gamma)^{1-\gamma}$, and define

¹³The home-production productivity shock captures the non-monetary value generated when the wife engages in home production (Becker, 1965).

the sector-specific female-to-male wage ratio faced by a couple in $\mathbf{d}$ as

$$r_{\mathbf{d},t} \equiv \frac{w_{j,n_f,t}^F}{w_{j,n_m,t}^M}.$$

Then, the couple's static indirect utility conditional on choosing $\mathbf{d}$ and drawing β is

$$\mathcal{U}_{\mathbf{d},t}(\beta) = \mathcal{A} \frac{w_{j,n_m,t}^M}{P_{j,t}} \begin{cases} (1 + r_{\mathbf{d},t}) & \text{if } e_{\mathbf{d},t} = 0 \\ \beta^\gamma & \text{if } e_{\mathbf{d},t} = 1 \end{cases}$$

Hence, the wife works in the market if and only if

$$\beta \leq \beta_{\mathbf{d},t}^*, \quad \beta_{\mathbf{d},t}^* = (1 + r_{\mathbf{d},t})^{1/\gamma}.$$

The market-work probability is therefore

$$\rho_{\mathbf{d},t} \equiv \Pr(e_{\mathbf{d},t} = 0) = 1 - \left(\frac{\chi}{\beta_{\mathbf{d},t}^*} \right)^\kappa = 1 - \chi^\kappa (1 + r_{\mathbf{d},t})^{-\kappa/\gamma}. \quad (22)$$

The couple's decision therefore compares her market work with home production. An interior participation margin in bundle $\mathbf{d}$ requires $0 < \rho_{\mathbf{d},t} < 1$. The upper bound, $\rho_{\mathbf{d},t} < 1$, holds for any $\chi > 0$, whereas the lower bound, $\rho_{\mathbf{d},t} > 0$, requires the lower support bound to lie below the market-work threshold: $\chi < (1 + r_{\mathbf{d},t})^{1/\gamma}$. Moreover, because the home-production productivity shock enters the couple's problem as $Q_{\mathbf{d},t} = \beta^{e_{\mathbf{d},t}} q_{\mathbf{d},t}$, the model imposes $\chi \geq 1$, ensuring that staying at home never reduces the effective level of the public good. For this condition to hold in every location-sector-time cell, the relevant restriction is $1 \leq \chi < \min_{\mathbf{d},t} (1 + r_{\mathbf{d},t})^{1/\gamma}$.

Given the wife's labor-supply rule, the ex ante expected utility from choosing $\mathbf{d}$ at time t is

$$\begin{aligned} W_{\mathbf{d},t}^c &\equiv \mathbb{E}[\mathcal{U}_{\mathbf{d},t}(\beta)] = \mathcal{A} \frac{w_{j,n_m,t}^M}{P_{j,t}} \left[(1 + r_{\mathbf{d},t}) F_\beta(\beta_{\mathbf{d},t}^*) + \int_{\beta_{\mathbf{d},t}^*}^\infty \beta^\gamma f_\beta(\beta) d\beta \right] \\ &= \frac{w_{j,n_m,t}^M}{P_{j,t}} \cdot \underbrace{\mathcal{A} \left[(1 + r_{\mathbf{d},t}) + \chi^\kappa \frac{\gamma}{\kappa - \gamma} (1 + r_{\mathbf{d},t})^{1-\kappa/\gamma} \right]}_{B_{\mathbf{d},t}}, \end{aligned}$$

where $B_{\mathbf{d},t}$ can be expressed entirely in terms of $\rho_{\mathbf{d},t}$ as $B_{\mathbf{d},t} = \mathcal{A} \chi^\gamma (1 - \rho_{\mathbf{d},t})^{-\gamma/\kappa} \frac{\kappa/\gamma - \rho_{\mathbf{d},t}}{\kappa/\gamma - 1}$. This term captures the quality-adjusted value of destination-sector bundle $\mathbf{d}$ for couples. Conditional on choosing $\mathbf{d}$, it combines the husband's sector-specific wage with the expected gain from the wife's endogenous labor-supply decision. Finiteness of the expected value requires $\kappa > \gamma$.

Comparative statics of the market-work probability. The probability that she works in the market in Equation 22 only depends on the relative wage r , the parameters of the Pareto distribution, χ and κ , and the public-good expenditure share, γ .

The derivative of ρ with respect to the relative wage between husband and wife is:

$$\frac{\partial \rho}{\partial r} = \frac{\kappa}{\gamma} \cdot \frac{1 - \rho}{1 + r} > 0. \quad (23)$$

This implies that a higher relative wage raises the opportunity cost of staying home by raising what the household gives up when the wife stays home. Thus, the threshold rises, and the draws that used to justify staying home no longer do, so that more wives work.

The derivative of ρ with respect to the parameters of the Pareto distribution are:

$$\frac{\partial \rho}{\partial \chi} = -\frac{\kappa}{\chi}(1 - \rho) < 0, \quad \frac{\partial \rho}{\partial \kappa} = -\frac{(1 - \rho) \log(1 - \rho)}{\kappa} > 0. \quad (24)$$

Raising χ implies that every couple is better off when she stays at home. Raising the lower bound χ scales the whole distribution of home productivity upward, with the threshold unchanged. More couples draw above β^* , so participation falls. Because χ is a scale parameter, the effect is proportional: $1 - \rho$ rises by the factor κ/χ . On the contrary, a larger shape parameter concentrates the draws near χ and thins the right tail. Staying home takes an exceptionally good draw, one above β^* , and those draws become rarer, so more wives work.

The derivative of ρ with respect to γ is:

$$\frac{\partial \rho}{\partial \gamma} = -\frac{\kappa(1 - \rho)}{\gamma^2} \log(1 + r) < 0. \quad (25)$$

Then, a higher public-good share raises the utility from any given β , since β^γ grows with γ when $\beta > 1$. The threshold $(1 + r)^{1/\gamma}$ therefore falls: a smaller home-productivity draw now matches the income gain from working. More couples choose home production. The effect is proportional to $\log(1 + r)$ because γ acts by discounting that income gain, and a larger gain gives it more to discount.

Indirect utility. Incorporating couples' ex ante expected utility conditional on choosing $\mathbf{d}$ at time t , the indirect utility relevant for household h 's dynamic migration problem in

Equation 16 is¹⁴

$$W_{o,t}^h = \begin{cases} \frac{w_{i,k,t}^M}{P_{i,t}} & \text{if } h = s_m \\ \frac{w_{i,k,t}^F}{P_{i,t}} & \text{if } h = s_f \\ \frac{B_{o,t} w_{i,k_m,t}^M}{P_{i,t}} & \text{if } h = c \end{cases}$$

Here, $P_{i,t}$ denotes the regional aggregate consumption price index defined in Equation 12.

3.2 Equilibrium

The economic geography at time t , denoted by Γ_t , consists of the time-varying exogenous sectoral productivities and the time-invariant shares of single individuals born in each region, migration costs, iceberg trade costs, and non-employment benefits: $\Gamma_t = \{A_{i,k,t}, \bar{s}_i, \mu_{od}, \tau_{ik,jn}, b_j\}_{i,j=1, k,n=1}^{L,K}$. The remaining time-invariant parameters are collected in the vector $\bar{\Gamma} = \{N, L, K, \delta, \chi, \kappa, \alpha, \gamma, \sigma, v, \vartheta, \{\psi_{k,m}, \psi_{k,f}, \eta_k, \iota_k\}_{k=1}^K\}$.

Let $N_{d,t}^{sm}$ and $N_{d,t}^{sf}$ denote, respectively, the masses of single men and single women living in location j and working in sector $n \in \{0, \dots, K\}$ at time t . Let $N_{d,t}^c$ denote the mass of couples living in location j in which the husband is assigned to sector $n_m \in \{0, \dots, K\}$ and the wife is assigned to productive market sector $n_f \in \{1, \dots, K\}$. The supplies of male and female labor to productive sectors $n = 1, \dots, K$ are therefore

$$M_{j,n,t} = N_{j,n,t}^{sm} + \sum_{n_f=1}^K N_{j,n,n_f,t}^c, \quad F_{j,n,t} = N_{j,n,t}^{sf} + \sum_{n_m=0}^K \rho_{j,n_m,n,t} N_{j,n_m,n,t}^c. \quad (26)$$

These expressions make the components of a couple's bundle explicit. The husband's sector is the second argument, and the wife's sector is the third. Accordingly, male labor supply in sector n sums over the wife's sector, whereas female labor supply in sector n sums over the husband's sector and weights each cell by the wife's probability of participation.

Let $N_t = \{\{N_{d,t}^{sm}\}, \{N_{d,t}^{sf}\}, \{N_{d,t}^c\}\}$ denote the complete distribution of household masses across locations and sectoral bundles at time t . Because the problems of intermediate-good firms, final-good firms, and married women's labor supply are static, the *temporary equilibrium* can be defined as follows.

Definition 1. *Given $(N_t, \Gamma_t, \bar{\Gamma})$, a **temporary equilibrium** is a vector of productive-sector wages $\{w_{j,n,t}^M, w_{j,n,t}^F\}$, sectoral prices $\{p_{j,n,t}, P_{j,n,t}\}$, regional price indices $\{P_{j,t}\}$, and alloca-*

¹⁴Singles have the same preferences over private consumption and the household good as spouses, $\gamma \log Q + (1 - \gamma) \log c$, but do not share Q with a spouse. Their indirect utility is therefore $\log(w/P)$ up to the constant $\gamma \log \gamma + (1 - \gamma) \log(1 - \gamma)$, which cancels from all welfare changes. This formulation assumes that welfare is comparable across marital statuses.

tions $\{\rho_{\mathbf{d},t}, M_{j,n,t}, F_{j,n,t}, Y_{j,n,t}, Q_{j,n,t}, Q_{j,t}, \pi_{ik,jn,t}\}$ such that, for every productive region-sector pair jn and period t :

- (i) Intermediate-good firms maximize profits subject to the CES production technology, and wages satisfy (10);
- (ii) Final-good firms maximize profits subject to sectoral CES and regional Cobb-Douglas aggregation. Sectoral price indices satisfy (11), and regional price indices satisfy (12);
- (iii) Couples optimally choose the wife's labor supply, and participation probabilities satisfy (22) for every sector pair (n_m, n_f) ;
- (iv) Labor markets clear: $p_{j,n,t}Y_{j,n,t} = w_{j,n,t}^M M_{j,n,t} + w_{j,n,t}^F F_{j,n,t}$;
- (v) Goods markets clear under balanced trade: $p_{i,k,t}Y_{i,k,t} = \sum_{j=1}^L \sum_{n=1}^K \pi_{ik,jn,t} \iota_n P_{j,t} Q_{j,t}$.

Given the path of exogenous fundamentals and an initial distribution of households across location-sector bundles, denoted by $N_0 = \left\{ \{N_{o,0}^{sm}\}, \{N_{o,0}^{sf}\}, \{N_{o,0}^c\} \right\}$, the model's *sequential competitive equilibrium* can be defined. Using the two-stage choice probabilities in Equations 17 and 19, the population of each household type evolves according to

$$N_{\mathbf{d},t+1}^h = \sum_{\mathbf{o} \in \mathcal{S}_h} \lambda_{\mathbf{o}\mathbf{d}|j,t}^h \lambda_{\mathbf{o},t}^h N_{\mathbf{o},t}^h, \quad \mathbf{d} \in \mathcal{S}_h(j), \quad \forall h. \quad (27)$$

Definition 2. Given $(N_0, \Gamma_t, \bar{\Gamma})$, a **sequential competitive equilibrium** is a sequence of productive-sector wages $\{w_{j,n,t}^M, w_{j,n,t}^F\}_{t=0}^\infty$, prices $\{p_{j,n,t}, P_{j,n,t}, P_{j,t}\}_{t=0}^\infty$, values $\{V_{\mathbf{o},t}^h\}_{t=0}^\infty$, and allocations $\{\rho_{\mathbf{d},t}, \pi_{ik,jn,t}, \lambda_{\mathbf{o},t}^h, \lambda_{\mathbf{o}\mathbf{d}|j,t}^h, N_{\mathbf{o},t}^h, M_{j,n,t}, F_{j,n,t}, Y_{j,n,t}, Q_{j,n,t}, Q_{j,t}\}_{t=0}^\infty$ for every household type h , such that, in each period t :

- (i) The temporary equilibrium holds;
- (ii) Couples and single individuals solve their respective migration problems: value functions satisfy (16), while location and conditional sectoral choice probabilities satisfy (17) and (18);
- (iii) The population of each household type evolves according to (27);
- (iv) Productive-sector male and female labor supplies satisfy (26).

3.3 Welfare Analysis of a Sectoral Productivity Shock

To quantify welfare, Equation 16 can be combined with $\mu_{oo} = 0$ to express the expected lifetime utility of household type h in bundle $\mathbf{o}$ at time t as

$$\begin{aligned} V_{\mathbf{o},t}^h &= \log(W_{\mathbf{o},t}^h) + \delta V_{\mathbf{o},t+1}^h - \vartheta \log(\lambda_{\mathbf{o}i,t}^h) - v \log(\lambda_{\mathbf{o}\mathbf{o}|i,t}^h) \\ &= \sum_{s=t}^{\infty} \delta^{s-t} \log \left(\frac{W_{\mathbf{o},s}^h}{(\lambda_{\mathbf{o}i,s}^h)^{\vartheta} (\lambda_{\mathbf{o}\mathbf{o}|i,s}^h)^v} \right). \end{aligned} \quad (28)$$

Here, $\lambda_{\mathbf{o}i,t}^h$ denotes the probability of remaining in the origin location, and $\lambda_{\mathbf{o}\mathbf{o}|i,t}^h$ denotes the probability of remaining in the origin sector bundle, conditional on remaining in the origin location.

I define a counterfactual economy by an alternative sequence of sectoral productivities $\{A_{j,n,t}\}_{t \geq 1}$. Following CDP's timing assumption, the change is unanticipated at $t = 0$, and households fully learn the new productivity path at $t = 1$. Let $\hat{x}_{t+1} = \hat{x}'_{t+1}/\hat{x}_{t+1}$ denote the ratio of changes over time in the counterfactual and baseline equilibria, and let $\tilde{x}_t \equiv x'_t/x_t = \prod_{s=1}^t \hat{x}_s$ denote the corresponding ratio of levels, under the normalization $x'_0 = x_0$.

I use Dynamic Hat Algebra¹⁵ to compute expected lifetime utility in the baseline economy, $V_{\mathbf{o},0}^h$, and in the counterfactual economy, $V_{\mathbf{o},0}'^h$. The consumption compensating variation for household type h in bundle $\mathbf{o}$ at time $t = 0$, denoted by $\zeta_{\mathbf{o}}^h$, is defined implicitly by

$$V_{\mathbf{o},0}'^h = V_{\mathbf{o},0}^h + \sum_{s=0}^{\infty} \delta^s \log(\zeta_{\mathbf{o}}^h) \iff \log(\zeta_{\mathbf{o}}^h) = (1 - \delta)(V_{\mathbf{o},0}'^h - V_{\mathbf{o},0}^h).$$

Because the temporary equilibrium and migration shares are identical across the two economies at $t = 0$, the welfare change in consumption-equivalent units is

$$Z_{\mathbf{o}}^h \equiv \log(\zeta_{\mathbf{o}}^h) = \sum_{s=1}^{\infty} \delta^s \log \left(\frac{\hat{W}_{\mathbf{o},s}^h}{(\hat{\lambda}_{\mathbf{o}i,s}^h)^{\vartheta} (\hat{\lambda}_{\mathbf{o}\mathbf{o}|i,s}^h)^v} \right), \quad (29)$$

where

$$\hat{W}_{\mathbf{o},s}^h = \begin{cases} \hat{w}_{i,k_m,s}^M / \hat{P}_{i,s} & \text{if } \mathbf{o} = (i, k_m), h = s_m, \\ \hat{w}_{i,k_f,s}^F / \hat{P}_{i,s} & \text{if } \mathbf{o} = (i, k_f), h = s_f, \\ \hat{B}_{\mathbf{o},s} \hat{w}_{i,k_m,s}^M / \hat{P}_{i,s} & \text{if } \mathbf{o} = (i, k_m, k_f), h = c. \end{cases}$$

Conditional on the observed staying shares, a larger ϑ , corresponding to a lower migration

¹⁵See Section C for more details.

elasticity, increases the welfare weight assigned to changes in the probability of remaining in the origin location. Similarly, a larger v increases the weight assigned to changes in the probability of remaining in the origin sector bundle conditional on staying in that location. Thus, declines in these probabilities enter the welfare measure with weights ϑ and v , respectively, where $\vartheta \geq v$.

Aggregating welfare changes across initial location-sector bundles, the aggregate welfare effect of a sectoral productivity shock is

$$Z = \underbrace{\sum_{i=1}^L \sum_{k=0}^K N_{o,0}^{s_m} Z_o^{s_m}}_{Z^{s_m}} + \underbrace{\sum_{i=1}^L \sum_{k=0}^K N_{o,0}^{s_f} Z_o^{s_f}}_{Z^{s_f}} + \underbrace{\sum_{i=1}^L \sum_{k_m=0}^K \sum_{k_f=1}^K N_{o,0}^c Z_o^c}_{Z^c}. \quad (30)$$

Aggregate welfare changes are measured in utility units. For couples, however, one dollar of market expenditure can generate more than one dollar of consumption-equivalent value because of economies of scale within marriage. Following Chiappori and Meghir (2015), a Money Measure Welfare Index (MMWI) is therefore constructed for singles and couples to obtain a comparable money-metric measure of welfare changes for the quantitative analysis. See Appendix B.2.

The model's mechanisms map directly onto the three empirical facts. Sectoral productivity shocks transmitted through trade generate differential changes in male and female wages across region-sector cells, consistent with Fact 1. Couples jointly choose a single location, but spouses enter this decision asymmetrically: the husband's wage $w_{j,n_m,t}^M$ scales the couple's entire payoff, whereas the wife's wage enters only through the relative wage $r_{d,t}$ in $B_{d,t}$. This asymmetry provides the mechanism underlying Fact 2. Finally, the participation threshold $\beta_{d,t}^*$ generates the adjustment margin in Fact 3, with the direction observed in the data: a decline in the husband's sectoral wage raises $r_{d,t}$, increases the threshold, and induces more married women to enter market work. Section 5 disciplines the model using the data.

4 Bringing the Model to the Data

This section applies the model developed above to the data to quantify the China Trade Shock in an economy with heterogeneous households. It first describes the quantitative environment, distinguishing parameters drawn from the literature from inputs observed directly in the data. It then presents the calibration of preference dispersion across locations and the couple-specific parameters, followed by the method for quantifying the China Trade Shock.

4.1 Quantitative Environment

The quantitative environment follows CDP. The economy comprises 87 locations: the 50 US states, with the District of Columbia included in Virginia, and 37 foreign trade partners, consisting of China, the 35 largest US trading partners, and the Rest of the World. It includes 22 productive sectors, comprising 12 manufacturing sectors, retail, construction, and eight service sectors, as well as a non-employment sector. Each model period corresponds to one quarter, and the model is solved for 200 quarters beginning in 2000Q1.

Adopting CDP’s quantitative environment serves two purposes. First, it provides a trade block that has already been disciplined by the data, ensuring that the productivity and trade fundamentals are not constructed specifically for this analysis. Second, it provides an exact benchmark: CDP’s model is a special case in which all households are single individuals, male and female labor are perfect substitutes, and $v = \vartheta$. Differences between the two sets of results can therefore be attributed to the household block rather than to the underlying quantitative environment. One caveat is the difference in timing: the empirical analysis in Section 2 follows ADH’s 2001–2012 window, whereas the quantitative exercise retains the earlier 2000–2007 window (Autor et al., 2013).

Because the trade side is taken directly from CDP, the quantitative exercise uses their full production block rather than the simplified version presented in Section 3. Intermediate-good firms use both labor and intermediate inputs from other sectors, and total factor productivity combines the Hicks-neutral component introduced in Section 3 with a variety-specific component. Implementing the household extension therefore requires replacing the labor input demanded by intermediate-good firms with the effective labor aggregate in Equation 7. All other inputs to the trade block are taken directly from CDP.¹⁶

δ and σ from previous literature. The discount factor is set to $\delta = 0.99$, following CDP. The within-sector elasticity of substitution between male and female labor, σ , is taken from Acemoglu et al. (2004) rather than estimated directly. This choice is motivated by the identification requirements implied by the CES technology in Section 3:

$$\log \left(\frac{w_{i,k,t}^M}{w_{i,k,t}^F} \right) = \log \left(\frac{\psi_{k,m}}{\psi_{k,f}} \right) - \frac{1}{\sigma} \log \left(\frac{M_{i,k,t}}{F_{i,k,t}} \right).$$

Identifying σ requires exogenous variation in the relative supply of male and female labor; without such variation, the estimated coefficient would be biased. Acemoglu et al. (2004)

¹⁶Sectoral trade elasticities $\{\eta_n\}$ are from Caliendo and Parro (2015); bilateral trade shares $\{\pi_{ik,jn,t}\}$, region-sector output and expenditure, and the path of sectoral productivity changes in China are from CDP; final-good expenditure shares $\{\iota_n\}$ come from the Bureau of Economic Analysis (BEA) input-output tables.

estimate an analogous relationship (their Equation 11), using the mobilization of men during World War II as an exogenous shift in women’s relative labor supply. Accordingly, I set σ to 1.4, the lower bound of their estimated range. This conservative choice reflects the fact that the cohorts examined in their study were almost entirely retired by 2000Q1, when the quantitative model begins.

Directly observed inputs. Household-side inputs are constructed from US microdata. The initial distribution of households across states and sector pairs, $\{N_{o,0}^{s,m}, N_{o,0}^{s,f}, N_{o,0}^c\}$,¹⁷ and the state-level share of singles, $\bar{s}_i$, are measured using the 5% IPUMS Census 2000. Male and female employment, wages, and wives’ labor-force participation are measured using the 2000 Census and the American Community Survey (ACS) for 2001–2007. The sector-specific gender-intensity parameters $\{\psi_{k,m}, \psi_{k,f}\}$ are recovered from hours worked by gender in each sector. Regional non-employment income, b_j , is measured by state unemployment-insurance benefits from the US Department of Labor and held fixed at its 2000 value. Table 4 summarizes these inputs and their sources.¹⁸

One discrepancy between the model and the data remains. In the model, couples first choose the wife’s productive sector and then determine her participation status, so the wife is always assigned to a productive sector, even when she does not participate in market work. In the data, however, the potential sector of a non-employed married woman is unobserved. A naive approach would assign non-employed wives to productive sectors in proportion to the sectoral distribution of employed wives. This approach creates a selection problem: sectors employing larger shares of wives would also receive larger shares of non-employed wives, even though wives assigned to those sectors are precisely the least likely to be out of the labor force. To recover each non-employed wife’s potential sector, the model is used to derive a selection correction.¹⁹ The implied mass of non-employed married women in state j and sector n is

$$NonF_{j,n,t}^c = \underbrace{F_{j,n,t}^c}_{\text{observed}} \cdot \underbrace{\frac{1 - \rho_{j,n,t}}{\rho_{j,n,t}}}_{\text{the model's odds of non-employment}}.$$

¹⁷The quantitative exercise holds household composition fixed: marriage, divorce, fertility, and population growth are excluded, so couples remain together throughout the transition. Figure A.4 provides supporting evidence for this assumption.

¹⁸Two conventions determine the interpretation of the participation margin. First, non-employment refers to being *out of the labor force*, rather than unemployed. Including unemployed individuals would roughly double the measured flows into non-employment and would confound the wife’s participation decision with the husband’s job loss. Second, following the convention in this literature, an unemployed worker retains the sector of their most recent job. Thus, a positive sector index does not necessarily indicate current employment, and participation is always measured using an explicit labor-force indicator.

¹⁹See Appendix D.1 for details.

Table 4: Observed Quantitative Inputs

Object	Description	Source
$\bar{s}_i, N_i$	State-level share of singles and population	IPUMS Census 2000 and ACS 2001–2007
$N_{o,0}^{s,m}, N_{o,0}^{s,f}$	Initial populations of single men and women by state $\times$ sector	IPUMS Census/ACS
$N_{o,0}^c$	Initial population of couples by state $\times$ husband-wife sector pair	IPUMS Census/ACS
$M_{o,t}, F_{o,t}$	Male and female employment by state $\times$ sector	IPUMS Census/ACS
$\psi_{k,m}, \psi_{k,f}$	Sector-specific gender intensities	Hours worked by gender, IPUMS Census/ACS
b_j	State-level non-employment income	US Department of Labor unemployment-insurance benefits, fixed at the 2000 value

Notes. The sample follows CDP and includes individuals aged 25–65 across 50 states, excluding widows. Non-employment refers to being out of the labor force rather than being unemployed. Sectors are classified according to CDP’s 22-sector partition.

Scaling observed employment by $1/\rho_{j,n,t}$ corrects for selection and assigns non-employed wives to their potential productive sectors.

The final data inputs are the migration matrices. Because the model is solved using Dynamic Hat Algebra, the time-invariant mobility costs μ_{od} cancel from the recursion governing the choice probabilities (Proposition C.2). The recursion therefore requires only two initial objects for each household type: the location shares $\lambda_{oj,0}^h$ defined in Equation 17 and the within-location shares $\lambda_{od|j,0}^h$ defined in Equation 18.

Following CDP, the initial unconditional transition probability $\lambda_{od,0}^h$ from Equation 19 is constructed as the product of an interstate migration probability from the ACS and a sectoral transition probability from the Current Population Survey (CPS). For interstate movers, the sectoral transition probability is imputed using the destination state’s sectoral transition matrix:

$$\lambda_{(i,k)(j,n),0}^h = \underbrace{f_{ij}^h}_{\text{ACS}} \times \underbrace{P_j^h(n | k)}_{\text{CPS}}.$$

This construction presents two challenges. First, the surveys differ in frequency: the ACS is annual, whereas the CPS is monthly. This mismatch is addressed following CDP. Second,

disaggregating CPS observations by household type produces sparsely populated cells. To address this issue, the 12 manufacturing sectors are temporarily aggregated into a single sector, as are the service sectors. The resulting transition probabilities are then disaggregated using national gender-specific sorting patterns. Appendix D.2 documents each step in the construction of the migration matrices.

4.2 Dispersion of Preferences across Sectors and Locations

The model features two dispersion parameters. The parameter v governs substitution across sectors within a location (Equation 18), whereas ϑ governs substitution across locations (Equation 17). The former is taken from the literature, and the latter is calibrated.

Following Artuc et al. (2010), CDP estimate their dispersion parameter using sectoral transitions among men in the March Current Population Survey (CPS) from 1975 to 2000. Because the CPS follows addresses, respondents who change residence are removed from the sample. The matched panel therefore records sectoral transitions only among individuals who remain in the same location and contains no transitions across locations. Consequently, the estimated parameter corresponds to applying Artuc et al. (2010)’s method to the within-location choice in Equation 18, which is precisely the role of v in the present model. The only distinction is that continuation values V^h are household-type specific, reflecting the economies of scale associated with marriage. Because the model does not impose differential rates of sectoral substitution across household types, v is set to CDP’s quarterly value of 5.34 for all household types.

I calibrate ϑ by indirect inference, following Rodríguez-Clare et al. (2026). The calibration targets the population response to import competition, as estimated by Autor et al. (2013), who regress the change in the log of the working-age population of a commuting zone on its exposure to Chinese imports and obtain a coefficient of -0.050 . The same regression is estimated using model-generated state-level data. The dependent variable is the percentage change in state population between 2000 and 2007, scaled to a decade as in Autor et al. (2013). The regressor is the state’s exposure to Chinese imports over 2000–2007, predicted using the Autor et al. (2013) instrument and rescaled to the mean of their commuting-zone measure so that the model and empirical coefficients have the same units.²⁰ The calibrated value, $\vartheta = 10.67$, generates a coefficient of -0.051 , compared with the target of -0.050 . Because $v < \vartheta$, households substitute more readily across sectors within a location than across locations.

²⁰The model sample is restricted to the 48 states in the Autor et al. (2013) estimation sample, excluding Alaska and Hawaii. Using first-stage fitted exposure makes the ordinary least squares coefficient in the model the counterpart of the two-stage least squares coefficient in the data.

Calibrating ϑ in an otherwise identical economy composed entirely of single individuals, in which each individual independently chooses a location and sector, yields a migration elasticity that is 5% higher under the same target and value of v . Given the model’s solution timing, this difference provides a lower bound on the quantitative importance of accounting for couples.²¹

4.3 Calibrating the Couple’s Maximization Problem

Four parameters govern the couple’s maximization problem and have no counterpart in CDP because they correspond to the wife’s endogenous labor supply. The parameters correspond to the wife’s weight in the household objective α , the public-good expenditure share γ , and the location and shape of the home-production shock, χ and κ . Since I solve the couple’s problem following the three-stage formulation in Chiappori and Mazzocco (2017), I choose to identify them *sequentially*, each from its own source, and each maps onto a distinct piece of variation. Table 5 summarizes the resulting values; Appendix D.3 describes the construction in detail.

The parameters α and γ are calibrated to exactly match observed expenditure shares. Because the couple’s utility function in Equation 20 is Cobb–Douglas, α corresponds to the wife’s share of total private consumption,²² while γ corresponds to the share of public-good expenditure in total household expenditure:

$$\alpha = \frac{c_f^*}{c_f^* + c_m^*}, \quad \gamma = \frac{q^*}{q^* + c_f^* + c_m^*}.$$

Both expenditure shares are measured using the 2000–2007 Consumer Expenditure Interview Survey. The public-good share γ is measured directly as the share of total expenditure devoted to market inputs into home production.²³ Estimating the bargaining weight α requires an additional step because the observed assignable private good, adult apparel, represents only a subset of private consumption. The spouses’ resource shares are therefore recovered using the method of Browning et al. (2013), which relies on expenditures by *singles*.

²¹In this quantification, the baseline economy is solved from 2008 onward, so differences in migration elasticities arise only through the counterfactual economy. If both the baseline and counterfactual economies were solved from $t = 0$, the difference in migration elasticities between the household model and the single-agent model would be larger.

²²The quantification holds α fixed, thereby ruling out shock-induced renegotiation of intrahousehold bargaining power. It also abstracts from the intensive margin of the wife’s labor supply: the wife either participates in market work or does not. The latter restriction appears empirically innocuous because, alongside the participation responses documented in Fact 3, wives’ hours exhibit no statistically significant response to either spouse’s exposure. These estimates are available upon request.

²³The home-production goods comprise shelter, utilities, food at home, and market substitutes for home time, such as domestic services and appliance repair.

This procedure yields $\alpha = 0.47$ and $\gamma = 0.49$.²⁴

The parameters κ and χ are calibrated using Equation 22. Expressing the equation in terms of the non-participation probability and taking logs yields

$$\log(1 - \rho(r)) = \underbrace{\kappa \log \chi}_{\text{intercept}} - \underbrace{(\kappa/\gamma)}_{\text{slope}} \log(1 + r). \quad (31)$$

Direct estimation of Equation 31 is complicated by endogeneity and sample selection. First, the relative wage r faced by a couple is an equilibrium outcome. Second, relative wage changes are observed only for couples in which the wife participates in market work, creating a truncation problem.

Then, I use the empirical results in Section 2 to provide exogenous variation through the China Trade Shock. When the husband is exposed, the shock directly affects his wage but not his wife's wage, influencing her participation decision through the relative wage r , as required by the exclusion restriction underlying Equation 31. The same shock simulated in the quantitative exercise therefore disciplines the model's participation margin.

Because Equation 31 is linear, its slope can be identified using the following Wald ratio:

$$\frac{\kappa}{\gamma} = -\frac{\Delta \log(1 - \rho)}{\Delta \log(1 + r)},$$

where both changes are measured within the same empirical experiment. Wives of exposed husbands increase their labor force participation by 4.2 percentage points from a baseline of 0.78 (Panel b of Table 3), implying $\Delta \log(1 - \rho) = -0.212$. The husband's exposure increases the relative wage r by 12.2 log points, corresponding to a 4.8 log-point increase in $1 + r$ (Table D.1). Together, these estimates imply

$$\frac{\kappa}{\gamma} = -\frac{-0.212}{0.048} = 4.4 \quad \implies \quad \kappa = \gamma \times \frac{\kappa}{\gamma} = 2.17.$$

The resulting calibration satisfies the model's theoretical restriction, $\kappa > \gamma$.

Conditional on the estimated slope, χ is calibrated to match the *level* of married women's labor force participation. Specifically, χ is chosen so that the model's mass-weighted average participation rate equals its empirical counterpart in $t_0 = 2000Q1$. The resulting value, $\chi = 1.92$, satisfies the support restriction $\chi \geq 1$ and yields an interior participation margin in every cell. This procedure also reconciles the two participation rates used in the calibration: the slope is identified from the PSID experiment using that sample's baseline participation rate of 0.78, whereas χ anchors the level to the model population's mass-weighted participation rate of 0.621.

²⁴Appendix D compares these estimates with those in the previous literature and discusses potential sources of discrepancy.

To interpret the calibrated parameters, consider the implied elasticity of married women’s labor force participation with respect to the female wage. Differentiating $\rho = 1 - \chi^\kappa(1+r)^{-\kappa/\gamma}$ with respect to the female wage and evaluating the resulting expression at the model’s anchor, $\bar{\rho} = 0.621$ and $r_0 = 0.607$, yields

$$e = \frac{\partial \log \rho}{\partial \log w_f} = \frac{\kappa}{\gamma} \cdot \frac{1 - \bar{\rho}}{\bar{\rho}} \cdot \frac{r_0}{1 + r_0} = 1.02$$

Thus, a 1% increase in the female wage raises married women’s labor force participation by 1.02%. This elasticity lies within the range for married women surveyed by Blundell and MaCurdy (1999) and exceeds the ≈ 0.25 – 0.30 aggregate extensive-margin elasticity preferred by Chetty et al. (2011), which is not specific to married women.

Table 5: Calibrated Household Parameters

Object	Name	Value
α	Wife’s weight in the household objective	0.47
γ	Public-good expenditure share	0.49
κ/γ	Slope of the participation rule	4.44
κ	Pareto shape parameter for home productivity	2.17
χ	Pareto lower bound for home productivity	1.92

Notes. See Appendix D for details.

4.4 Quantifying the China Trade Shock

To quantify the effects of the China Trade Shock, the shock must first be defined within the model. Following CDP, it is represented as an increase in total factor productivity in *Chinese* manufacturing sectors. Thus, the shock is neither a reduction in US productivity nor an intervention imposed directly on US states. Instead, it is a foreign supply shock that is transmitted through the trade block as a demand shock to US producers: cheaper Chinese imports reduce both the prices and quantities at which US manufacturers sell in domestic and third-country markets. The path of Chinese sectoral productivity is taken from CDP and calibrated so that the resulting growth in US imports from China reproduces the first stage in Autor et al. (2013). The model therefore defines the shock at the industry level, consistent with the reduced-form evidence in Section 2.

The shock is quantified by comparing the two economies defined for productivity shocks in Section 3.3. As in CDP, the *baseline* economy includes the observed growth in Chinese productivity, whereas the *counterfactual* economy eliminates this growth. Formally,

$\hat{A}_{\text{China},k,t} = 1$ for every manufacturing sector k and period t in the counterfactual economy.

All results reported below are differences between the baseline and counterfactual economies. Accordingly, a positive employment effect indicates that the China Trade Shock increased employment relative to a world without the shock, while a positive welfare effect indicates that the relevant household type was better off. Throughout the remainder of the paper, the *baseline economy* refers to the calibrated specification described in this section, whereas *counterfactual exercises* refer to specifications that modify either the construction of directly observed inputs or selected parameter values.

5 Quantitative Consequences of the Shock Under Joint Decision-Making

In this section, I present the results of the quantitative exercise. First, I examine the effects of the China Trade Shock on manufacturing output, total employment, population changes, participation responses, and welfare. I then compare the effects of the baseline model with those of a single-agent economy. Secondly, I perform three counterfactual exercises to quantify the importance of each margin of adjustment by shutting down, one by one, the response of migration, participation, and the gender wage gap.

5.1 Quantitative Effects of the China Trade Shock

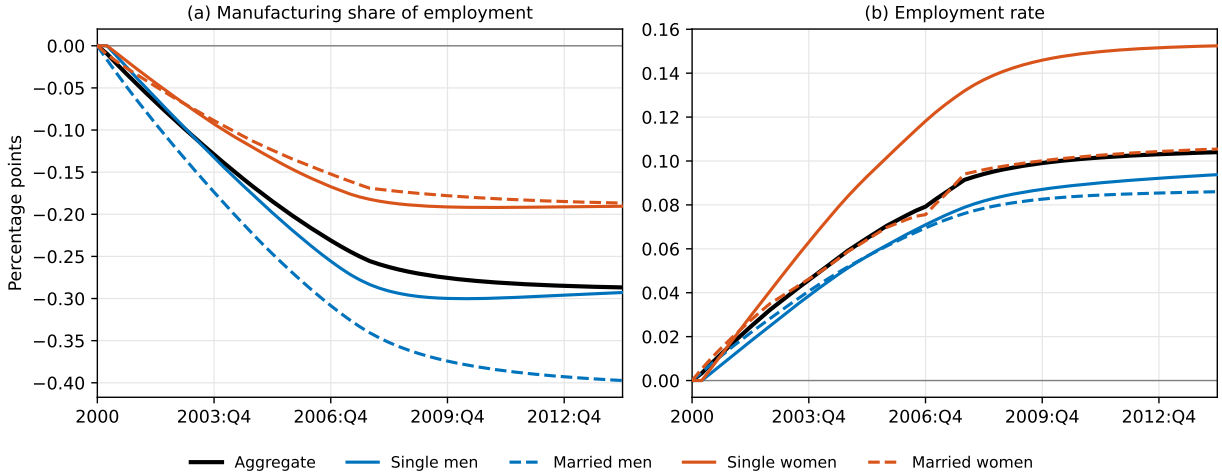
The China Trade Shock, as expected, reduces aggregate manufacturing employment and lowers it for every group (Panel A of Figure 3). The decline observed over the first 13 years does not reverse in the long run (Panel A of Table E.1). There are important gender differences in the manufacturing decline: women’s decline is about half that of men’s and varies little by marital status. Married men experience the largest manufacturing decline, -0.41 p.p. in $t = 199$, compared with -0.29 p.p. for single men. When manufacturing decline is measured as manufacturing jobs per capita, a metric comparable to CDP, both quantification exercises agree to the second decimal place (-0.20 vs -0.22).

Most of the decline in manufacturing employment does not translate into non-employment. In fact, the opposite occurs: the employment rate increases for all groups, with a particularly large increase for single women (Panel B of Figure 3). The reason is that the shock is a boom for workers. The increase in productivity in Chinese manufactured goods implies a cheaper consumption basket in the US, thereby reducing the regional price index. The relative real wage differential between working in productive sectors and non-employment then increases, incentivizing single women and all men to work more. In fact,

services grow by more than manufacturing shrinks (+0.25 vs −0.21).

Importantly, because married men are already working, they experience the largest manufacturing losses yet gain the least, so most of their adjustment comes from reallocation across sectors rather than exiting the workforce. For married women, the employment gain is the added-worker effect, because an increase in her employment rate reflects the couple’s decision. The shock hits male-intensive manufacturing, raising $r = w_f/w_m$ and drawing wives into the labor force, as shown in Equation 23.

Figure 3: Employment effect of the China Trade Shock



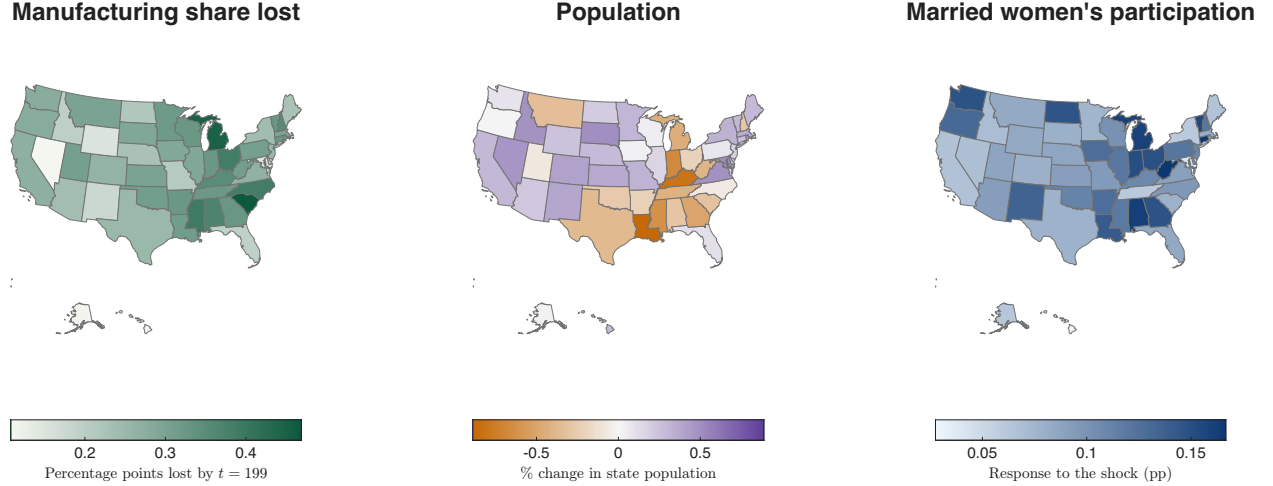
Notes. Panel A is the manufacturing share of employment, in percentage points of employment; Panel B is the employment rate, in percentage points of population.

In Figure 4, I plot the long-run decline in manufacturing employment, the change in population, and the change in married women’s labor force participation by state. South Carolina, Michigan, Mississippi, Ohio, and North Carolina are the five states that lose the most manufacturing share of employment. At the same time, Louisiana (17th in manufacturing decline), Kentucky (8th), Indiana (13th), Mississippi (3rd), and Georgia (10th) are the five states that lose the most population. In fact, across states, the correlation between population change and manufacturing share loss is −0.58, indicating that the states hit hardest by the shock were also the ones that lost population.

Regarding participation responses, even though participation increases in all states, the increases are greatest in West Virginia (20th in manufacturing decline), Connecticut (12th), Alabama (6th), Michigan (2nd), and Vermont (9th). Interestingly, the states with the lowest participation responses are also the ones that lose the least manufacturing. Across states, the correlation between manufacturing share loss and married women’s participation response is +0.51: wives go to work where their husbands’ jobs went. At the same time, the correlation

between participation response and population change is -0.44 , implying that wives work more in states where people leave.

Figure 4: Manufacturing Employment Decline and Migration and Participation Responses



Notes. Effect of the China Trade Shock by $t = 199$. The left panel plots the decline in manufacturing's share of the state's employment, in percentage points, shown as a positive number. The middle panel plots the change in the state's population as a percentage of its no-shock level. The national population is fixed, so this panel is pure reallocation across states: it sums to zero, and one state's gain is another's loss. The right panel plots the change in married women's participation rate, in percentage points, averaged over the couples living in the state at $t = 199$.

The increase in the employment rate, out-migration, and participation responses translates into welfare gains across all groups, with the average individual in the US experiencing a 0.18% increase in welfare due to the shock, as measured by the Money Metric Welfare Index (MMWI) developed in Appendix B.2.²⁵ These gains are nevertheless fairly evenly distributed across household types: single women gain the most, at 0.23, compared with single men at 0.16 and couples at 0.17. Single women are concentrated in expanding sectors, are underrepresented in the 12 manufacturing sectors affected by the shock, and do not face joint location decisions. Couples fall in between the two single groups, but their gains are much closer to those of single men. The reason is that a couple's indirect utility is $\frac{Bw^M}{P}$, so the husband's wage drives the household's gain, while the wife's contribution operates only through her participation.

Within couples, the welfare gain decreases monotonically with exposure and is ordered

²⁵One potential concern would be that incorporating couples also changes welfare in the rest of the world. In Figure E.2, I compare the welfare gains for the 37 other countries in my model and in CDP. The two series correlate at 0.997, and the same 30 of 36 regions gain in both; the large difference is in the UK, which earns 0.048 less in my model.

by the husband. When neither spouse starts in manufacturing, couples have the highest welfare gain (0.18), followed by couples in which only the wife does, then couples in which only the husband does, and finally couples in which both do. Since his wage enters utility directly and hers only through the expected value of her productivity, his exposure reduces household income, whereas hers affects the option value of her working.

Table 6: Welfare Effects of the China Trade Shock

	Welfare effect
Aggregate	0.18
Single men	0.16
Single women	0.23
Couples	0.17
Both in manufacturing (3.1%)	0.13
Husband only (13.7%)	0.14
Wife only (6.6%)	0.16
Neither (76.6%)	0.18

Notes. Consumption-equivalent gain, weighted by each person’s money metric at 2000Q1, corresponding to the wage for a single person, and to $\alpha_g^{1-\gamma} B w_m$ for a spouse. When splitting couples, they are tagged by the exposure cell at $t = 1$.

In Figure E.1, I compare welfare gains across states and find that the geography of gains is nearly identical across household types. As expected, the main losers are the manufacturing states: Louisiana, Kentucky, Indiana, and Mississippi are among the four states furthest below their national average. The states that gain most are those with the least manufacturing to lose. Couples experience the greatest spatial inequality in welfare. This reflects the joint-location friction: a couple cannot take one job in one state and the other in another.

My previous results highlight how the shock affected each household type, showing that aggregate values hide substantial heterogeneity in winners and losers, but they cannot quantify the importance of abstracting away from couples. Thus, I compare the previous results with the effect of the shock in a counterfactual world in which everyone is single; I call this world the *single-agent economy*.

To build the single-agent economy, I reconstruct all inputs from the data, split couples into singles, and estimate the migration elasticity in this economy to match the same moment as in the baseline economy. I then solve the model again under this structure. In Table 7, I compare manufacturing decline, employment rate, and welfare effects in both economies. I find that the aggregate decline in manufacturing employment is almost the same in both

economies (the difference is 0.9%).

Table 7: Comparison of the Effects in the Baseline and in a Single-Agent Model

	Baseline	Single-agent model	% diff.
<i>Panel A. Manufacturing share of employment</i>			
Aggregate	−0.29	−0.29	+0.9
Men	−0.38	−0.38	+0.9
Women	−0.19	−0.19	+2.3
<i>Panel B. Employment rate</i>			
Aggregate	0.11	0.13	−15.4
Men	0.01	0.09	+7.0
Women	0.13	0.17	−26.7
<i>Panel C. Welfare effects</i>			
Aggregate	0.175	0.177	−1.1
Single men	0.16	0.17	−5.4
Single women	0.23	0.19	+20.5
Married men	0.17	0.16	+3.8
Married women	0.17	0.19	−12.3

Notes. Panel A is the manufacturing share of employment, in percentage points of employment; Panel B is the employment rate, in percentage points of population. % diff. is $100 \times (\text{baseline/single-agent} - 1)$. Each row is the same people in both economies. The single-agent economy dissolves every couple into two singles, so its married men and women are the baseline’s spouses, each weighted by their own wage; its single men and women are the baseline’s singles.

The baseline economy nevertheless generates 15% smaller employment gains, primarily because the employment response is weaker for women. This difference arises because female labor supply is 34% less elastic in the baseline economy. A wife’s labor supply decision depends on the relative wage $r = w_f/w_m$, rather than on her own wage alone, so her employment response is dampened by her husband’s employment. The two economies also differ in the speed of adjustment in total employment (Figure E.3). In the short run, the baseline model front-loads the adjustment because couples can adjust the wife’s labor supply immediately, whereas mobility takes time. Over longer horizons, however, the single-agent economy adjusts more rapidly, reaching 95% of its long-run employment value 10 quarters

earlier than the baseline economy.

Although the two economies generate similar aggregate welfare gains, they differ in how those gains are distributed. Relative to the single-agent economy, married women’s welfare gains are 12.3% lower, whereas married men’s gains are 3.8% higher. Because a wife’s decision to participate depends on her wage relative to her husband’s, she responds to the shock partly through his exposure and cannot fully capture the expansion in her own sector; moreover, both spouses share a single household-level gain. The distribution also differs among singles: in the baseline economy, single women gain 20.5% more, while single men gain 5.4% less. Because female employment responds 27% less, wages absorb more of the shock in the baseline economy. Specifically, the female wage response is 12% larger, whereas the male wage response is 9% smaller. Single women therefore supply labor to a tighter market, while single men supply labor to a weaker one.

Joint decision-making manifests as spatial dispersion. The spatial distribution of welfare gains is nearly identical across the two economies (Figure E.4), with a cross-state correlation of +0.96. However, welfare gains are roughly one-third more dispersed in the baseline economy: the coefficient of variation is 0.27, compared with 0.20 in the single-agent economy. The baseline economy generates the largest relative losses in highly exposed states, including Louisiana, Ohio, Texas, and Georgia.

Taken together, the results indicate that incorporating couples changes *who* gains from the China Trade Shock far more than it changes *how much* the economy gains. In both the baseline and the single-agent economies, the shock is a boom: manufacturing’s share of employment falls by about 0.29 p.p., workers are drawn from non-employment into the expanding service sectors, and aggregate welfare rises by almost the same amount. The two economies also agree on the geography, with state welfare correlated at +0.96, but they disagree on the incidence. Because the shock falls on male-intensive manufacturing, it raises the relative wage $r = w_f/w_m$, incentivizing wives to enter the labor force, most where manufacturing contracted most (a cross-state correlation of +0.51), and most in couples where only the husband is exposed. Then, couples use the added-worker effect as a mechanism to adjust to the shock. Through general equilibrium effects, this creates a redistribution between single men and single women. Given that married women respond only to relative wages, female labor supply is about a third less elastic than in the single-agent economy, and the female wage therefore does more of the adjusting. Single women sell into a scarcer female labor market and gain more than they would in a world without couples; single men gain less.

5.2 Counterfactual Exercises

In this section, I perform three counterfactual exercises to quantify the importance of each margin of adjustment. In the first one, I set $1/\vartheta \rightarrow 0$ to shut down migration responses. In the second one, I set $\kappa/\gamma \rightarrow 1$ and recalibrate χ accordingly to dampen participation responses. In the last one, I set $1/\sigma \rightarrow 0$ to reduce gender-wage responses. In Table E.2, I summarize the exact value used in each exercise. Notably, each exercise is a full re-solve of the model with the corresponding parameter changed, while all other parameters are held at their baseline values.

Counterfactual 1: No migration response ($\vartheta \rightarrow \infty$)

Given that $1/\vartheta$ represents the migration elasticity in the model, $1/\vartheta \rightarrow 0$ shuts down migration responses to the shock by imposing the elasticity to be 0. Table 8 compares employment effects, the participation response of married women, and welfare effects in the baseline and in the counterfactual economies. In both cases, the decline in manufacturing employment is similar, but the employment effect is 11.7% smaller in the counterfactual economy. Given that migration across state lines is not allowed, individuals cannot access better employment opportunities elsewhere.

I do not find that participation compensates for the lack of spatial mobility. Even if this result could seem counterintuitive, it is driven by how couples decide on her participation. Given that ρ is homogeneous of degree zero in wages, it depends only on $r = w_f/w_m$ and not on the level, and this counterfactual moves both spouses' wages by nearly the same amount, leaving the wedge intact to the third decimal. But the exposed lose the least in participation response (-2.8% , -3.0%) and the unexposed the most (-3.5%).

Aggregate welfare moves $+1.0\%$; the aggregate employment rate falls 11.7%. Moving is how people reach jobs, not how they capture gains, and since sector switching is still allowed, a displaced worker still has their primary adjustment margin. For couples, the null difference hides heterogeneity by exposure. Shutting the margin costs the two husband-in-manufacturing cells 2.7% and 3.4% of their gains, and leaves wife-only and unexposed couples *better* off by 0.7% and 1.6%. In fact, unexposed couples gain because they do not have to pay the cost of the exposed's mobility. When the exposed relocate, they disproportionately move to areas where the unexposed already live; their arrival increases local labor supply and pushes local wages down. With the response shut, nobody arrives, and the unexposed keep a tighter local market.

Table 8: Effects of the China Trade Shock under Counterfactual 1

	Baseline	No migration resp.	% change
<i>Panel A. Employment effects at $t = 199$, pp</i>			
Manufacturing share of employment	-0.29	-0.29	-2.2
Employment rate	0.11	0.1	-11.7
<i>Panel B. Married women's participation at $t = 199$, pp</i>			
All couples	0.100	0.097	-3.4
Both in manufacturing	0.103	0.100	-3.0
Husband only	0.104	0.102	-2.8
Wife only	0.101	0.097	-3.4
Neither	0.1	0.096	-3.5
<i>Panel C. Welfare, money-metric consumption-equivalent %</i>			
Aggregate	0.18	0.18	+1.0
Single men	0.16	0.16	+2.9
Single women	0.23	0.24	+0.9
Couples	0.17	0.17	+0.7
Both in manufacturing	0.13	0.12	-2.7
Husband only	0.14	0.14	-3.4
Wife only	0.16	0.16	+0.7
Neither	0.18	0.18	+1.6

Notes. Individuals are tagged at $t = 1$ based on the sector in which they are employed and the state in which they live. % diff. is $100 \times (\text{CF}/\text{Baseline} - 1)$. In Panel A, the first row corresponds to the manufacturing share of employment, in percentage points of employment; and the second one to the employment rate, in percentage points of population. Panel B is the quarterly participation rate, mass-weighted over the cohort that started in each cell and carried with the household through every later transition.

Counterfactual 2: No participation response ($\kappa/\gamma \rightarrow 1$)

To perform this counterfactual, I take the derivative of the participation probability with respect to $\ln(1 + r)$:

$$\frac{\partial \rho}{\partial \ln(1 + r)} = \frac{\kappa}{\gamma}(1 - \rho),$$

so, once the participation *level* is held, κ/γ is the only free multiplier on the response. The model requires $\kappa > \gamma$, so the most the response can be damped is $\kappa_0/\gamma = 4.44\times$; at

$\kappa = 0.65$, the response is damped $3.39\times$. I also re-pin χ to keep the anchor participation rate unchanged.

Table 9 compares employment effects, the participation response of married women, and welfare effects in the baseline and in the counterfactual economies. In terms of employment effects, while the manufacturing decline is only 3% smaller in the counterfactual, the employment rate falls by 24.3%, which is the largest employment effect of the three counterfactual exercises. If wives cannot respond to the shock by increasing their participation, the overall employment effect is lower.

While in the previous counterfactual, couples did not substitute migration for participation, in this counterfactual, they substitute participation for migration. The difference in migration rates between couples in manufacturing and other sectors increases by 3.7% for the husband's sector and by 6.7% for the wife's sector.²⁶ The substitution runs therefore one way only: couples answer a damped participation margin by moving more, but they cannot answer a closed spatial margin by working more. Single women are less likely to migrate in the counterfactual economy because married women cannot enter the market, allowing single women to take full advantage of higher wages in booming sectors.²⁷

In terms of welfare gains, the aggregate gain falls by 4.6% while single women gain 8.8% more and couples lose 7.2%. It is the only one of the three exercises that *widens* the single-women-minus-single-men welfare gap. It is due to the same mechanism as the single-agent comparison, pushed further. Damping κ/γ takes married women further out of the labor market; female labor gets scarcer still, and the single women who are free to respond capture the wage.

For couples, the welfare gain changes support the added-worker mechanisms. First, the worst type of couple is the one in which only the husband is exposed: a wife whose husband is exposed and who is not herself exposed has room to respond, and damping the response is what takes it away. When she is in manufacturing too, her own wage is falling, and there was less of a response to dampening. Even unexposed couples lose 6.9%, unlike in the previous counterfactual. The added-worker margin is insurance every couple holds; the spatial margin is insurance only the exposed use.

²⁶Even if, in this exercise, the migration response differential when she is exposed is twice as large as when he is exposed, in the baseline economy, couples respond by leaving their states 4.9 times more when he is exposed, consistent with the evidence in Fact 2.

²⁷Table E.3 provides the migration response for manufacturing workers and non-manufacturing workers in detail. Given the results in the empirical analysis, I focus on migration rate differentials.

Table 9: Effects of the China Trade Shock under Counterfactual 2

	Baseline	No participation resp.	% change
<i>Panel A. Employment response at $t = 199$, pp</i>			
Manufacturing share of employment	-0.293	-0.285	-2.9
Employment rate	0.11	0.09	-24.3
<i>Panel B. Migration response differential between manufacturing and non-manuf.</i>			
Single men	+0.0206	+0.0210	+1.8
Single women	+0.0115	+0.0096	-16.0
Couples — husband's sector	+0.0259	+0.0269	+3.7
Couples — wife's sector	+0.0053	+0.0057	+6.7
<i>Panel C. Welfare, money-metric consumption-equivalent %</i>			
Aggregate	0.1748	0.1667	-4.6
Single men	0.1571	0.1533	-2.4
Single women	0.2341	0.2547	+8.8
Couples	0.1697	0.1574	-7.2
Both in manufacturing	0.1249	0.1154	-7.6
Husband only	0.1399	0.1257	-10.2
Wife only	0.1611	0.1531	-5.0
Neither	0.1796	0.1673	-6.9

Notes. Individuals are tagged at $t = 1$ based on the sector in which they are employed and the state in which they live. In Panel B, couples are tagged twice, by each spouse's $t = 1$ sector; a non-working wife's sector is her potential sector. % diff. is $100 \times (\text{CF}/\text{Baseline} - 1)$. In Panel A, the first row corresponds to the manufacturing share of employment, in percentage points of employment; and the second one to the employment rate, in percentage points of population. In Panel B, a mover is defined as anyone who has changed state at least once by $t = 199$. In Panel C, re-pinning χ and κ roughly doubles B , so this economy's own money metrics weight couples more heavily; on its own, the aggregate falls 6.9% rather than 4.6%, and the type rows are unchanged to the first decimal.

Counterfactual 3: No gender-wage response ($\sigma \rightarrow \infty$)

Men and women enter production through a CES nest with elasticity σ , and the wage ratio moves only as $\hat{r} = (\hat{M}/\hat{F})^{1/\sigma}$. If $1/\sigma \rightarrow 0$, $\hat{r} \rightarrow 1$, thus shutting down responses through the relative wage. Given that it is impossible to reach the limit, I set $1/\sigma = 0.01$ against a calibrated 0.714, so 1.4% of the gender-wage response survives.

Table 10 compares the effects of the China Trade Shock in the baseline economy and in this counterfactual economy. Manufacturing's share moves up by 0.7% overall, while

underneath it, men’s declines deepen (single men +6.0%, married men +4.4%) and women’s become shallower (single women −8.5%, married women −3.3%). Since wages can no longer do the adjusting, quantities have. The shock hits male-intensive manufacturing and relative male labor falls, so in the baseline economy $r = w_f/w_m$ rises and part of the burden is absorbed by the relative wage and by the wives it pulls in. At $\sigma = 100$, the relative wage barely moves, so the same contraction has to show up as who leaves manufacturing. So, it also shuts down the participation response, thereby reducing the employment effect for women and increasing it for men. It also translates into a reduction of the migration rate of exposed households across all groups (Table E.4).

In terms of welfare, aggregate welfare moves by 5.8%; single men move by +18.8% and single women by −10.4%, in the same direction as the employment effects. In fact, this is the only counterfactual in which single women are worse off. The gendered incidence of this shock is carried almost entirely by $r = w_f/w_m$; when this channel is shut down, single women are no longer able to take advantage of the shock. On the other hand, all couples benefit from the shock, but the more exposed ones win the most, because removing the relative wage movement helps the couples who were absorbing most of it.

The three exercises on the map

To compare the three counterfactuals on the same scale, I plot the percentage change in state money-metric welfare relative to the baseline economy in Figure 5. The first thing to notice is that the three exercises bite in the same places: the losers are always Louisiana, Michigan, Indiana, and Kentucky, which are the manufacturing states. The reason is that none of the three exercises changes where manufacturing falls: the largest state-level difference in manufacturing share lost is 0.036 pp against a decline of 0.28 pp; thus, they do not move where the shock lands; they move what it costs to be there.

Shutting the spatial response moves aggregate welfare by +1.0% (Table 8), and that near-null hides everything: welfare becomes twelve times more spatially unequal, and the correlation between a state’s welfare change and its population response in the baseline is +0.98. Migration redistributes: the losers are the exposed states that were losing population; the winners are the destinations that were gaining it, which no longer have their gains diluted by arrivals. Counterfactuals 2 and 3 are nearly uniform by comparison, but in opposite directions. Damping participation not only reduces aggregate welfare gains by 4.6% (Table 9), but it also makes every state worse off. Flattening the gender wage response increases the welfare gains by 5.8% (Table 10) while, at the same time, almost every state is better off.

Taken together, the margin shut-downs rank the mechanisms, but the ranking depends

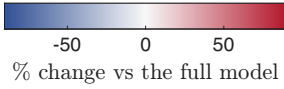
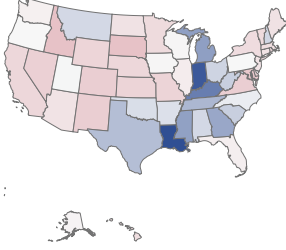
Table 10: Effects of the China Trade Shock under Counterfactual 3

	Baseline	No gender-wage resp.	% change
<i>Panel A. Employment response at $t = 199$, pp</i>			
Manufacturing share of employment	-0.2933	-0.2954	+0.7
Men	-0.3835	-0.4015	+4.7
Women	-0.1906	-0.1812	-5.0
Employment rate	0.1121	0.0972	-13.3
Men	0.0988	0.1266	+28.1
Women	0.1245	0.0696	-44.1
<i>Panel B. Married women's participation (quarterly rate), by exposure cell</i>			
All couples	+0.10063	+0.03689	-63.3
Both in manufacturing	+0.10320	+0.03871	-62.5
Husband only	+0.10461	+0.04032	-61.5
Wife only	+0.10067	+0.03666	-63.6
Neither	+0.09981	+0.03623	-63.7
<i>Panel C. Ever moved state, by type and by the affected person's own origin</i>			
Single men	+0.0206	+0.0202	-2.1
Single women	+0.0115	+0.0111	-3.3
Couples — husband's sector	+0.0259	+0.0251	-3.3
Couples — wife's sector	+0.0053	+0.0051	-5.1
<i>Panel D. Welfare, money-metric consumption-equivalent %</i>			
Aggregate	0.1748	0.1850	+5.8
Single men	0.1571	0.1866	+18.8
Single women	0.2341	0.2097	-10.4
Couples	0.1697	0.1817	+7.1
Both in manufacturing	0.1249	0.1367	+9.4
Husband only	0.1399	0.1516	+8.4
Wife only	0.1611	0.1732	+7.5
Neither	0.1796	0.1916	+6.7

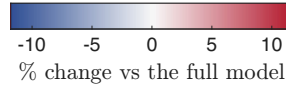
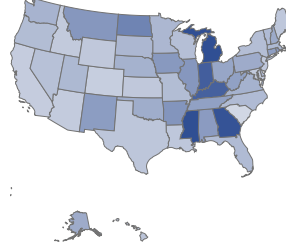
Notes. Individuals are tagged at $t = 1$ based on the sector in which they are employed and the state in which they live. % diff. is $100 \times (\text{CF}/\text{Baseline} - 1)$. Panel B is the quarterly participation rate. In Panel C, a mover is defined as anyone who has changed state at least once by $t = 199$.

Figure 5: Distribution of Welfare Gain Changes in each Counterfactual

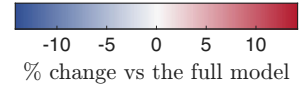
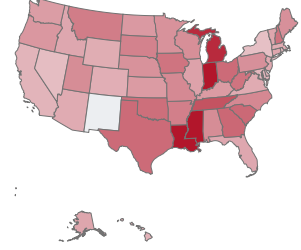
No migration response



No participation response



No gender-wage response



Notes. Each map plots the percentage change in state money-metric welfare, comparing each exercise against the baseline economy. Each panel carries its own scale, read off its own color bar. Magnitudes are therefore not comparable to the naked eye across panels.

on whether one looks at the size of the gain or at its distribution. In terms of distribution, migration is the dominant margin: shutting the spatial response moves aggregate welfare by only 1%, yet it lowers the employment response by 12% and makes state welfare twelve times more unequal, from best to worst. The states that lose are the exposed ones that were losing population; the states that win are the destinations whose gains had been diluted by arrivals. Within couples, the loss falls on those whose husband starts in manufacturing, which makes mobility insurance for the exposed. However, given that the participation rule depends only on the ratio of the spouses' wages, and closing the spatial margin moves both wages by almost the same amount, participation does not rise to fill the gap. The substitution runs one way only: when participation is damped, couples move more, but when migration is shut, they do not work more. Nevertheless, the gender-wage channel carries nearly all of the gendered incidence: single men gain 18.8% more, single women 10.4% less, and the manufacturing decline is shifted from women onto men. In terms of size, damping participation is the one experiment that lowers the aggregate gain (-4.6%) and cuts the employment response the most (-24%). It hurts every couple, including unexposed ones, which is why the added-worker margin is insurance that every couple holds, not just the exposed.

6 Conclusion

Workers migrate less in response to local shocks than theory predicts, and the conventional explanation is that moving is costly. This paper argues that the explanation also lies in how spatial models of local labor markets represent households: they typically assume a single worker chooses where to live, yet more than half of households do not fit this framework. In these households, two people jointly choose a location by weighing their respective labor-market exposures, a fundamentally different problem from the one faced by a single worker. This distinction affects both how much of a shock the economy absorbs and which households ultimately bear its costs.

Using PSID data to compare individuals who were and were not exposed to the China Trade Shock within the same commuting zone, I document three facts. First, exposed individuals experienced worse labor-market outcomes and were more likely to leave their commuting zones. Second, couples responded asymmetrically: out-migration increased when the husband was exposed but did not increase when the wife was exposed. Third, wives' labor supply exhibited a similar asymmetry, rising when the husband was exposed—an added-worker effect—and falling sharply when the wife herself was exposed. Thus, couples use both spatial and intra-household insurance to mitigate the shock.

To quantify the relative importance of these margins and the general-equilibrium consequences of joint decision-making, I extend Caliendo et al. (2019) to incorporate heterogeneous households: single men, single women, and couples. Couples jointly choose a location and a pair of sectors, with substitution across sectors and locations occurring at different rates, as in Rodríguez-Clare et al. (2026). After choosing where to live and work, they decide whether the wife works in the market or at home. The couple's problem therefore introduces two margins that do not exist for singles and that operate in opposite directions: joint location choice makes moving more difficult, whereas endogenous labor-force participation makes staying more bearable.

The quantitative exercise shows that the trade shock is, overall, a boom for US workers: cheaper imports lower the price index, employment rates rise across all household types, and the average individual gains 0.18% in consumption-equivalent terms. Manufacturing employment falls by 0.29 p.p., and individuals adjust through both migration and increased labor-force participation. The states experiencing the largest manufacturing-employment losses also lose the most population and exhibit the largest increases in married women's labor supply. Moreover, participation increases most in the states with the largest population losses. Overall, single women gain the most from the shock, single men lose the most, and couples fall in between. This difference between single men and single women is driven

by couples' decisions. Because couples determine the wife's market participation based on relative wages, female labor supply is less elastic than men's. Single women, by contrast, are concentrated in expanding sectors and face no joint location constraint, allowing them to take full advantage of the shock.

For couples, spatial and intrahousehold insurance are not symmetric substitutes. When the participation response is dampened, couples move more; when migration is shut down, however, wives do not work more. The participation decision depends only on the ratio of spouses' wages, and closing the spatial margin changes both wages in the same way. The two margins also insure different households. Participation provides insurance to every couple: dampening this response reduces the gains of all couples, including those with no direct exposure, and lowers aggregate welfare by 4.6%. Migration, by contrast, insures only exposed households. Shutting it down has little effect on aggregate welfare, but it makes welfare across states twelve times more unequal, with the losses concentrated among couples whose husbands initially work in manufacturing.

The policy implication is that aggregate gains provide an incomplete picture of who benefits from the shock. Trade adjustment assistance is designed around individuals, targeting workers or industries that are exposed to trade shocks. Yet the binding mobility constraint operates at the household level, and a transfer that replaces a displaced husband's earnings does not restore the option his family has lost. Policies that reduce the cost of moving jointly or increase the destination's value to the spouse who is not driving the move would help alleviate this constraint. Moreover, because part of the shock's incidence operates through women's wages, it can also affect single workers who were never employed in an exposed industry.

The direction of the gender asymmetry in my results is a feature of *this* shock rather than a general property of the model. Because the China Trade Shock affects manufacturing, a male-intensive sector, it exposes husbands more than wives and draws wives into the labor market. Holding the gender wage ratio nearly fixed eliminates about 63% of wives' participation response and closes most of the gap between single women's and single men's gains. A shock to a female-intensive sector would reverse the asymmetry, although its welfare consequences need not be a mirror image. Thus, this paper opens a broader research agenda on how the economy adjusts to local shocks when couples make decisions jointly.

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Appendix

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A Additional Results in the Empirical Analysis

A.1 Robustness Exercises

A.1.1 The China Trade Shock across Sample Commuting Zones

Given the PSID sample restrictions and the size of the PSID, I include 375 commuting zones out of the 722 that ADH use in their analysis. My results could therefore be driven by which zones are included in the PSID rather than by differential exposure to the China Trade Shock. To assess this concern, this section replicates ADH’s main results using the 375 CZs represented in the sample.²⁸²⁹

Table A.1 first compares the CZs in the sample with the full set used by ADH. Because the restricted sample is a subset of the full sample, the two groups are not independent, and a conventional difference-in-means test is therefore ill-defined. The final column instead reports the standardized difference, defined as the difference in means between the restricted and full samples divided by the square root of the average of their variances. This measure is expressed in standard-deviation units, so it is comparable across variables measured on different scales. Following Imbens and Rubin (2015), absolute standardized differences below 0.25 indicate comparability. All standardized differences in Table A.1 are 0.13 or smaller, including the difference for the change in import penetration. By this criterion, the CZs in the PSID sample are comparable to the full ADH sample with respect to both the control variables and the China Trade Shock itself.

Next, I estimate ADH’s Equation (2) for the CZs in the sample

$$\Delta Y_{i,t+h} = \alpha_t + \beta_{1,h} \Delta IP_i^{China,US} + X'_{i,t} \beta_2 + \varepsilon_{i,t+h},$$

where $\Delta Y_{i,t+h}$ denotes the change in outcome Y for CZ i between $t = 2000$ and year $h > 2000$. As in ADH, $\Delta IP_i^{China,US}$ is instrumented with $\Delta IP_i^{China,Others}$, following the identification logic described in Section 2. The analysis considers ADH’s three main outcome categories: employment, income, and population headcount.

Figure A.1 replicates ADH’s employment results for the manufacturing employment-to-population ratio in panel a, the non-manufacturing employment-to-population ratio in panel b, and the total wage-and-salary employment-to-population ratio in panel c. Across all three outcomes, the estimates based on the full set of CZs closely align with those based only on the CZs in the PSID sample. Figure A.2 reports the corresponding results for income: personal income per capita in panel a, government transfers per capita in panel b, and total

²⁸Because the analysis uses the geocoded restricted version of the PSID, the identities of the commuting zones in the sample cannot be disclosed, nor can a map be provided at any geographic level below the state.

²⁹This exercise uses the replication package provided by David Dorn on his website.

labor compensation per worker in panel c. The two sets of estimates again track each other closely. The only visible divergence is a somewhat larger increase in government transfers among the sample CZs, although the coefficients are not statistically distinguishable. Finally, Figure A.3 reports population headcounts, which ADH use to measure migration responses. The estimates are similar for the total population and for individuals ages 40–64. However, the population decline among individuals ages 25–39 is larger in the sample CZs, and the estimates indicate a statistically significant decline among individuals ages 18–24 that is absent from ADH’s full-sample results. Because the PSID sample has an average age of 37, these differences among younger age groups are unlikely to systematically bias the main results.

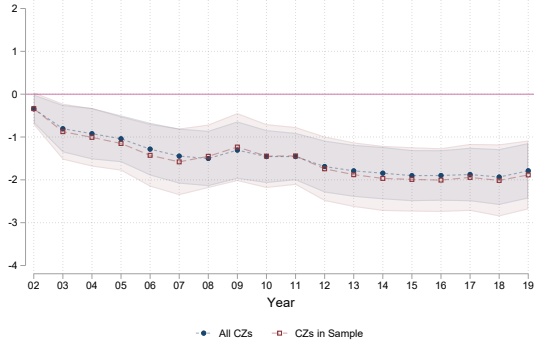
Taken together, the two exercises yield consistent evidence. The CZs represented in the PSID are comparable to the full ADH sample in terms of observable characteristics, including exposure to the China Trade Shock, and estimating ADH’s specification on these CZs produces similar coefficients. Thus, although the PSID sample restrictions reduce geographic coverage, they do not appear to select local labor markets in which the China Trade Shock had systematically different effects.

Table A.1: Comparing CZs in the Sample versus all CZs

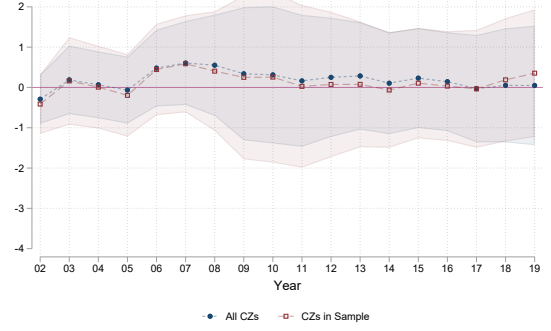
	Full		Restricted			Standardized
	Mean	SD	Mean	SD	Diff	Difference
Control variables (in 2000)						
Manufacturing emp./total emp.	16.18	7.46	15.68	6.91	-0.49	-0.07
Female emp./total emp.	64.38	5.51	64.54	5.21	0.16	0.03
Routine occupation emp./total emp.	31.90	2.37	32.20	2.16	0.30	0.13
Offshorability index	0.00	0.51	0.06	0.48	0.07	0.13
College educated/total pop.	53.61	7.47	54.41	6.88	0.80	0.11
Foreign born/total pop.	14.76	12.81	15.74	12.89	0.97	0.08
Non-white population/total pop.	18.18	10.95	18.81	10.36	0.63	0.06
Population 65+/total pop.	12.37	2.91	12.15	2.81	-0.21	-0.07
Population 40-64/total pop.	30.10	1.87	30.02	1.79	-0.08	-0.04
Population 0-17/total pop.	25.63	2.22	25.65	2.14	0.02	0.01
Δ log population 1970-1990/2	12.31	12.25	12.77	12.35	0.47	0.04
Change in CZ import penetration (2000 to 2012)						
Imports by US	0.89	0.59	0.88	0.55	-0.01	-0.02
Imports by other high-inc. countries	1.22	0.69	1.21	0.64	-0.01	-0.02
Number of Commuting Zones	722		375			

Figure A.1: China Trade Shock Impact on Employment

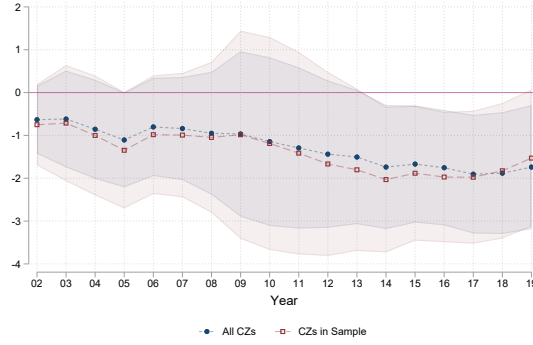
(a) Manufacturing Emp./Working-age pop.



(b) Nonmanufacturing Emp./Working-age pop.



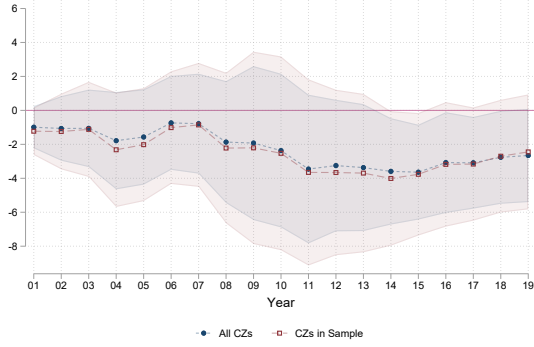
(c) Total Wage and Salary Emp./Working-age pop.



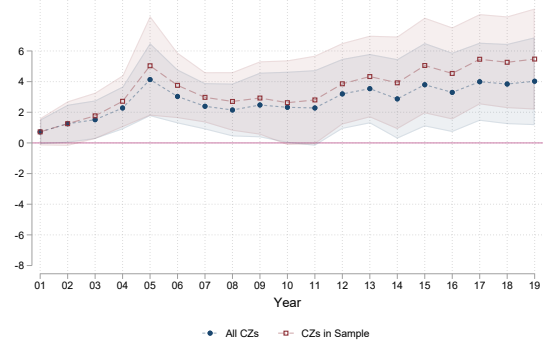
Notes. In each panel, I plot the original ADH result for that outcome variable in blue and, in red, the result when I consider only the CZs in the sample. A CZ is considered in the sample if I can estimate its fixed effect when estimating Equation 4.

Figure A.2: China Trade Shock Impact on Log Income

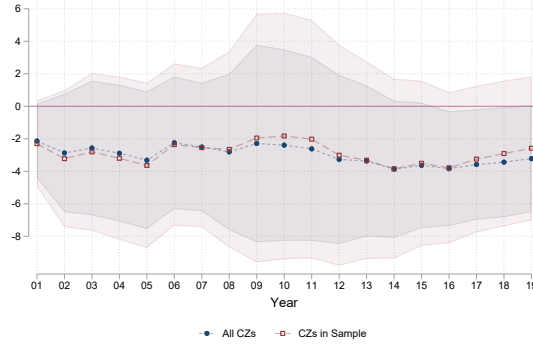
(a) Log(Personal Income per capita)



(b) Log(Government Transfers per capita)

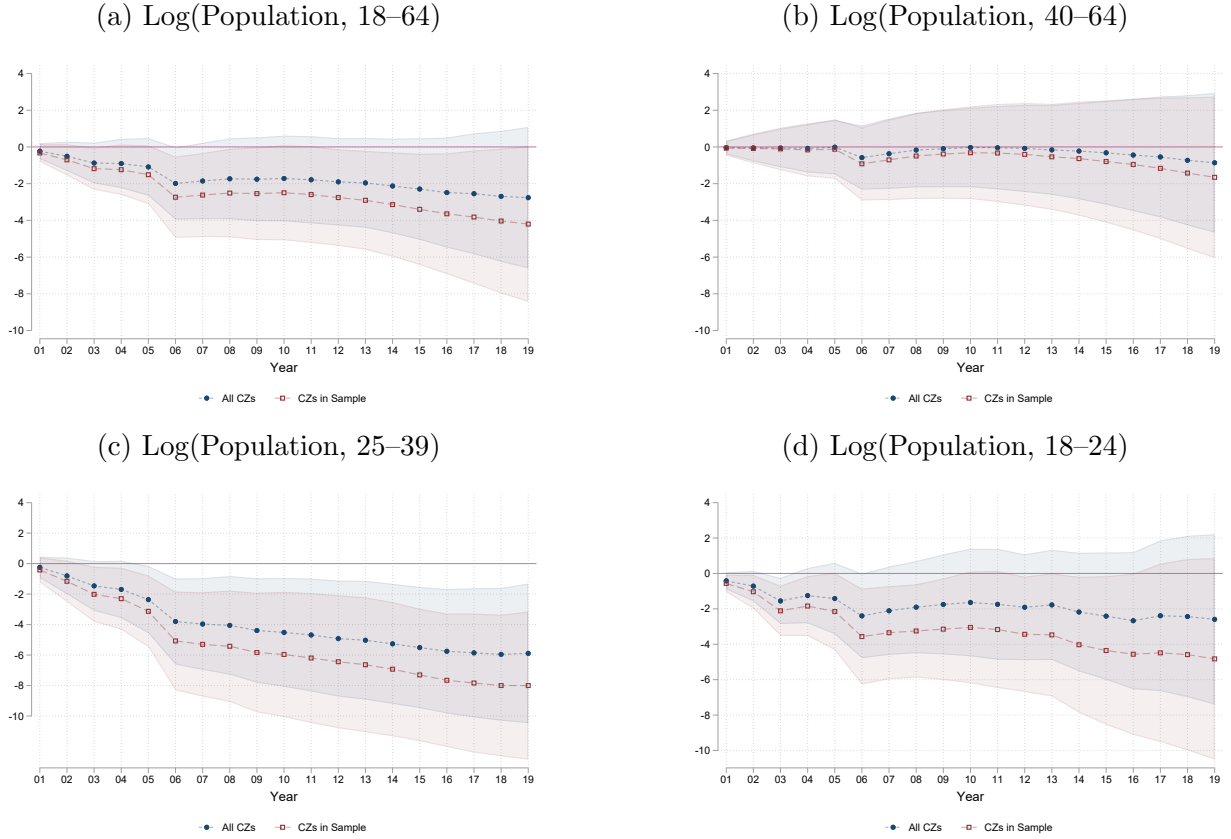


(c) Log(Total Labor Compensation per worker)



Notes. In each panel, I plot the original ADH result for that outcome variable in blue and, in red, the result when I consider only the CZs in the sample. A CZ is considered in the sample if I can estimate its fixed effect when estimating Equation 4.

Figure A.3: China Trade Shock Impact on Log Population Headcounts



Notes. In each panel, I plot the original ADH result for that outcome variable in blue and, in red, the result when I consider only the CZs in the sample. A CZ is considered in the sample if I can estimate its fixed effect when estimating Equation 4.

A.1.2 The China Trade Shock and Family Structure

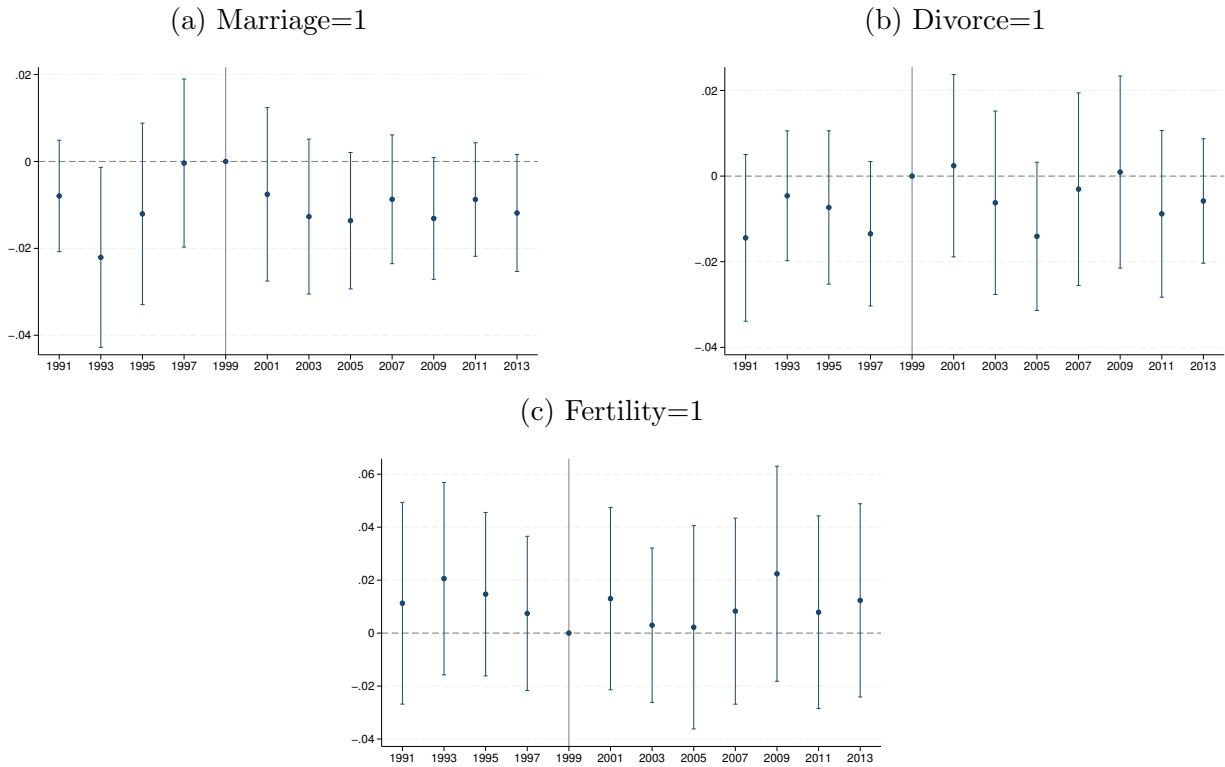
One potential concern is that the China Trade Shock alters household structure. Such changes could bias the results through two channels: the shock may affect marriage, divorce, or fertility decisions that subsequently shape out-migration responses, and couples who remain together after the shock may differ systematically from those who divorce. To assess this concern, I use the full sample of individuals and estimate Equation 4 for the probabilities of marrying, divorcing, and having a new child. Figure A.4 shows that, between 2001 and 2013, exposed individuals were no more likely than non-exposed individuals to marry, divorce, or have a child.

At first glance, this result may appear inconsistent with Autor et al. (2019), who find that a one-unit trade shock reduces the share of previously married women by 0.21 percentage points and increases the share of never-married women by 1.2 percentage points. The apparent discrepancy reflects differences in how marital status is measured. In the ACS,

marital status records legal status and does not separately identify cohabiting women. Consequently, women who transition from single to cohabiting are classified as married in my data but remain single in Autor et al. (2019). In addition, because the ACS is a repeated cross-section, a woman's prior marital status cannot be recovered once she is observed as married. Women who transition from divorced to married are therefore classified simply as married in their analysis, whereas they are excluded from my sample.

I also find no robust effect of trade shocks on fertility. One likely explanation is that I allow for the exposure of both men and women, whereas their analysis focuses exclusively on women. My estimates therefore capture a broader set of household shocks and adjustment margins.

Figure A.4: Exposure to the China Trade Shock and Family Outcomes



Notes. Each panel plot uses the title as the outcome variable to estimate Equation 4. Robust standard errors are two-way clustered at the industry and individual levels. Confidence intervals are constructed at the 95% level. In the equivalent static difference-in-differences model, the coefficient of interest is not statistically significant for any of the three outcomes.

A.1.3 Other Migration Drivers

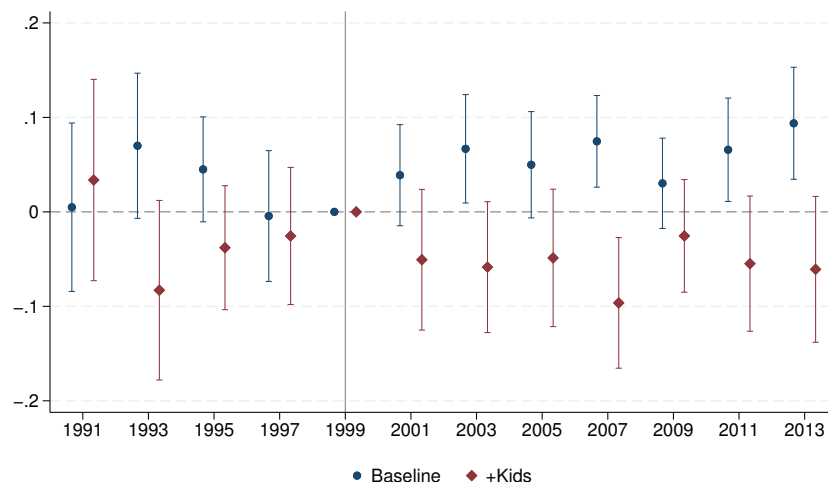
My results raise two potential questions. First, the behavioral differences I attribute to joint decision-making may instead reflect the presence of children rather than couple status itself.

To address this concern, I extend Equation 4 to a triple-difference specification that accounts for whether an individual has a child younger than 18. If individuals with children respond differently from those without children, the coefficient on the triple interaction should be statistically different from zero. Figure A.5 shows that this coefficient is not statistically significant in any year except 2007. Thus, exposure does not generate systematically different out-migration responses among individuals with and without children. Individuals with children nevertheless have a baseline out-migration probability that is 4.2 percentage points lower than that of individuals without children. Therefore, the two groups differ in their intercepts, even though their responses to exposure are similar.

Second, the estimated differences could reflect gender-specific responses to the shock. To address this concern, I extend Equation 4 to a triple-difference specification that accounts for whether the individual is a woman. If women and men respond differently to the shock, the coefficient on the triple interaction should be statistically different from zero. Figure A.6 shows that this coefficient is not statistically significant in any year during the shock. Thus, exposed women do not exhibit a systematically different out-migration response from exposed men.

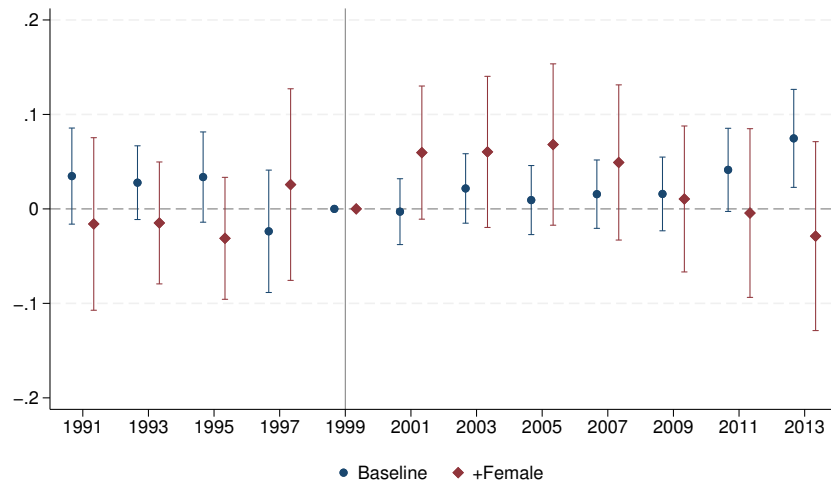
Taken together, these results indicate that the asymmetric migration response among couples, depending on which partner is exposed, is driven neither by the presence of children in the household nor by general gender differences in responses to the shock.

Figure A.5: Exposure to the China Trade Shock and Out-Migration Responses, by Parental Status



Notes. Robust standard errors are two-way clustered at the industry and individual levels. Confidence intervals are constructed at the 95% level. In blue, I plot the coefficient on the interaction between exposure and year. In red, I plot the coefficient on the triple interaction among exposure, year, and an indicator for having a child younger than 18.

Figure A.6: Exposure to the China Trade Shock and Out-Migration Responses, by Gender



Notes. Robust standard errors are two-way clustered at the industry and individual levels. Confidence intervals are constructed at the 95% level. In blue, I plot the coefficient on the interaction between exposure and year. In red, I plot the coefficient on the triple interaction among exposure, year, and an indicator for being a woman.

A.2 Additional Tables

Table A.2: Exposure to the China Trade Shock and Labor Market Outcomes

	(1)	(2)	(3)
	Employed=1	Unemployed=1	Lab Force=1
Exposed=1	-0.007 (0.019)	-0.002 (0.008)	-0.008 (0.017)
(Exposed=1)x(Post=1)	-0.063*** (0.017)	0.023** (0.010)	-0.039*** (0.012)
Baseline Mean	0.90	0.04	0.94
	(1)	(2)	(3)
	Weekly Hours	Hourly Wage	Labor Income
Exposed=1	-0.454 (1.644)	-0.677 (4.011)	1,978.003 (8,565.422)
(Exposed=1)x(Post=1)	-2.058** (0.889)	-2.581** (1.160)	-5595.397** (2,273.988)
Baseline Mean	40.62	24.26	50,236.60
Observations	38,634	38,634	38,634

Notes. Robust standard errors are two-way clustered at the industry and individual levels.

* for $p < .10$, ** for $p < 0.05$, *** for $p < 0.01$.

Table A.3: Binary versus Continuous Exposure to the China Trade Shock

	(1)	(2)	(3)
	Employed=1	Unemployed=1	Labor Force=1
Exposed=1	-0.007 (0.019)	-0.002 (0.008)	-0.008 (0.017)
(Exposed=1)x(Post=1)	-0.063*** (0.017)	0.023** (0.010)	-0.039*** (0.012)
Shock	0.003* (0.002)	-0.000 (0.001)	0.003** (0.001)
(Shock)x(Post=1)	-0.007*** (0.002)	0.003*** (0.001)	-0.003*** (0.001)
Observations	38,634	38,634	38,634
R2	0.08	0.05	0.06
Baseline Mean	0.90	0.04	0.94

Notes. Robust standard errors are two-way clustered at the industry and individual levels.

* for $p < .10$, ** for $p < 0.05$, *** for $p < 0.01$.

Table A.4: Exposure to the China Trade Shock and Out-Migration Responses

	(1)
	Out-Migration=1
Exposed=1	-0.064
	(0.042)
(Exposed=1)x(Post=1)	0.047**
	(0.021)
Observations	37,443
R^2	0.09
Baseline Mean	0.07

Notes. Robust standard errors are two-way clustered at the industry and individual levels. * for $p < .10$, ** for $p < 0.05$, *** for $p < 0.01$.

Table A.5: Wife's Exposure to the China Trade Shock and Husband's Labor Market Outcomes

	(1)	(2)
	Husband Employed=1	Husband's Hourly Wage
Exposed Wife=1	-0.021 (0.020)	-1.527 (1.355)
(Exposed Wife=1)x(Post=1)	-0.030 (0.026)	-9.349*** (1.557)
Observations	14,083	14,083
R^2	0.10	0.33
Baseline Mean	0.95	30.42

Notes. All monetary values are in 2017 USD, using the PCE index, and include zeros. Robust standard errors are two-way clustered at the wife's industry and individual levels. * for $p < .10$, ** for $p < 0.05$, *** for $p < 0.01$.

Table A.6: Couple's Exposure to the China Trade Shock and Household Income Responses

	(1)	(2)
	Household Income	
Exposed Husband=1	7,357.215 (11692.540)	
(Exposed Husband=1)x(Post=1)	3,930.551 (6,330.419)	
Exposed Wife=1		529.105 (12003.901)
(Exposed Wife=1)x(Post=1)		-2348.701 (4,316.531)
Observations	14,083	14,083
R^2	0.35	0.35
Baseline Mean	120226.29	120600.92

Notes. All monetary values are in 2017 USD, using the PCE index, and include zeros. Robust standard errors are two-way clustered at the husband's industry and household levels in Column 1 and the wife's industry and household levels in Column 2. * for $p < .10$, ** for $p < 0.05$, *** for $p < 0.01$.

B Model Derivations

This appendix derives the choice probabilities and value function under the nested Gumbel shock and then develops the money-metric measure used to compare welfare across singles and couples.

B.1 Choice Probabilities and Expected Values

The derivation follows Appendix B.2 of Rodríguez-Clare et al. (2026). Fix h , $\mathbf{o}$, and t , and suppress these indices whenever doing so creates no ambiguity. Define $x_{\mathbf{d}} \equiv \delta V_{\mathbf{d},t+1} - \mu_{\mathbf{o}\mathbf{d}}$. The household chooses $\mathbf{d}$ if and only if $x_{\mathbf{d}} + \varepsilon_{\mathbf{d}} \geq x_{\mathbf{z}} + \varepsilon_{\mathbf{z}}$ for every $\mathbf{z}$.

Choice probabilities. Let $G_{\mathbf{d}}(\cdot)$ denote the partial derivative of F in Equation 14 with respect to its $\mathbf{d}$ -th argument. For any $\mathbf{d} \in \mathcal{S}_h(j)$,

$$G_{\mathbf{d}}(\varepsilon) = \frac{1}{\vartheta} \left(\sum_{\mathbf{z} \in \mathcal{S}_h(j)} e^{-\varepsilon_{\mathbf{z}}/\vartheta} \right)^{v/\vartheta-1} e^{-\varepsilon_{\mathbf{d}}/\vartheta} F(\varepsilon).$$

The choice probability for $\mathbf{d}$ is $\int G_{\mathbf{d}}(\varepsilon_{\mathbf{d}}, \{\varepsilon_{\mathbf{d}} + x_{\mathbf{d}} - x_{\mathbf{z}}\}_{\mathbf{z} \neq \mathbf{d}}) d\varepsilon_{\mathbf{d}}$. Evaluating each remaining argument at $\varepsilon_{\mathbf{z}} = \varepsilon_{\mathbf{d}} + x_{\mathbf{d}} - x_{\mathbf{z}}$ yields $e^{-\varepsilon_{\mathbf{z}}/\vartheta} = e^{-\varepsilon_{\mathbf{d}}/\vartheta} e^{(x_{\mathbf{z}} - x_{\mathbf{d}})/\vartheta}$ and therefore

$$\sum_{\mathbf{z} \in \mathcal{S}_h(\ell)} e^{-\varepsilon_{\mathbf{z}}/\vartheta} = e^{-\varepsilon_{\mathbf{d}}/\vartheta} e^{-x_{\mathbf{d}}/\vartheta} \Omega_{\ell}, \quad \Omega_{\ell} \equiv \sum_{\mathbf{z} \in \mathcal{S}_h(\ell)} e^{x_{\mathbf{z}}/\vartheta},$$

where $\Omega_{\mathbf{o}j,t}^h \equiv \sum_{\mathbf{z} \in \mathcal{S}_h(j)} \exp[\delta V_{\mathbf{z},t+1}^h - \mu_{\mathbf{o}\mathbf{z}}]^{1/\vartheta}$. Substituting, and writing $T \equiv e^{-x_{\mathbf{d}}/\vartheta} \sum_{\ell} \Omega_{\ell}^{v/\vartheta}$,

$$\Pr(\mathbf{d}) = \frac{e^{x_{\mathbf{d}}/\vartheta}}{\Omega_j} \cdot \frac{\Omega_j^{v/\vartheta}}{\sum_{\ell} \Omega_{\ell}^{v/\vartheta}} \int_{-\infty}^{\infty} \frac{1}{\vartheta} T e^{-\varepsilon_{\mathbf{d}}/\vartheta} \exp(-T e^{-\varepsilon_{\mathbf{d}}/\vartheta}) d\varepsilon_{\mathbf{d}}.$$

The integrand is the density of a univariate Gumbel distribution with scale ϑ , so the integral equals one. The first fraction corresponds to $\lambda_{\mathbf{o}\mathbf{d}|j,t}^h$ in Equation 18, while the second corresponds to $\lambda_{\mathbf{o}j,t}^h$ in Equation 17.

Expected value. The joint probability that $x_{\mathbf{d}} + \varepsilon_{\mathbf{d}} \leq a$ and that $\mathbf{d}$ is chosen is obtained by integrating the same expression up to $a - x_{\mathbf{d}}$. This yields $\lambda_{\mathbf{o}\mathbf{d},t}^h \exp(-\tilde{T} e^{-a/\vartheta})$, where $\tilde{T} \equiv \sum_{\ell} \Omega_{\ell}^{v/\vartheta}$. Conditional on $\mathbf{d}$ being chosen, the maximized payoff therefore follows a Gumbel distribution with location $\vartheta \log \tilde{T}$ and scale ϑ , implying a mean of $\vartheta \log \tilde{T} + \vartheta \bar{e}$.

Because this expression does not depend on $\mathbf{d}$,

$$\mathbb{E} \left[\max_{\mathbf{d}} \{x_{\mathbf{d}} + \varepsilon_{\mathbf{d}}\} \right] = \vartheta \log \left(\sum_{\ell=1}^L \Omega_{\mathbf{o}\ell,t}^{v/\vartheta} \right) + \vartheta \bar{e},$$

which, after dropping the constant, delivers Equation 16.

Welfare Recursion From Equation 17 evaluated at the origin location i , $\sum_{\ell} \Omega_{\mathbf{o}\ell,t}^{v/\vartheta} = (\lambda_{\mathbf{o}i,t}^h)^{-1} \Omega_{\mathbf{o}i,t}^{v/\vartheta}$. From Equation 18 evaluated at the origin bundle and $\mu_{\mathbf{o}\mathbf{o}} = 0$, $\Omega_{\mathbf{o}i,t} = (\lambda_{\mathbf{o}\mathbf{o}|i,t}^h)^{-1} \exp(\delta V_{\mathbf{o},t+1}^h)^{1/v}$. Combining,

$$\vartheta \log \left(\sum_{\ell} \Omega_{\mathbf{o}\ell,t}^{v/\vartheta} \right) = \delta V_{\mathbf{o},t+1}^h - \vartheta \log (\lambda_{\mathbf{o}i,t}^h) - v \log (\lambda_{\mathbf{o}\mathbf{o}|i,t}^h),$$

which, substituted into Equation 16 and iterated forward, gives Equation 28.

B.2 The Money-Metric Welfare Index

I follow Chiappori and Meghir (2015) to construct a Money-Metric Welfare Index (MMWI), defined as the income each spouse would require to attain their utility level if they faced the full price of public consumption. For notational simplicity, I suppress the indices $\mathbf{d}$ and t .

Let $u_g \equiv \gamma \log Q + (1 - \gamma) \log c_g$, for $g \in \{f, m\}$, denote the log utility of spouse g . Substituting Equation 21 into u_g , I obtain

$$\begin{aligned} u_f &= \log \frac{I(e)}{P} + e\gamma \log \beta + K + (1 - \gamma) \log \alpha, \\ u_m &= \log \frac{I(e)}{P} + e\gamma \log \beta + K + (1 - \gamma) \log(1 - \alpha), \end{aligned}$$

where $K \equiv \gamma \log \gamma + (1 - \gamma) \log(1 - \gamma)$. In levels, $U_g = \mathcal{A}_g \beta^{e\gamma} I(e)/P$ with $\mathcal{A}_f = e^K \alpha^{1-\gamma}$ and $\mathcal{A}_m = e^K (1 - \alpha)^{1-\gamma}$. Ex ante utility corresponds to

$$\mathbb{E}[U_g] = (\mathcal{A}_g/\mathcal{A}) \mathbb{E}[\mathcal{U}]. \tag{B.1}$$

Thus, each spouse's expected utility is a fixed multiple of the couple's expected utility. The ratio of private consumption is $c_f^*/c_m^* = \alpha/(1 - \alpha)$ and is independent of wages, β , and e . Because the model contains no distribution factors, an increase in the wife's wage raises household income without changing her share of it.

1. Conditional sharing rule (CSR). The CSR allocates between the spouses the income remaining after expenditure on the public good:

$$\tilde{\rho}_f = Pc_f^* = \alpha(1 - \gamma)I, \quad \tilde{\rho}_m = Pc_m^* = (1 - \alpha)(1 - \gamma)I, \quad \tilde{\rho}_f + \tilde{\rho}_m = (1 - \gamma)I.$$

The CSR therefore excludes both the γI spent on the public good and the increase in public-good production generated by home production.

2. Lindahl prices. One unit of Q costs the couple $P_Q \equiv P\beta^{-e}$. Thus, when the wife remains at home, home production reduces the effective price of the public good. Each spouse's personalized price equals their marginal willingness to pay for Q , expressed in monetary units:

$$P_g = P \frac{\partial u_g / \partial Q}{\partial u_g / \partial c_g} = \frac{\gamma}{1 - \gamma} \cdot \frac{Pc_g^*}{Q^*} \implies P_f = \alpha P_Q, \quad P_m = (1 - \alpha)P_Q.$$

These personalized prices satisfy the Samuelson condition $P_f + P_m = P_Q$. Because both spouses value Q identically, each bears a share of its cost equal to their bargaining weight.

3. Generalized sharing rule (GSR). The GSR adds each spouse's valuation of public expenditure at their personalized price to their private expenditure:

$$\rho_f^* = Pc_f^* + P_f Q^* = \alpha(1 - \gamma)I + \alpha\gamma I = \alpha I, \quad \rho_m^* = (1 - \alpha)I.$$

A wife who remains at home ($e = 1$, $I = w^M$) has no labor earnings but is allocated resources equivalent to αw^M . The home-production shock does not directly affect the GSR; instead, it operates through the Lindahl price.

4. Money Metric Welfare Index (MMWI). The MMWI of spouse g , denoted by m_g , is the income that spouse g would need to attain utility u_g when facing a reference price vector rather than personalized prices. An individual with preferences $Q^\gamma (c_f^\alpha c_m^{1-\alpha})^{1-\gamma}$ and income m , facing price P for private consumption and $\bar{P}_Q$ for the public good, attains

$$v(m) = \log m - (1 - \gamma) \log P - \gamma \log \bar{P}_Q + K.$$

Equating $v(m_g)$ with u_g and solving for m_g yields the general formula $m_g = \rho_g^* (\bar{P}_Q / P_g)^\gamma$.

The remaining question is which reference price $\bar{P}_Q$ to use. If I use the couple's own effective price, $\bar{P}_Q = P\beta^{-e}$, then $m_f = \alpha^{1-\gamma}I$ and $m_m = (1 - \alpha)^{1-\gamma}I$. Under this reference price, the index does not capture home production: for a wife who stays at home, it equals $\alpha^{1-\gamma}w^M$ regardless of her draw of β . I therefore use the market price, $\bar{P}_Q = P$, so that the index explicitly accounts for home production. In this case, $v(m) = \log(m/P) + K$, and

setting $v(m_f) = u_f$ yields

$$m_f = \alpha^{1-\gamma} \beta^{e\gamma} I, \quad m_m = (1 - \alpha)^{1-\gamma} \beta^{e\gamma} I. \quad (\text{B.2})$$

Rewriting the wife's index as $m_f = \alpha I \cdot \alpha^{-\gamma} \cdot \beta^{e\gamma}$ separates three forces. The first term, αI , is her GSR share. The second term, $\alpha^{-\gamma}$, corresponds to the Lindahl-price adjustment. The third, $\beta^{e\gamma}$, captures the value of home production for spouses.

Couples solve the mobility problem in Equation 15 before β is realized. The relevant ex ante index is therefore the certain income $\bar{m}_g$ that allows a person with the same preferences to attain $\mathbb{E}[U_g]$. Because income m yields level utility $e^K m/P$, and because $\mathbb{E}[U_g] = e^K \alpha^{1-\gamma} \mathbb{E}[\beta^{e\gamma} I]/P$,

$$\bar{m}_f = \alpha^{1-\gamma} \Phi w^M, \quad \bar{m}_m = (1 - \alpha)^{1-\gamma} \Phi w^M. \quad (\text{B.3})$$

Summing Equation (B.2) across spouses, I can express the gains from the public nature of Q as

$$\frac{m_f + m_m}{\beta^{e\gamma} I} = G(\alpha, \gamma) \equiv \alpha^{1-\gamma} + (1 - \alpha)^{1-\gamma} > 1 \quad \text{whenever } 0 < \gamma < 1.$$

Thus, the spouses would jointly require more income than the household possesses if each had to purchase the public good separately at its full market price.

5. MMWI for singles. The central assumption is that singles have the same preferences as spouses but cannot share the public good with another household member. A single of gender g with wage w^g therefore solves

$$\max \gamma \log Q^g + (1 - \gamma) \log c^g \quad \text{s.t. } P(c^g + Q^g) = w$$

This problem implies $c^g = (1 - \gamma)w^g/P$ and $Q^g = \gamma w^g/P$, with indirect utility $u_s^g = \log \frac{w^g}{P} + K$. The constant K does not affect choices and cancels out in every welfare change. Thus, the model's $W^{s_g} = w^g/P$ is an exact representation of these preferences, and no quantitative result changes. What a single lacks is not the public good itself but its *publicness*: a single person must pay its full price. Accordingly, the MMWI for a single of gender g is

$$m^g = \frac{w^g}{P}$$

6. Money-Metric-Weighted Aggregate MMWI. Starting from Equation 30, I can express the per-person consumption-equivalent change in utility as

$$\bar{Z} = \frac{\sum N^{s_m} Z^{s_m} + \sum N^{s_f} Z^{s_f} + \sum N^c Z^c}{\sum N^{s_m} + \sum N^{s_f} + \sum N^c},$$

The measure $\bar{Z}$ averages percentage changes defined relative to different bases: for a single, the base is w/P ; for a spouse, it is a share of the household's full income, together with a shared public good. To express the per-person consumption-equivalent change in monetary units, I therefore construct a money-metric-weighted measure that aggregates across singles and couples.

Because each spouse's expected utility is a constant multiple of the couple's expected utility, the consumption-equivalent change of each spouse is identical to that of the couple when expressed in hat notation. The money-metric-weighted measure is therefore

$$\bar{Z}^{\$} = \frac{\sum m_i Z_i}{\sum m_i} = \bar{Z} + \frac{\text{Cov}(m_i, Z_i)}{\bar{m}}.$$

The covariance term captures whether the gains from the shock accrue to individuals with larger money metrics, such as spouses, or to those with smaller money metrics, such as singles.

C Dynamic Hat Algebra

This appendix develops the Dynamic Hat Algebra used to solve and simulate the model. It expresses the equilibrium entirely in terms of changes over time and counterfactual ratios, so that I never have to recover the levels of the fundamentals. The appendix proceeds in two parts. I first state the two propositions that characterize the baseline economy: one for the temporary equilibrium at a given point in time and another for the full sequential equilibrium, together with the algorithm used to solve them. I then derive the corresponding results for a counterfactual path of sectoral productivities.

C.1 Solving the Model

Notation. Let $\dot{x}_{t+1} = x_{t+1}/x_t$ denote the proportional change in any variable x between t and $t + 1$ in the baseline economy. Let $\hat{x}_{t+1} = \dot{x}'_{t+1}/\dot{x}_{t+1}$, where $\dot{x}'_{t+1} = x'_{t+1}/x'_t$, denote the ratio of the counterfactual changes. Primes index counterfactual levels throughout. For every feasible bundle $\mathbf{o}$, define $u_{\mathbf{o},t}^h = \exp(V_{\mathbf{o},t}^h)$, and define regional expenditure as $E_{j,t} \equiv P_{j,t}Q_{j,t}$. Because local sectoral productivities $\{A_{j,n,t}\}$ are the only fundamentals that vary over time, a quantitative application begins with a path of productivity changes $\{\dot{A}_{j,n,t}\}$ and solves the equilibrium system below.

For couples, define the quality-adjusted value of destination $\mathbf{d}$ as

$$B_{\mathbf{d},t} = \mathcal{A} \left[(1 + r_{\mathbf{d},t}) + \chi^\kappa \frac{\gamma}{\kappa - \gamma} (1 + r_{\mathbf{d},t})^{1-\kappa/\gamma} \right].$$

Because $\mathcal{A}$ is constant, it cancels from all proportional changes and counterfactual ratios of B . I retain it in the level expressions below only for completeness. The proportional change in B is

$$\dot{B}_{\mathbf{d},t+1} \equiv \frac{B_{\mathbf{d},t+1}}{B_{\mathbf{d},t}} = \frac{(1 + r_{\mathbf{d},t+1}) + \chi^\kappa \frac{\gamma}{\kappa - \gamma} (1 + r_{\mathbf{d},t+1})^{1-\kappa/\gamma}}{(1 + r_{\mathbf{d},t}) + \chi^\kappa \frac{\gamma}{\kappa - \gamma} (1 + r_{\mathbf{d},t})^{1-\kappa/\gamma}}. \quad (\text{C.1})$$

Proposition C.1. *Given the temporary-equilibrium allocation at t , the temporary equilibrium at $t + 1$ can be computed without knowing the levels of the fundamentals. It is characterized by the following system of equations:*

$$\begin{aligned} \dot{w}_{j,n,t+1}^M &= \dot{p}_{j,n,t+1} \dot{A}_{j,n,t+1}^{(\sigma-1)/\sigma} \dot{M}_{j,n,t+1}^{-1/\sigma} \dot{Y}_{j,n,t+1}^{1/\sigma}, \\ \dot{w}_{j,n,t+1}^F &= \dot{p}_{j,n,t+1} \dot{A}_{j,n,t+1}^{(\sigma-1)/\sigma} \dot{F}_{j,n,t+1}^{-1/\sigma} \dot{Y}_{j,n,t+1}^{1/\sigma}, \\ \dot{Y}_{j,n,t+1} &= \dot{A}_{j,n,t+1} \left(s_{j,n,m,t} \dot{M}_{j,n,t+1}^{(\sigma-1)/\sigma} + s_{j,n,f,t} \dot{F}_{j,n,t+1}^{(\sigma-1)/\sigma} \right)^{\sigma/(\sigma-1)}, \end{aligned}$$

$$\begin{aligned}
\dot{P}_{j,n,t+1} &= \left(\sum_{i=1}^L \sum_{k=1}^K \pi_{ik,jn,t} (\dot{p}_{i,k,t+1})^{1-\eta_n} \right)^{1/(1-\eta_n)}, \\
\pi_{ik,jn,t+1} &= \pi_{ik,jn,t} \left(\frac{\dot{p}_{i,k,t+1}}{\dot{P}_{j,n,t+1}} \right)^{1-\eta_n}, \\
\dot{P}_{j,t+1} &= \prod_{n=1}^K \dot{P}_{j,n,t+1}, \quad \dot{E}_{j,t+1} = \dot{P}_{j,t+1} \dot{Q}_{j,t+1}, \\
\dot{p}_{i,k,t+1} \dot{Y}_{i,k,t+1} &= \sum_{j=1}^L \sum_{n=1}^K \varphi_{ik,jn,t} \frac{\pi_{ik,jn,t+1}}{\pi_{ik,jn,t}} \dot{E}_{j,t+1},
\end{aligned}$$

for all region-sector pairs (j, n) , where $s_{j,n,m,t} \equiv \frac{w_{j,n,t}^M M_{j,n,t}}{p_{j,n,t} Y_{j,n,t}}$ and $s_{j,n,f,t} \equiv \frac{w_{j,n,t}^F F_{j,n,t}}{p_{j,n,t} Y_{j,n,t}}$ are the male and female shares of sectoral revenue, and $\varphi_{ik,jn,t} \equiv \frac{\pi_{ik,jn,t} \iota_n P_{j,t} Q_{j,t}}{p_{i,k,t} Y_{i,k,t}}$ is the share of origin-sector revenue from (i, k) sold to destination sector (j, n) . Moreover, for each destination bundle relevant to couples, $\dot{r}_{d,t+1} = \dot{w}_{j,n_f,t+1}^F / \dot{w}_{j,n_m,t+1}^M$, $r_{d,t+1} = r_{d,t} \dot{r}_{d,t+1}$, $\rho_{d,t+1} = 1 - \chi^\kappa (1 + r_{d,t+1})^{-\kappa/\gamma}$, and $\dot{B}_{d,t+1}$ follows from Equation C.1.

Building on Proposition C.1, I next characterize the sequential equilibrium in time differences. Equation 18 implies that the ratio of within-location shares between $t + 1$ and t is

$$\frac{\lambda_{od|j,t+1}^h}{\lambda_{od|j,t}^h} = \frac{\exp [\delta (V_{d,t+2}^h - V_{d,t+1}^h)]^{1/v}}{\Omega_{oj,t+1}^h / \Omega_{oj,t}^h}, \quad \frac{\Omega_{oj,t+1}^h}{\Omega_{oj,t}^h} = \sum_{z \in \mathcal{S}_h(j)} \lambda_{oz|j,t}^h (\dot{u}_{z,t+2}^h)^{\delta/v},$$

where the second equality follows from the time invariance of μ_{oz} . Define the proportional change in the inclusive value of location j as

$$D_{oj,t}^h \equiv \sum_{z \in \mathcal{S}_h(j)} \lambda_{oz|j,t}^h (\dot{u}_{z,t+2}^h)^{\delta/v}. \quad (\text{C.2})$$

The location shares in Equation 17 and the value function in Equation 16 depend on continuation values only through the inclusive values. Their time differences can therefore be expressed entirely in terms of $D_{oj,t}^h$.

Proposition C.2. *Given the economy's initial allocation, including the initial location shares $\lambda_{oj,0}^h$ and within-location shares $\lambda_{od|j,0}^h$, together with an anticipated convergent sequence of productivity changes, the sequential equilibrium in time differences can be solved without recovering the levels of the fundamentals. The solution satisfies the following system of*

nonlinear equations:

$$\begin{aligned}
\lambda_{\mathbf{o}\mathbf{d}|j,t+1}^h &= \frac{\lambda_{\mathbf{o}\mathbf{d}|j,t}^h (\dot{u}_{\mathbf{d},t+2}^h)^{\delta/v}}{\sum_{z \in \mathcal{S}_h(j)} \lambda_{\mathbf{o}z|j,t}^h (\dot{u}_{z,t+2}^h)^{\delta/v}}, \quad \forall h, \\
\lambda_{\mathbf{o}j,t+1}^h &= \frac{\lambda_{\mathbf{o}j,t}^h (D_{\mathbf{o}j,t}^h)^{v/\vartheta}}{\sum_{\ell=1}^L \lambda_{\mathbf{o}\ell,t}^h (D_{\mathbf{o}\ell,t}^h)^{v/\vartheta}}, \quad \forall h, \\
\dot{u}_{\mathbf{o},t+1}^h &= \dot{W}_{\mathbf{o},t+1}^h \left(\sum_{\ell=1}^L \lambda_{\mathbf{o}\ell,t}^h \left(\sum_{z \in \mathcal{S}_h(\ell)} \lambda_{\mathbf{o}z|\ell,t}^h (\dot{u}_{z,t+2}^h)^{\delta/v} \right)^{v/\vartheta} \right)^{\vartheta}, \quad \forall h, \\
N_{\mathbf{d},t+1}^h &= \sum_{\mathbf{o} \in \mathcal{S}_h} \lambda_{\mathbf{o}\mathbf{d}|j,t}^h \lambda_{\mathbf{o}j,t}^h N_{\mathbf{o},t}^h, \quad \mathbf{d} \in \mathcal{S}_h(j), \quad \forall h, \\
M_{j,n,t+1} &= N_{j,n,t+1}^{s_m} + \sum_{n_f=1}^K N_{j,n,n_f,t+1}^c, \\
F_{j,n,t+1} &= N_{j,n,t+1}^{s_f} + \sum_{n_m=0}^K \rho_{j,n_m,n,t+1} N_{j,n_m,n,t+1}^c,
\end{aligned}$$

for all feasible bundles and all sectors at each t , where $D_{\mathbf{o}\ell,t}^h$ is defined in Equation C.2 and the sectoral prices, trade shares, and participation objects solve the temporary-equilibrium system in Proposition C.1.

Because the migration costs $\mu_{\mathbf{o}\mathbf{d}}$ are time-invariant, they cancel from all ratios and therefore need not be recovered in levels. The flow term in the couple's continuation value is the only one that contains an additional factor. In particular, because the couple's indirect utility is $W_{\mathbf{o},t}^c = B_{\mathbf{o},t} w_{i,k_m,t}^M / P_{i,t}$, its proportional change is $\dot{B}_{\mathbf{o},t+1} \dot{w}_{i,k_m,t+1}^M / \dot{P}_{i,t+1}$. Thus, the relevant term is the *ratio* defined in Equation C.1, rather than the level $B_{\mathbf{o},t+1}$.

Algorithm for Solving the Sequential Competitive Equilibrium. Given an initial equilibrium at $t = 0$ and an anticipated convergent sequence of sectoral productivity changes $\left\{ \dot{A}_{j,n,t} \right\}_{j=1, n=1, t=0}^{L,K,T}$, the algorithm is as follows:

1. Fix a sufficiently large terminal horizon T and guess a path for the changes in continuation values $\left\{ \dot{u}_{\mathbf{o},t+1}^{h,(0)} \right\}_{t=0}^T$, subject to the terminal condition $\dot{u}_{\mathbf{o},T+1}^{h,(0)} = 1$ for every feasible bundle $\mathbf{o}$.
2. Decompose the initial migration matrix into location shares $\lambda_{\mathbf{o}j,0}^h = \sum_{\mathbf{d} \in \mathcal{S}_h(j)} \lambda_{\mathbf{o}\mathbf{d},0}^h$ and within-location shares $\lambda_{\mathbf{o}\mathbf{d}|j,0}^h = \lambda_{\mathbf{o}\mathbf{d},0}^h / \lambda_{\mathbf{o}j,0}^h$. Using Proposition C.2, update these objects forward in the following order: the within-location shares, $D_{\mathbf{o}j,t}^h$, and then the location shares.

3. Starting from the initial allocation, apply the law of motion to recover the entire path of household masses $\left\{N_{\mathbf{d},t}^{s,m,(0)}, N_{\mathbf{d},t}^{s,f,(0)}, N_{\mathbf{d},t}^{c,(0)}\right\}_{t=0}^T$, and aggregate these masses into sectoral male labor supplies $\left\{M_{j,n,t}^{(0)}\right\}_{t=0}^T$.
4. Solve the temporary equilibrium sequentially for each period $t \geq 0$:
 - (a) Guess the participation rates $\{\rho_{\mathbf{d},t+1}^{(0)}\}$ and use them to construct the sectoral female labor supplies $\{F_{j,n,t+1}^{(0)}\}$.
 - (b) Conditional on the labor aggregates and the path of fundamentals, solve the production block for $\{\dot{Y}_{j,n,t+1}^{(0)}, \dot{w}_{j,n,t+1}^{M,(0)}, \dot{w}_{j,n,t+1}^{F,(0)}\}$, the trade block for $\{\dot{p}_{j,n,t+1}^{(0)}, \dot{P}_{j,n,t+1}^{(0)}, \pi_{ik,jn,t+1}^{(0)}\}$, and the regional aggregation block for $\{\dot{P}_{j,t+1}^{(0)}, \dot{E}_{j,t+1}^{(0)}\}$.
 - (c) Recover $\{\dot{r}_{\mathbf{d},t+1}^{(0)}\}$, construct the levels $r_{\mathbf{d},t+1}^{(0)} = r_{\mathbf{d},t} \dot{r}_{\mathbf{d},t+1}^{(0)}$, and update $\{\rho_{\mathbf{d},t+1}^{(1)}, \dot{B}_{\mathbf{d},t+1}^{(1)}\}$.
 - (d) Iterate over steps (a) through (c) until the participation block converges.
5. Given the temporary-equilibrium path and the guessed continuation values, solve backward from T for the updated path $\{\dot{u}_{\mathbf{o},t+1}^{h,(1)}\}_{t=0}^T$ using the $\dot{u}^h$ equation in Proposition C.2. This backward solution requires both the location shares and the within-location shares.
6. Use the updated path as the new guess and repeat steps 2 through 5 until the sequential equilibrium converges.

C.2 Solving the Counterfactual Equilibrium

A counterfactual path for sectoral productivity fundamentals $\{A'_{j,n,t}\}_{t \geq 1}$ defines the corresponding productivity hats $\{\hat{A}_{j,n,t}\}_{t \geq 1}$. Under CDP's timing, households do not anticipate the change at $t = 0$. Therefore, $\lambda'_{\mathbf{o}j,0} = \lambda_{\mathbf{o}j,0}^h$, $\lambda'_{\mathbf{od}|j,0} = \lambda_{\mathbf{od}|j,0}^h$, and $N'_{\mathbf{o},1} = N_{\mathbf{o},1}^h$.

Dividing each counterfactual migration equation in Proposition C.2 by its baseline counterpart, and using $\dot{\lambda}_{\mathbf{od}|j,t}^h = \lambda_{\mathbf{od}|j,t}^h / \lambda_{\mathbf{od}|j,t-1}^h$ and $\dot{\lambda}_{\mathbf{o}j,t}^h = \lambda_{\mathbf{o}j,t}^h / \lambda_{\mathbf{o}j,t-1}^h$, yields the following counterfactual analogue of Equation C.2:

$$\hat{D}_{\mathbf{o}j,t}^h \equiv \sum_{\mathbf{z} \in \mathcal{S}_h(j)} \lambda'_{\mathbf{oz}|j,t-1} \dot{\lambda}_{\mathbf{oz}|j,t}^h (\hat{u}_{\mathbf{z},t+1}^h)^{\delta/v}. \quad (\text{C.3})$$

Proposition C.3. *Given the baseline economy and a convergent sequence of counterfactual productivity changes relative to the baseline path, the counterfactual sequential equilibrium*

can be solved without recovering the fundamental levels. For $t \geq 2$, it is characterized by the following system of nonlinear equations:

$$\begin{aligned}
\lambda'_{od|j,t} &= \frac{\lambda'^h_{od|j,t-1} \dot{\lambda}^h_{od|j,t} (\hat{u}^h_{d,t+1})^{\delta/v}}{\hat{D}^h_{oj,t}}, \quad \forall h, \\
\lambda'_{oj,t} &= \frac{\lambda'^h_{oj,t-1} \dot{\lambda}^h_{oj,t} (\hat{D}^h_{oj,t})^{v/\vartheta}}{\sum_{\ell=1}^L \lambda'^h_{o\ell,t-1} \dot{\lambda}^h_{o\ell,t} (\hat{D}^h_{o\ell,t})^{v/\vartheta}}, \quad \forall h, \\
\hat{u}^h_{o,t} &= \hat{W}^h_{o,t} \left(\sum_{\ell=1}^L \lambda'^h_{o\ell,t-1} \dot{\lambda}^h_{o\ell,t} \left(\sum_{\mathbf{z} \in \mathcal{S}_h(\ell)} \lambda'^h_{o\mathbf{z}|\ell,t-1} \dot{\lambda}^h_{o\mathbf{z}|\ell,t} (\hat{u}^h_{\mathbf{z},t+1})^{\delta/v} \right)^{v/\vartheta} \right)^{\vartheta}, \quad \forall h, \\
N'_{d,t+1} &= \sum_{o \in \mathcal{S}_h} \lambda'^h_{od|j,t} \lambda'^h_{oj,t} N'_{o,t}, \quad \mathbf{d} \in \mathcal{S}_h(j), \quad \forall h, \\
M'_{j,n,t+1} &= N'^{s,m}_{j,n,t+1} + \sum_{n_f=1}^K N'^{c}_{j,n,n_f,t+1}, \\
F'_{j,n,t+1} &= N'^{s,f}_{j,n,t+1} + \sum_{n_m=0}^K \rho'_{j,n_m,n,t+1} N'^{c}_{j,n_m,n,t+1}.
\end{aligned}$$

In period $t = 1$, when households first learn the new path, the migration equations instead become

$$\begin{aligned}
\lambda'_{od|j,1} &= \frac{\vartheta^h_{od|j} (\hat{u}^h_{d,2})^{\delta/v}}{\sum_{\mathbf{z} \in \mathcal{S}_h(j)} \vartheta^h_{o\mathbf{z}|j} (\hat{u}^h_{\mathbf{z},2})^{\delta/v}}, \\
\lambda'_{oj,1} &= \frac{\lambda^h_{oj,1} \left(\sum_{\mathbf{z} \in \mathcal{S}_h(j)} \vartheta^h_{o\mathbf{z}|j} (\hat{u}^h_{\mathbf{z},2})^{\delta/v} \right)^{v/\vartheta}}{\sum_{\ell=1}^L \lambda^h_{o\ell,1} \left(\sum_{\mathbf{z} \in \mathcal{S}_h(\ell)} \vartheta^h_{o\mathbf{z}|\ell} (\hat{u}^h_{\mathbf{z},2})^{\delta/v} \right)^{v/\vartheta}}, \\
\hat{u}^h_{o,1} &= \hat{W}^h_{o,1} \left(\sum_{\ell=1}^L \lambda^h_{o\ell,1} \left(\sum_{\mathbf{z} \in \mathcal{S}_h(\ell)} \vartheta^h_{o\mathbf{z}|\ell} (\hat{u}^h_{\mathbf{z},2})^{\delta/v} \right)^{v/\vartheta} \right)^{\vartheta}, \quad \vartheta^h_{od|j} \equiv \lambda^h_{od|j,1} (\hat{u}^h_{d,1})^{\delta/v}.
\end{aligned}$$

The counterfactual temporary equilibrium is characterized by

$$\begin{aligned}
\hat{w}^M_{j,n,t} &= \hat{p}_{j,n,t} \hat{A}_{j,n,t}^{(\sigma-1)/\sigma} \hat{M}_{j,n,t}^{-1/\sigma} \hat{Y}_{j,n,t}^{1/\sigma}, \\
\hat{w}^F_{j,n,t} &= \hat{p}_{j,n,t} \hat{A}_{j,n,t}^{(\sigma-1)/\sigma} \hat{F}_{j,n,t}^{-1/\sigma} \hat{Y}_{j,n,t}^{1/\sigma}, \\
\hat{Y}_{j,n,t} &= \hat{A}_{j,n,t} \left(s_{j,n,m,t} \hat{M}_{j,n,t}^{(\sigma-1)/\sigma} + s_{j,n,f,t} \hat{F}_{j,n,t}^{(\sigma-1)/\sigma} \right)^{\sigma/(\sigma-1)},
\end{aligned}$$

$$\begin{aligned}
\hat{P}_{j,n,t} &= \left(\sum_{i=1}^L \sum_{k=1}^K \pi_{ik,jn,t} (\hat{p}_{i,k,t})^{1-\eta_n} \right)^{1/(1-\eta_n)}, \\
\pi'_{ik,jn,t} &= \pi_{ik,jn,t} \left(\frac{\hat{p}_{i,k,t}}{\hat{P}_{j,n,t}} \right)^{1-\eta_n}, \\
\hat{P}_{j,t} &= \prod_{n=1}^K \hat{P}_{j,n,t}^{\iota_n}, \quad \hat{E}_{j,t} = \hat{P}_{j,t} \hat{Q}_{j,t}, \\
\hat{p}_{i,k,t} \hat{Y}_{i,k,t} &= \sum_{j=1}^L \sum_{n=1}^K \varphi_{ik,jn,t} \frac{\pi'_{ik,jn,t}}{\pi_{ik,jn,t}} \hat{E}_{j,t}, \\
\hat{r}_{\mathbf{d},t} &= \frac{\hat{w}_{j,n_f,t}^F}{\hat{w}_{j,n_m,t}^M},
\end{aligned}$$

and the corresponding counterfactual level objects satisfy

$$r'_{\mathbf{d},t} = r'_{\mathbf{d},t-1} \dot{r}_{\mathbf{d},t} \hat{r}_{\mathbf{d},t}, \quad \rho'_{\mathbf{d},t} = 1 - \chi^\kappa (1 + r'_{\mathbf{d},t})^{-\kappa/\gamma}, \quad \hat{B}_{\mathbf{d},t} = \frac{B'_{\mathbf{d},t}}{B_{\mathbf{d},t}},$$

with $B'_{\mathbf{d},t}$ evaluated at $r'_{\mathbf{d},t}$. The baseline revenue shares $s_{j,n,m,t}$, $s_{j,n,f,t}$ and revenue-allocation shares $\varphi_{ik,jn,t}$ are those of Proposition C.1, taken from the baseline equilibrium.

Solution Algorithm for the Counterfactual Equilibrium. Given the baseline sequential equilibrium and a convergent sequence of counterfactual changes in fundamentals $\{\hat{A}_{j,n,t}\}_{t \geq 1}$, the counterfactual equilibrium is computed as follows:

1. Because the temporary equilibrium and migration shares at $t = 0$ are unchanged, initialize all counterfactual objects at their baseline values.
2. Fix a sufficiently large horizon T and guess a path $\{\hat{u}_{\mathbf{o},t}^{h,(0)}\}_{t=1}^{T+1}$, imposing $\hat{u}_{\mathbf{o},T+1}^{h,(0)} = 1$ for all feasible bundles $\mathbf{o}$ and household types h .
3. Use the guessed continuation values to compute the path of counterfactual migration shares. At $t = 1$, construct $\vartheta_{\mathbf{o}\mathbf{d}|j}^h$ using the baseline within-location shares and $\hat{u}_{\mathbf{d},1}^{h,(0)}$, and apply the period-1 equations in Proposition C.3. For $t \geq 2$, update, in sequence, the within-location shares, $\hat{D}_{\mathbf{o}j,t}^h$ from Equation C.3, and the location shares.
4. Starting from the same initial allocation as in the baseline economy, use the counterfactual shares to compute the implied path of counterfactual household populations $\{N'_{\mathbf{d},t}{}^{s,m,(0)}, N'_{\mathbf{d},t}{}^{s,f,(0)}, N'_{\mathbf{d},t}{}^{c,(0)}\}_{t=1}^{T+1}$.
5. Given the counterfactual household masses, recover the path of counterfactual sectoral male labor supply $\{M'_{j,n,t}{}^{(0)}\}_{t=1}^{T+1}$.

6. Solve the counterfactual temporary equilibrium period by period. For each $t \geq 0$, proceed as follows:
 - (a) Guess the counterfactual participation rates $\left\{ \rho_{\mathbf{d},t+1}'^{(0)} \right\}$.
 - (b) Use these participation rates to compute counterfactual female labor supply $\left\{ F_{j,n,t+1}'^{(0)} \right\}$.
 - (c) Construct $\left\{ \hat{M}_{j,n,t+1}^{(0)} \right\}$ and $\left\{ \hat{F}_{j,n,t+1}^{(0)} \right\}$. Given these objects and the counterfactual changes in fundamentals, recover $\left\{ \hat{Y}_{j,n,t+1}^{(0)}, \hat{r}_{\mathbf{d},t+1}^{(0)} \right\}$ from the production block. Next, solve the goods-market-clearing block for $\left\{ \hat{p}_{j,n,t+1}^{(0)}, \hat{P}_{j,n,t+1}^{(0)}, \hat{P}_{j,t+1}^{(0)}, \pi_{ik,jn,t+1}'^{(0)} \right\}$ and recover $\left\{ \hat{w}_{j,n,t+1}^{M,(0)}, \hat{w}_{j,n,t+1}^{F,(0)} \right\}$ from the firms' first-order conditions.
 - (d) Recover $\left\{ r_{\mathbf{d},t+1}'^{(0)}, \rho_{\mathbf{d},t+1}'^{(0)}, B_{\mathbf{d},t+1}'^{(0)} \right\}$. If the updated $\rho_{\mathbf{d},t+1}'^{(0)}$ differs from its initial guess, return to step (b) and iterate until the participation block converges.
 - (e) Repeat steps (a) through (d) sequentially for each period t .
7. Given the counterfactual temporary-equilibrium path and the guessed continuation values, solve backward from T for the updated path $\left\{ \hat{u}_{\mathbf{o},t}^{h,(1)} \right\}_{t=1}^{T+1}$ using the $\hat{u}^h$ equations in Proposition C.3. At $t = 1$, $\hat{u}_{\mathbf{o},1}^h$ appears on both sides through ϑ , so solve this equation as a fixed point.
8. Use the updated path as the new guess and repeat steps 3 through 7 until the continuation-value path converges.
9. Compute welfare using Equation 29, with $\hat{\lambda}_{\mathbf{o}i,s}^h$ and $\hat{\lambda}_{\mathbf{o}\mathbf{o}|i,s}^h$ constructed from the counterfactual and baseline location and within-location staying shares.

D Quantification of the Model

This section details three important elements in the model's quantification. First, I explain how I assign each non-employed married woman to their potential sector. Then, I describe how I build the migration matrices by household type. Finally, I describe the calibration of the couple's maximization problem.

D.1 Assignment of Non-Employed Wives to their Potential Sector

When using the ACS to construct the initial distribution of couples across states and sector pairs, $\{N_{o,0}^c\}$, a married woman who is out of the labor force is considered non-employed. Thus, I do not observe the counterfactual sector in which she would have worked if they jointly decided she would participate in the market. In the model, couples choose the wife's productive sector first and her participation second, so she is never assigned to the non-employment sector itself. Then, I need to distribute the non-employed wives across sectors.

If I were to distribute them across productive sectors in the same proportions as the wives employed, I would be abstracting from a selection issue. The sectors in which married women are more likely to be employed are also the ones in which they are less likely to be non-employed. Then, following the distribution of employed wives, I would assign a larger share of non-employed wives to sectors less likely to be their potential sector.

So, to recover the counterfactual sector properly, I return to the model and write, for every period, which I drop from the indices, the labor supply of married women in each state j and sector n :

$$F_{j,n}^c = \rho_{j,n} N_{j,n}^c.$$

Then, the non-employed women, which I denote by $\text{Non-}F_{j,n}^c$, in that same state j and sector n cell are:

$$\text{Non-}F_{j,n}^c = (1 - \rho_{j,n}) N_{j,n}^c.$$

So, the ratio of non-employed women to employed women corresponds to

$$\frac{\text{Non-}F_{j,n}^c}{F_{j,n}^c} = \frac{(1 - \rho_{j,n})}{\rho_{j,n}},$$

which implies that the mass of non-employed wives in that cell is

$$\text{Non-}F_{j,n}^c = \underbrace{F_{j,n}^c}_{\text{observed}} \times \underbrace{\frac{1 - \rho_{j,n}}{\rho_{j,n}}}_{\text{the model's odds of non-employment}}.$$

Dividing by $\rho_{j,n,t}$ therefore corrects the selection problem and assigns each non-employed wife to her potential productive sector.

These cell-level masses are in shares rather than headcounts, so nothing in the model's odds forces them to sum to the number of non-employed married women actually observed in state j . I therefore normalize them within each state, rescaling the $\{\text{Non-}F_{j,n}^c\}_n$ proportionally so that they add up to the observed state total.

D.2 Building the Migration Matrix

Since I solve the model with Dynamic Hat Algebra, the time-invariant mobility costs μ_{od} cancel from the recursion that governs the choice probabilities (Proposition C.2). The recursion requires only two initial objects for each household type: the location shares $\lambda_{oj,0}^h$ of Equation 17 and the within-location shares $\lambda_{od|j,0}^h$ of Equation 18.

Following CDP's Appendix E, I build it as the product of an interstate probability from the ACS and a sectoral transition probability from the Current Population Survey (CPS), where an interstate mover's sectoral transition is imputed from the destination state's sector matrix:

$$\lambda_{(i,k)(j,n),0}^h = \underbrace{f_{ij}^h}_{\text{ACS}} \times \underbrace{P_j^h(n | k)}_{\text{CPS}}.$$

Summing over the destination sectors of location j gives $\lambda_{oj,0}^h = f_{ij}^h$, and dividing by that sum gives $\lambda_{od|j,0}^h = P_j^h(n | k)$.³⁰

To build the matrices, I follow these steps. First, interstate flows are derived from the ACS migration question for 2001–2007, which records the state of residence one year earlier. The resulting state-to-state matrix is therefore annual, whereas the model period is quarterly. Following CDP's Appendix E, I assume that interstate moves are evenly distributed throughout the year and that a household makes at most one such move annually, so the quarterly matrix retains one-fourth of the moves and imputes the remaining three-fourths as non-moves:

$$f_{i \rightarrow j}^q = \frac{1}{4} f_{i \rightarrow j} \quad (j \neq i), \quad f_{i \rightarrow i}^q = \frac{1}{4} f_{i \rightarrow i} + \frac{3}{4}.$$

I apply this adjustment to every quarter from 2000 to 2007. It is worth mentioning that the adjustment is quantitatively important. Combining the *annual* interstate probability

³⁰Where an origin puts no mass on a location, the within-location share is undefined. I set it to zero rather than a uniform distribution, since the two factors only ever enter as a product, and a uniform distribution would place mass on destinations that the data indicate are unreachable once the location share moves away from zero.

with the *quarterly* sector matrix without it would inflate mobility by roughly a factor of six, implying a quarterly own-state retention rate of 0.959 for single men against 0.994 in CDP. The error would fall entirely on the location margin, which is calibrated to the population’s response to the shock.

Next, when looking at sectoral transitions from the CPS, instead of considering transitions across 22 sectors plus non-employment, I pool sectors together because, otherwise, when disaggregating by household type, I would have too few observations per cell. I therefore pool the 12 manufacturing sub-sectors into a single manufacturing sector and the 8 service sub-sectors into a single service sector, leaving an aggregated five-sector structure: non-employment, manufacturing, retail, construction, and services. I estimate transitions on this aggregation separately for single men, single women, and couples, with couples moving on the joint space of both spouses’ groups.

Combining the first two steps gives a transition matrix over five sectors and 50 states, by household type. Let $\tilde{\mathbf{o}}$ denote the origin bundle in that grid and $\tilde{\mathbf{d}}$ denote the corresponding destination bundle. However, the production side considers 22 productive sectors in 50 states. Applying the same sectoral aggregation to the production side would imply losing the dispersion across manufacturing sectors in the China Trade Shock. Then, I need to map the matrices to the full grid. I do this by factorizing each transition probability into a group-level move and a within-group destination. For an individual of gender $g \in \{m, f\}$ in a household of type h ,

$$\underbrace{P_{h,g}(\mathbf{d} \mid \mathbf{o})}_{\text{model's grid}} = \underbrace{P_h(\tilde{\mathbf{d}} \mid \tilde{\mathbf{o}})}_{\text{ACS + CPS, by household type}} \times \underbrace{P_g(\mathbf{d} \mid \tilde{\mathbf{d}}, \mathbf{o})}_{\text{within-group composition, by gender}},$$

so that household type governs in *which* broad sector, and in which state, a household moves to, and the within-sector composition governs *where within* that sector it lands. Because this factorizes a single conditional distribution, the rows of the resulting matrix sum to one by construction. The within-group composition only redistributes mass among sectors of the same group *in the same state*, so it leaves the location shares unchanged and acts only on the within-location shares.

To build the within-group composition, I split the CPS sample by sex and compute, within manufacturing and within services, the share of each group going to every sub-sector. This step matters: men and women sort differently within both groups, with men concentrated in heavy and durable manufacturing and women in lighter, non-durable goods. Without it, the model would miss one of the channels through which the shock affected men and women differently.

D.3 Calibration of the Couple’s Maximization Problem

This appendix documents the calibration of the couple’s maximization problem, with $\beta \sim \text{Pareto}(\chi, \kappa)$ described in Equation 20. There are four parameters that govern it: the wife’s weight in the household objective, α ; the public-good expenditure share, γ ; and the location and shape of the home-production shock, χ and κ . Given that I follow Chiappori and Mazzocco (2017)’s method to solve the couple’s problem, I identify the four parameters *sequentially*, each from its own source, and each maps onto a distinct piece of variation.

D.3.1 α and γ From Expenditure Data

The optimal allocations for private and public consumption in Equation 21 are Cobb-Douglas shares. So, both parameters map onto expenditure objects exactly: α maps onto the share of female private consumption in total private consumption, and γ maps onto the share of public expenditure in total expenditure:

$$\alpha = \frac{c_f^*}{c_f^* + c_m^*}, \quad \gamma = \frac{q^*}{q^* + c_f^* + c_m^*}.$$

Data. I use the Consumer Expenditure Interview Survey (CEX), 2000–2007, to construct three inputs.³¹ First, the *assignable private goods* c_m and c_f are adult apparel and footwear.³² Second, the *public good* q is the market input to home production: because $Q = \beta^e q$, the object I need is what the household combines with the wife’s home time. It comprises shelter, fuels and utilities, food at home, cable and internet, do-it-yourself repair materials, and household operations such as domestic service, lawn care, and appliance repair. Household operations are the *market substitutes* for home time, so omitting them would measure q selectively on precisely the margin the model is about. Third, I construct *Total expenditure* Y as consumption only.³³

The sample consists of household-year pairs with complete twelve-month expenditure records. Couples are intact married couples with both spouses present, aged 25–65; singles are one-person consumer units in the same age band. I drop households containing a 16–17 year-old, because the CEX books apparel for ages 16 and over under the adult men’s and women’s codes, so a teenager’s clothing would otherwise be read as a spouse’s assignable private consumption. The couples and singles samples are forced disjoint.

³¹The complete list of expenditure codes entering each of the three objects is available upon request.

³²Children’s, boys’, girls’, and infants’ apparel are excluded, since the model has no children.

³³The CEX expenditure file also carries asset stocks, debt balances, loans, principal repayments, and asset-sale receipts. All of these are excluded, as are transfers out of the household and gifts.

Estimating α . I start by estimating the wife’s weight in the household objective α , using expenditures that can be assigned to each spouse. In the CEX, apparel is the only assignable good category, but it is only a *subset* of private consumption, so the raw apparel ratio, 0.60 in this sample, is not α . The reason is that women spend, on average, more on apparel than men, so if I consider the raw ratio, I would assume the wife has the larger Pareto weight. Thus, to obtain the true value of α , I follow Browning et al. (2013) and recover the resource shares ρ_f and ρ_m , and estimate an Engel curve for apparel among *singles* of the same gender. Then, evaluated at their own resources $\rho_g Y$, the resource shares must reproduce the aggregate apparel spending observed in couples. Using both ρ_f and ρ_m , I recover $\alpha = \rho_f / (\rho_f + \rho_m)$. The correction is substantial; it moves α from 0.60 to 0.47, which *reverses* which spouse has the larger bargaining power.

Voena (2015) estimate a value of 0.3, which is one-third smaller than the one I estimate. The two numbers do not align because they represent different inputs in each paper. In Voena (2015)’s world, α prices the wife’s outside option given changes in divorce laws across US states, while in my model, α reflects her observed consumption. In addition, she estimates α using data from the PSID between 1980 and 1990, years when female labor force participation was lower than in the 2000s, so it is expected that as women enter the labor force, their bargaining power increases.

Estimating γ . Then, I move to estimate the public-good expenditure share, γ . One approach to estimate γ would be to assume that it corresponds to the residual $1 - \rho_f - \rho_m$. The residual is 0.13, implying the couple spends roughly \$5,400 a year on public goods, while couples are *observed* spending \$22,300 a year on shelter, utilities, and food alone. The reason is that the apparel slopes in the Engel curves are essentially flat ($R^2 = 0.004$). Browning et al. (2013)’s method allows me to recover the *ratio*, but it carries almost no information about the *level* of resources, and the residual absorbs the error. So, I measure γ directly as the share of total spending devoted to market inputs for home production, including shelter, utilities, food at home, and market substitutes for home time, such as domestic services and appliance repair. Then, $\gamma = 0.4887$.

Del Boca and Flinn (1995) estimate γ to be 0.204. The reason the two values differ is that 0.204 is an estimate based on divorced parents. The only possibility that divorced parents have for joint consumption concerns children; they cannot enjoy economies of scale in housing, utilities, transportation, among others. Therefore, when considering couples’ joint consumption while together, it is expected to account for a significant share of total consumption.

Standard errors. The CEX is a stratified, multistage, clustered sample, and the BLS does not publish the design variables needed for an analytic variance. Thus, following the BLS recommendation, I re-run the *entire* estimator on each of the 44 replicate weight vectors and take the spread across replicates as the sampling variance. Then, the standard errors for α correspond 0.021 and to 0.006 for γ .

D.3.2 κ/γ from the China Trade Shock in the PSID

As mentioned in Section 4,

$$\log(1 - \rho(r)) = \underbrace{\kappa \log \chi}_{\text{intercept}} - \underbrace{(\kappa/\gamma)}_{\text{slope}} \log(1 + r). \quad (\text{D.1})$$

is a structural relation, not a regression. Cross-sectional variation in the relative wage r is not exogenous: r is an equilibrium object. In addition, I observe r only when the wife participates in the market. Then, I use the empirical design in Section 2 as the identification strategy for the participation block.

In particular, I identify κ/γ from the experiment in which the *husband's* industry is exposed. The wife is not directly displaced, so the shock affects her participation decision only through r , which is exactly the exclusion restriction required by Equation D.1.

Because the equation is linear, the parameter is a Wald ratio:

$$\frac{\kappa}{\gamma} = -\frac{\Delta \log(1 - \rho)}{\Delta \log(1 + r)}. \quad (\text{D.2})$$

To construct the numerator of Equation D.2, I use the estimates in Table A.5. Wives of exposed husbands raise labor-force participation by 4.2 percentage points from a baseline of 0.78, so ρ increases from 0.78 to 0.822. Then, the numerator is:

$$\Delta \log(1 - \rho) = \log(1 - 0.822) - \log(1 - 0.78) = -0.212.$$

To construct the denominator, I estimate the static version of Equation 4 on $\log(\text{Hourly Wage})$ for each spouse. The sample consists of individuals who have at least one observation with positive wages before and during the shock. Table D.1 shows that when the husband is exposed to the shock, his hourly wage decreases by 16.4% while her wage decreases by 4.2%. Since the husband's wage falls by more than his wife's, the relative wage she faces rises. Using the estimates in Table D.1, r increases by 12.2 log points, which implies an increase of 13%. The r implied by the baseline mean is 0.88. However, this number includes only employed individuals, while the numerator considers all individuals in the sample. Then, the correct value for r_0 is 0.607. Thus, the denominator of Equation D.2

is

$$\Delta \log(1 + r) = \log(1 + 0.686) - \log(1 + 0.607) = 0.0477.$$

Replacing both the numerator and the denominator in Equation D.2,

$$\frac{\kappa}{\gamma} = -\frac{-0.212}{+0.048} = 4.44.$$

Replacing $\gamma = 0.4887$, I obtain $\kappa = \gamma \cdot \frac{\kappa}{\gamma} = 2.17$, which satisfies the integrability restriction $\kappa > \gamma$.

Table D.1: Husband's Exposure to the China Trade Shock and log(Hourly Wage)

	(1) Husband	(2) Wife
Exposed Husband=1	-0.078*	0.150***
	(0.042)	(0.052)
(Exposed Husband=1)x(Post=1)	-0.164*	-0.042
	(0.086)	(0.070)
Observations	12,023	9,979
R^2	0.35	0.31
Baseline Mean	3.31	2.91

Notes. Robust standard errors are two-way clustered at the husband's industry and individual levels. Given that I restrict the sample to individuals who have at least one observation with positive wages before and during the shock, the number of observations in this table does not match the number of observations in Fact 3. * for $p < .10$, ** for $p < 0.05$, *** for $p < 0.01$.

D.3.3 χ As a Level Restriction

The participation rule in Equation D.1 is log-linear in $1 - \rho$: its *slope* in $\log(1 + r)$ identifies κ/γ , and its *intercept* identifies χ . With κ/γ estimated above and $\kappa = \gamma \cdot (\kappa/\gamma)$, χ is the single remaining free parameter, and I set it so that the model reproduces the observed *level* of married-women participation in 2000Q1, $\rho_0 = 0.621$. Married women's participation is therefore a *targeted* moment and is not reported as an untargeted success of the model.

To do so, let m_o be the couples mass in state o in $t = 0$ and $r_{o,0} = w_{o,0}^F/w_{o,0}^M$ its wage ratio. Then, I set χ such that the mass-weighted average participation rate the model generates

equals the one in the data:

$$\sum_{\mathbf{o}} m_{\mathbf{o},0} \rho(r_{\mathbf{o},0}; \chi) = \rho_0. \quad (\text{D.3})$$

Since ρ is strictly decreasing in χ , Equation D.3 has a unique root, and it yields $\chi = 1.92$. This satisfies the support restriction $\chi \geq 1$ and leaves the participation margin interior in every cell.

There is an untargeted moment the model misses, and it is worth explaining why. Since ρ is monotone in r , and a non-employed husband's income is the benefit b_j , the ratio $r = w_f/b_j$ is mechanically the largest anywhere in the model. The model must therefore predict that wives of non-employed husbands work *more*. However, in the data, they work *less*. The reason is that there is strong positive assortative matching in the marriage market (Browning et al., 2014), and the model abstracts from it. So, when estimating χ , I restrict to bundles in which the husband is employed.

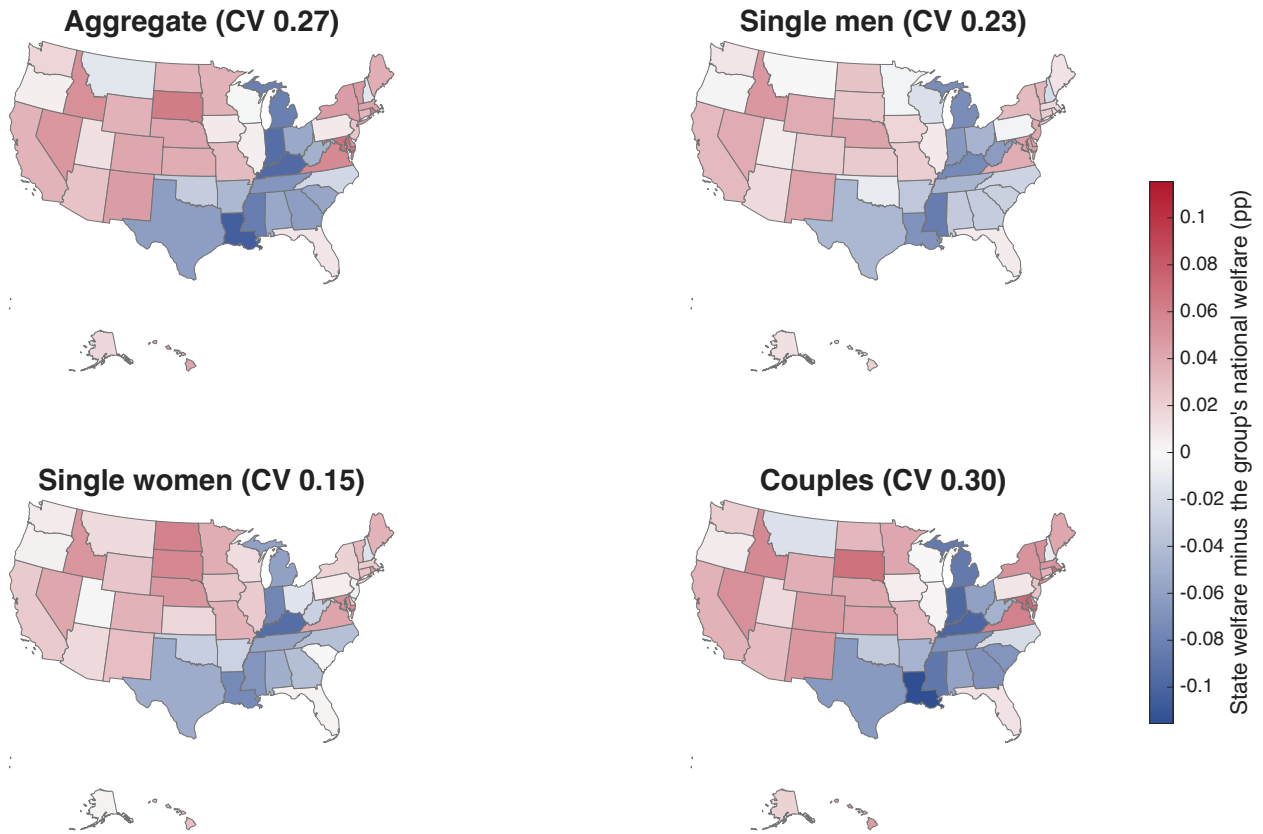
E Additional Results in the Quantitative Exercises

Table E.1: Long Run Employment Effects of the China Trade Shock

	Employment effect
<i>Panel A. Manufacturing share of employment (p.p.)</i>	
Aggregate	−0.2933
Single men	−0.2869
Married men	−0.4122
Single women	−0.1880
Married women	−0.1920
<i>Panel B. Employment rate (p.p.)</i>	
Aggregate	0.1121
Single men	0.1064
Married men	0.0964
Single women	0.1582
Married women	0.1106

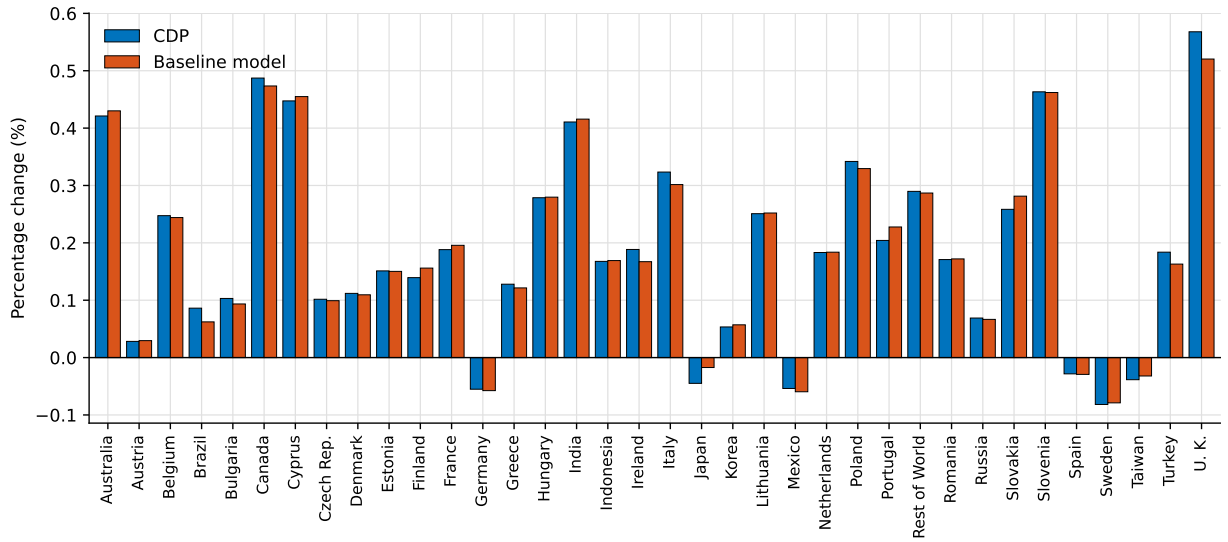
Notes. Panel A is the manufacturing share of employment, in percentage points of employment; Panel B is the employment rate, in percentage points of population.

Figure E.1: Distribution of Welfare Effects



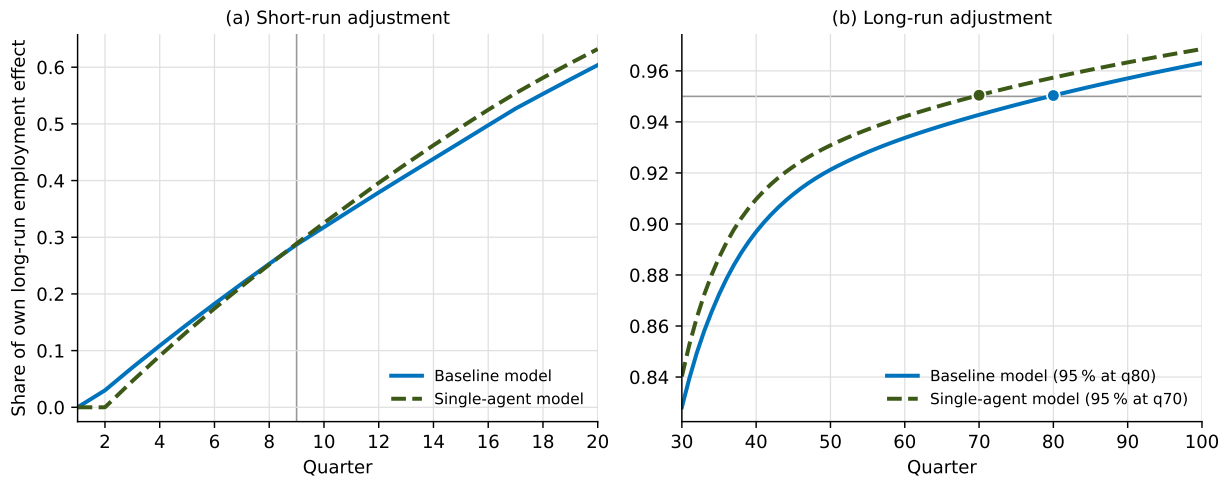
Notes. State welfare minus the group's own national welfare, money-metric consumption-equivalent in percentage points. For each group, I report the coefficient of variation in the title.

Figure E.2: Comparison of Welfare Effects in the Rest of the World



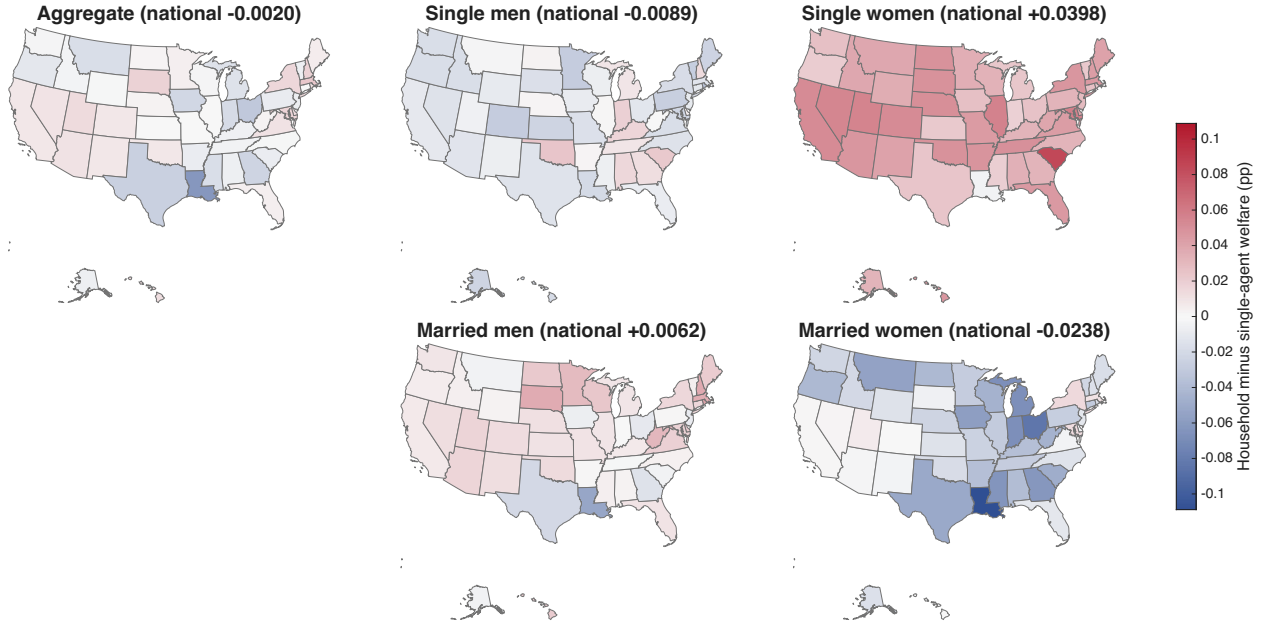
Notes. Consumption-equivalent welfare gains calculated in both models as in Caliendo et al. (2019). Across the 36 regions, the two series correlate at 0.997, with mean gains of 0.187% (CDP) and 0.186% (baseline model).

Figure E.3: Speed of Adjustment of Total Employment



Notes. In each panel, I plot how much of the adjustment of total employment in $t = 199$ has taken place in each quarter. The gray vertical line in Panel A shows the quarter in which both lines cross. The gray horizontal line in Panel B shows the quarter in which each economy reaches 95% of its long-run value.

Figure E.4: Distribution of Welfare Effects Gaps Across Economies



Notes. In each map I plot, the baseline model state money-metric welfare minus single-agent model, in percentage points, for the same people located at their $t = 1$ state. One shared scale; the national gap is in each title. In the baseline, a husband and wife carry the couple's welfare, so the two married panels differ only through the single-agent side.

Table E.2: Parameter Values to Shut Down Margins

Counterfactual	Instrument	Value
C1. No migration response	$1/\vartheta \rightarrow 0$	$\vartheta : 10.67 \rightarrow 10^6$
C2. No participation response	$\kappa/\gamma \rightarrow 1^+$	$\kappa : 2.1695 \rightarrow 0.65$
C3. No gender-wage response	$1/\sigma \rightarrow 0$	$\sigma : 1.4 \rightarrow 100$

Notes. The first counterfactual is the only one that fully shuts down the margin. For counterfactual 2, since the model requires $\kappa > \gamma$, the most the response can be damped is $\kappa_0/\gamma = 4.44\times$; at $\kappa = 0.65$, the response is damped $3.39\times$. I also re-pin χ to keep the anchor participation rate unchanged. For the third counterfactual, I set $1/\sigma = 0.01$ against a calibrated 0.714, so 1.4% of the gender-wage response survives.

Table E.3: Ever Moved State by $t = 199$ in Counterfactual 2

	Baseline	No participation resp.	Difference	% change
<i>Single men</i> — own sector				
Manufacturing	+0.0155	+0.0163	+0.0008	+4.9
Other sector	−0.0051	−0.0047	+0.0004	−7.6
Manuf. − other	+0.0206	+0.0210	+0.0004	+1.8
<i>Single women</i> — own sector				
Manufacturing	+0.0119	+0.0108	−0.0011	−9.2
Other sector	+0.0004	+0.0012	+0.0007	—
Manuf. − other	+0.0115	+0.0096	−0.0018	−16.0
<i>Couples</i> — husband's sector				
Manufacturing	+0.0130	+0.0134	+0.0004	+2.9
Other sector	−0.0129	−0.0135	−0.0006	+4.6
Manuf. − other	+0.0259	+0.0269	+0.0010	+3.7
<i>Couples</i> — wife's sector				
Manufacturing	−0.0018	−0.0019	−0.0002	+9.2
Other sector	−0.0071	−0.0076	−0.0005	+7.3
Manuf. − other	+0.0053	+0.0057	+0.0004	+6.7

Notes. Individuals are tagged at $t = 1$ based on the sector in which they are employed and the state in which they live. Couples are tagged twice, by each spouse's $t = 1$ sector; a non-working wife's sector is her potential sector. Every wife has one, so the wife's rows partition couples, while the husband's rows exclude the 27.2% of couples whose husbands are not in the labor force. A mover is defined as anyone who has changed state at least once by $t = 199$. "Difference" is the exercise minus the baseline, so a negative value means the shut-down shrank the response.

Table E.4: Ever Moved State by $t = 199$ in Counterfactual 3

	Baseline	No gender-wage resp.	Difference	% change
<i>Single men</i> — own sector (17.5% manufacturing, 72.2% other)				
Manufacturing	+0.0155	+0.0165	+0.0010	+6.6
Other sector	−0.0051	−0.0036	+0.0015	−28.6
Manuf. − other	+0.0206	+0.0202	−0.0004	−2.1
<i>Single women</i> — own sector (9.7% manufacturing, 77.2% other)				
Manufacturing	+0.0119	+0.0128	+0.0009	+7.8
Other sector	+0.0004	+0.0017	+0.0013	—
Manuf. − other	+0.0115	+0.0111	−0.0004	−3.3
<i>Couples</i> — husband's sector (16.8% manufacturing, 56.0% other)				
Manufacturing	+0.0130	+0.0125	−0.0005	−3.5
Other sector	−0.0129	−0.0125	+0.0004	−3.0
Manuf. − other	+0.0259	+0.0251	−0.0008	−3.3
<i>Couples</i> — wife's sector (9.7% manufacturing, 90.3% other)				
Manufacturing	−0.0018	−0.0016	+0.0001	−7.0
Other sector	−0.0071	−0.0067	+0.0004	−5.6
Manuf. − other	+0.0053	+0.0051	−0.0003	−5.1

Notes. Individuals are tagged at $t = 1$ based on the sector in which they are employed and the state in which they live. Couples are tagged twice, by each spouse's $t = 1$ sector; a non-working wife's sector is her potential sector. Every wife has one, so the wife's rows partition couples, while the husband's rows exclude the 27.2% of couples whose husbands are not in the labor force. A mover is defined as anyone who has changed state at least once by $t = 199$. "Difference" is the exercise minus the baseline, so a negative value means the shut-down shrank the response.